Fair Car – AI-Based Used Car Price Prediction

A Project Report

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Project Declaration

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I hereby declare that the project entitled, "**Fair Car – AI-Based Used Car Price Prediction**", is my own work, based on my personal study and/or research and that I have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication.

I certify that I have not copied, in part or whole, nor otherwise plagiarized the work of any other person.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (DATA SCIENCE)** to be submitted as Semester V project as part of our curriculum.

Student's Signature

ABSTRACT

In today's growing automobile market, determining a fair resale value for used vehicles has become a major challenge for both buyers and sellers. Traditional methods often rely on guesswork or unverified listings, leading to inaccurate pricing and lack of transparency.

This project, "Fair Car – AI-Based Used Car Price Prediction," aims to build a web-based application that uses machine learning algorithms to predict the fair market value of used cars based on factors such as brand, model, year, mileage, fuel type, and condition. By leveraging data-driven insights, the system ensures accuracy, fairness, and trust in vehicle valuation.

This solution promotes transparency and consumer empowerment in the second-hand automobile market by providing reliable, AI-based price estimation. It bridges the gap between technology and real-world needs, aligning with the vision of building impact-driven, intelligent systems for modern society.

The project highlights the integration of artificial intelligence and web technologies to deliver an efficient, accessible, and user-friendly platform. It not only simplifies the process of determining car prices but also educates users about key factors influencing vehicle depreciation and value trends.

Furthermore, the implementation of this project strengthens practical understanding of machine learning, data pre-processing, model evaluation, and real-time deployment. It demonstrates how AI-driven solutions can transform traditional market systems into transparent and data-centric environments.

Looking ahead, the project can be further enhanced by integrating real-time market data and user feedback to improve model accuracy. Incorporating features such as dynamic price adjustment, vehicle image analysis, and sentiment analysis of online listings could make the system more adaptive and intelligent. These future enhancements will strengthen the model's reliability and expand its use across multiple regions and vehicle categories.

Overall, Fair Car exemplifies the potential of technology in addressing real-world economic challenges while fostering trust, fairness, and sustainability within the used car ecosystem.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

The automotive industry stands as one of the most influential pillars of the modern global economy. Since the inception of the assembly line by Henry Ford in the early 20th century, the automobile has transformed from a luxury novelty into an indispensable utility for billions of people worldwide. It has reshaped cities, influenced geopolitical strategies regarding oil, and fundamentally altered the social fabric by enabling personal mobility on an unprecedented scale. Within this vast ecosystem, the market for pre-owned or "used" vehicles has quietly grown to become a titan, often eclipsing the new car market in terms of sheer transaction volume and economic velocity.

In many developed and developing economies, the ratio of used car sales to new car sales is significantly high. For instance, in markets like the United States and parts of Europe, used car transactions can outnumber new car sales by a factor of three to one. This dominance is driven by a confluence of economic and practical factors. Firstly, the depreciation curve of a new vehicle is notoriously steep; a new car can lose anywhere from 20% to 30% of its value within the first year of ownership. This rapid devaluation makes new cars a poor financial investment for many. In contrast, a used vehicle, having already undergone this initial depreciation cliff, offers a more stable asset value and a lower entry price, making car ownership accessible to a broader demographic, including students, young professionals, and lower-income families.

Screenshot 1.1 : Used Car Price Prediction

However, the transition from a new car-centric market to a vibrant used car economy has not been without its structural challenges. Unlike the market for new commodities, where prices are standardized by manufacturers and enforced through authorized dealerships (MSRP), the used car market is inherently fragmented and unregulated regarding pricing. Each used car is a unique asset, its value etched by its specific

history—how it was driven, where it was stored, how well it was maintained, and the specific accidents or repairs it has undergone. This uniqueness makes the valuation of used cars a complex, multi-dimensional problem that has puzzled economists and consumers alike for decades.

Historically, the valuation process was an opaque art form. It relied heavily on the subjective judgment of experts—mechanics, dealers, and seasoned car enthusiasts. These individuals would inspect a vehicle, listen to the engine, check the paintwork, and offer a price based on their intuition and local market knowledge. While often effective on a small scale, this manual approach is fraught with inconsistency. It is susceptible to human bias, limited by the individual's specific experience, and unable to process the vast, dynamic trends of the macro market. A dealer in a rural area might value a pickup truck significantly differently than a dealer in a metropolitan centre, leading to price dispersion that inefficiencies in the market.

The late 20th and early 21st centuries saw the rise of "Blue Books" and printed valuation guides. These publications attempted to standardize pricing by collecting average transaction data and publishing it periodically. While a step forward, these guides were static. They could not account for the rapid fluctuations in the market caused by external shocks—such as a sudden rise in fuel prices shifting demand from SUVs to compact cars, or supply chain disruptions (like the semiconductor shortage of 2020-2022) causing used car prices to skyrocket. Furthermore, they offered a "one-size-fits-all" price range that often failed to account for the granular details of a specific vehicle, such as a rare trim package or a specific colour combination that might command a premium.

The dawn of the Digital Age and the proliferation of the internet brought the next wave of transformation. Online marketplaces like eBay Motors, Craigslist, and specialized auto portals allowed buyers and sellers to connect directly, bypassing traditional intermediaries. This explosion of digital listings created a treasure trove of data—millions of data points representing car attributes and their asking prices. However, the sheer volume of this data overwhelmed the human capacity to process it. A buyer could now see thousands of cars, but determining which one was "fairly" priced became an even more daunting task due to information overload.

This is where the Fourth Industrial Revolution, specifically the domain of Artificial Intelligence (AI) and Machine Learning (ML), enters the narrative. AI offers the computational power and algorithmic sophistication required to make sense of this chaotic market. Unlike human analysts who might look at three or four variables (Year, Make, Model, Mileage), Machine Learning algorithms can ingest dozens, even hundreds, of variables simultaneously. They can identify non-linear relationships and subtle interaction effects that are invisible to the human eye. For example, an algorithm might learn that for a specific brand of sports car, the colour "Red" commands a 5% premium, but only if the car is less than 5 years old; or that a sunroof adds value to a sedan but has negligible impact on a convertible.

Screenshot 1.2 : AI – Driven Valuations

The "Fair Car" project is conceived at this intersection of automotive economics and advanced data science. It represents a shift from subjective, heuristic-based valuation to objective, data-driven prediction. By leveraging state-of-the-art regression algorithms, "Fair Car" aims to ingest historical transaction data, learn the latent patterns governing price formation, and output a precise, scientifically backed price prediction. This is not merely a technical exercise; it is an economic intervention designed to bring transparency, efficiency, and trust to a market that has historically lacked all three.

1.2 Problem Statement

The used car market, despite its vast size and liquidity, is classically cited in economic theory as an example of a market plagued by "Information Asymmetry." In his seminal 1970 paper, "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism," economist George Akerlof described how the quality of goods in a market can degrade if buyers cannot distinguish between high-quality goods ("peaches") and low-quality goods ("lemons"). This theoretical framework perfectly encapsulates the core problem that the "Fair Car" project seeks to address.

1. Information Asymmetry and the "Lemons" Problem: In a typical used car transaction, the seller possesses a significant informational advantage. They know the vehicle's complete history—every cold start, every pothole hit, every delayed oil change, and every minor accident that might not show up on a standard history report. The buyer, on the other hand, sees only the cosmetic exterior and the odometer reading. Because the buyer cannot be certain of the car's quality, they are unwilling to pay a premium price, fearing they might be buying a "lemon." This forces sellers of high-quality cars to lower their prices to match the market average. Eventually, sellers of high-quality cars may exit the market because they cannot get a fair price, leaving the market dominated by lower-quality vehicles. This adverse selection leads to market failure. "Fair Car" aims to reduce this asymmetry by providing an objective, data-based valuation that serves as a proxy for quality and fair market value, bridging the trust gap between buyer and seller.

2. The Subjectivity and Inconsistency of Manual Valuation: Current valuation methods are heavily reliant on human judgment, which is inherently flawed. Cognitive Bias: Humans are susceptible to anchoring bias (relying too heavily on the first piece of information offered), confirmation bias (interpreting data to confirm pre-existing beliefs), and emotional bias (overvaluing a car due to sentimental attachment).Inconsistency: A car valued at $10,000 by one dealership might be valued at $8,500 by another just down the street. This variance creates confusion for consumers and opportunities for arbitrage that do not reflect true value creation. Lack of Scalability: Manual valuation is slow and labour intensive. A professional appraiser can only evaluate a limited number of vehicles per day. In a digital marketplace with millions of listings, manual appraisal is impossible.

3. The Complexity of Pricing Variables: The price of a used car is not a simple linear function of its age. It is a complex, multi-dimensional surface determined by the interaction of numerous variables. On-Linear Depreciation: Cars do not depreciate at a constant rate. A luxury car might lose 40% of its value in year one, while a reliable economy car might only lose 15%. Brand Perception: The value of a "brand" is intangible but powerful. A Toyota with 100,000 miles might be worth more than a Fiat with 50,000 miles due to the market's perception of reliability. Regional Variance: A convertible is worth more in California than in Alaska. A 4x4 truck commands a premium in rural areas compared to urban centres. Macroeconomic Factors: Inflation, fuel prices, and interest rates all exert dynamic pressure on used car prices. The average consumer, and even the average dealer, lacks the computational capacity to

weigh all these factors simultaneously and accurately. They rely on heuristics (rules of thumb) that are often outdated or oversimplified.

4. The "Black Box" Nature of Existing Tools: While there are online valuation tools available (e.g., Kelley Blue Book, Edmunds), their methodologies are often proprietary and opaque. Users input data and get a number, but they are given little insight into why that number was generated. Furthermore, these tools often provide a wide "range" (e.g., "Your car is worth between $12,000 and $15,000") which, while statistically safe, is often too broad to be actionable for a specific negotiation. There is a lack of tools that offer a precise point estimate backed by transparent, explainable AI logic.

5. Time and Effort Costs: For a buyer to ensure they are getting a fair deal, the current "best practice" involves hours of research: browsing multiple websites, creating spreadsheets to compare similar listings, calling dealers, and visiting lots. This high "search cost" is a friction in the market that discourages participation and reduces overall economic efficiency. In summary, the problem is that the used car market is inefficient, opaque, and risky due to information asymmetry and the cognitive limitations of human valuation. There is a critical need for an automated, intelligent system that can instantly synthesize vast amounts of market data to provide a precise, objective, and reliable price prediction.

1.3 Research Gap

The application of machine learning to economic forecasting is a well-established field, yet within the specific niche of used car price prediction, several significant research gaps persist that this project aims to address.

1. Limitations of Standard Regression Models: Much of the existing academic literature on car price prediction utilizes basic linear regression models. While these models are easy to interpret, they fundamentally fail to capture the complexity of the real world. Linear regression assumes a straight-line relationship between dependent and independent variables. However, car pricing is inherently non-linear. For example, the relationship between mileage and price is not a straight line; the depreciation impact of mileage accelerates at certain thresholds (e.g., crossing 100,000 miles) and plateaus at others. Existing studies often overlook these non-linearities, leading to models with high error rates (high RMSE). "Fair Car" seeks to close this gap by employing ensemble methods like Random Forest and Gradient Boosting (XG Boost), which can model complex, non-linear decision boundaries.

2. Data Granularity and Feature Engineering: Many previous studies rely on limited datasets, often scraped from a single website or covering a short time. These datasets frequently lack critical granular details. For instance, many models consider "Model" (e.g., Honda Civic) but ignore "Trim Level" (e.g., LX vs. Touring), which can account for a price difference of thousands of dollars. Similarly, features like "Service History" or "Number of Previous Owners" are often excluded due to data availability issues, despite their massive impact on value. This project aims to utilize a more comprehensive dataset and perform rigorous feature engineering to extract maximum signal from the available data, such as

deriving the "age" of the car from the "year" and the current date or creating interaction terms between "Brand" and "Segment."

3. Lack of Regional Contextualization: Most of the high-quality research and open-source models are tailored to the North American or European markets. These markets have specific characteristics (e.g., high prevalence of automatic transmissions, dominance of leasing). Developing markets, such as India or Southeast Asia, have vastly different dynamics (e.g., preference for high fuel efficiency, manual transmissions, different brand hierarchies). A model trained on US data is virtually useless in an Indian context. There is a significant gap in robust, publicly available research focused on developing markets. "Fair Car" addresses this by training on data specific to the target region, ensuring the predictions reflect local market realities.

4. The Gap Between Academia and Industry: There is a "deployment gap." Sophisticated models developed in academic papers often remain as code in a repository or equations in a PDF. They are rarely deployed into user-friendly, accessible applications that non-technical users can interact with. Conversely, commercial tools are user-friendly but scientifically opaque. "Fair Car" aims to bridge this divide by not only building a high-performance model but also wrapping it in a modern, intuitive web interface, making advanced data science accessible to the everyday car buyer.

5. Explainable and Trust: A major gap in AI adoption is the "Black Box" problem. Users are sceptical of algorithmic outputs if they don't understand them. Few existing car price prediction systems offer "explainable"—telling the user why the price is what it is (e.g., "This price is lower than average because the mileage is 20% higher than the norm for this year"). While the primary scope of this project is prediction, the research acknowledges this gap and aims to use models (like Decision Trees/Random Forests) that offer inherent feature importance metrics, paving the way for more explainable AI in future iterations.

1.4 Objectives of the Study

1. To develop an AI-based system for predicting used car prices.
   The system will use important vehicle details such as brand, model, manufacturing year, fuel type, transmission type, and kilometres driven to estimate the approximate market price of a used car.

2. To improve the accuracy and consistency of car price estimation.
   The study aims to use a machine learning model (Random Forest Regression) to analyse vehicle data and generate more accurate and consistent price predictions compared to traditional manual or rule-based pricing methods.

3. To provide a fair and reliable price suggestion for users.
   The system will help buyers and sellers understand the approximate value of a car, making the pricing process more transparent, data-driven, and helpful for better decision-making in the used car market.

1.5 Scope of the Project

Defining the scope is essential to manage expectations and ensure the project remains focused and achievable.

1. Data Scope: The project relies on a structured dataset containing historical records of used car transactions. Attributes Included: The model will consider the following key attributes:
   - Name/Model: The specific make and model of the car (e.g., Maruti Swift).
   - Location: The city or region where the car is being sold.
   - Year: The year of manufacture.
   - Kilometres Driven: The total distance the car has travelled.
   - Fuel Type: Petrol, Diesel, CNG, LPG, Electric.
   - Transmission: Manual, Automatic.
   - Owner Type: First Owner, Second Owner, Third Owner, etc.
   - Mileage (Fuel Efficiency): The rated fuel economy of the car (km/Liter).
   - Engine Capacity: The displacement of the engine (CC).
   - Power: The maximum power output (bhp).
   - Seats: The number of seats.

   Target Variable: The dependent variable to be predicted is the Price (e.g., Lakhs).

Screenshot 1.3 : Car Features

2. Algorithmic Scope: The project is strictly confined to Supervised Machine Learning using Regression techniques. In Scope: Linear Regression, Lasso/Ridge Regression, Decision Trees, Random Forest, Gradient Boosting (XG Boost/Light GBM). Out of Scope: Unsupervised learning (clustering), Deep Learning (Neural Networks/CNNs/RNNs) - unless traditional ML fails to yield acceptable results. Time-series forecasting is also out of scope; the model treats each transaction as an independent event rather than a sequence in time.

3. Application Scope: The final artifact will be a software application. Platform: Web Application (accessible via a browser). Functionality: Input form for car details -> "Predict" button -> Display of estimated price. Exclusions: The app will not be a marketplace. It will not allow users to list cars for sale, contact sellers, or process payments. It is a calculator/estimator tool only.

4. User Scope: The system is designed for three primary user personas: The Novice Buyer: A person with little knowledge of cars who wants a quick check to see if a price is fair. The Private Seller: An individual wanting to know what their car is worth before posting an ad. The Small Dealer: A local used car dealer who needs a quick reference tool for valuation.

5. Limitations: Visual Inspection: The model cannot "see" the car. It assumes the car is in "average" condition for its age and mileage. It cannot account for scratches, dents, torn upholstery, or a pristine, showroom-quality maintenance record unless these are explicitly captured in a "Condition Grade" variable (which is often missing in standard datasets). Market Shocks: The model is trained on historical data. It may lag in responding to sudden, black-swan events (e.g., a sudden change in tax law, a pandemic, a recall). Rare Vehicles: The model will perform poorly on rare, vintage, or exotic cars (e.g., a 1960s Mustang or a Ferrari) because there are too few data points to learn a reliable pattern. It is optimized for mass-market consumer vehicles.

1.6 Significance of the Study

The "Fair Car" project is not merely an academic requirement; it has profound implications for the real-world automotive ecosystem. Its significance can be understood through the lens of various stakeholders.

1. Economic Significance: The used car market is a multi-billion-dollar industry. Inefficiencies in this market result in "Deadweight Loss"—economic value that is lost due to transactions that fail to happen because buyers and sellers cannot agree on a price. By providing a trusted pricing standard, "Fair Car" reduces friction, encourages more transactions, and increases the overall liquidity and velocity of the market. It helps align the market closer to the ideal of "Perfect Competition" where information is freely available.

2. Significance for the Consumer (Buyer): For most households, a car is the second most expensive purchase they will ever make, after a home. Overpaying for a car by even 10-15% represents a significant financial loss—money that could have been saved for education, healthcare, or investment. "Fair Car" acts as a financial safeguard. It provides the buyer with leverage during negotiations. Instead of relying on the seller's word, the buyer can point to the AI prediction as an objective

benchmark. This shifts the power dynamic from the seller (who usually knows more) to a more balanced state.

3. Significance for the Seller: Sellers often struggle with the "Endowment Effect"—the psychological tendency to value an object higher simply because they own it. This leads to unrealistic asking prices. A car listed too high will sit unsold for months, depreciating further and incurring maintenance/storage costs. "Fair Car" provides a reality check. It helps sellers set a competitive price that attracts buyers while ensuring they don't leave money on the table. It speeds up the sales cycle, converting the asset back into cash more quickly.

4. Significance for the Industry (Dealers & Fintech): Dealers: For used car dealerships, time is money. The faster they can evaluate a trade-in and get it on the lot, the better. An automated tool streamlines the acquisition process. Fintech & Insurance: Banks need to know the value of a car to issue a loan (Loan-to-Value ratio). Insurers need it to calculate premiums and pay-outs for total losses. Currently, these industries spend millions on manual appraisal services. An AI model offers a scalable, instant, and low-cost alternative for these institutional assessments.

1.7 Structure of the Report

The report for the “Fair Car” project is structured into six comprehensive chapters, each designed to guide the reader through the entire lifecycle of the project, from understanding the problem to implementing and evaluating the AI solution.

Chapter 1: Introduction establishes the foundation by presenting the background of the used car market, highlighting the challenges of fair price estimation, and identifying gaps in existing research. It also clearly defines the objectives, scope, and significance of the project, setting the stage for the work that follows.

Chapter 2: Review of Literature provides an extensive survey of at least twenty to twenty-five research studies. This chapter examines economic theories relevant to pricing, such as Akerlof’s “Market for Lemons,” as well as machine learning applications in automotive price prediction. It compares methodologies like Linear Regression, Random Forest, Gradient Boosting, and Neural Networks, while identifying the limitations of previous approaches that the “Fair Car” project seeks to address.

Chapter 3: Research Methodology explains the technical and conceptual framework of the study. It details the research design, hypothesis formulation, and the alignment of objectives with the hypotheses. This chapter also describes the dataset, including its source, size, and attributes, and discusses the selection of machine learning models and the tools and technologies used, such as Python, Pandas, Scikit-Learn, and Stream lit or Flask. Additionally, it justifies the methodology, providing reasoning for the chosen models and techniques.

Chapter 4: Exploratory Data Analysis (EDA) focuses on understanding and preparing the dataset for modeling. It includes an overview of the data, preprocessing and cleaning steps to handle missing values and outliers, and feature engineering to create meaningful predictors for car prices. Statistical analysis and visualization are

used to identify patterns, trends, and relationships within the data, while the key insights drawn from these analyses help guide the modeling process.

Chapter 5: Implementation documents the development of the "Fair Car" AI system. It describes the overall workflow, the system architecture and model pipeline, and includes code snippets or pseudocode for clarity. The chapter also details the model training process, hyperparameter optimization for performance enhancement, the software stack utilized, and the practical implications of the implemented system for real-world price estimation.

Chapter 6: Results and Discussion present the evaluation of the models using metrics such as Accuracy, R-squared, MAE, and RMSE. It interprets the experimental results, compares the performance of different algorithms, and demonstrates improvements over existing baseline methods. Graphs, charts, and tables are provided to visually summarize findings and facilitate understanding of the outcomes.

CHAPTER 2: REVIEW OF LITERATURE

2.1 Vaneesha K. H. et al., 2024 [21] The authors conducted a comparative analysis of K-Nearest Neighbor (KNN) and Support Vector Machine (SVM) regression techniques for predicting used car prices. The study utilized a dataset of used car sales to evaluate the performance of both models. Results indicated that the KNN algorithm achieved higher accuracy and lower error rates compared to SVM in estimating vehicle prices. The researchers emphasized that KNN's ability to find similar historical examples made it particularly effective for this domain, though they noted the computational cost increases with dataset size. The study also analyzed the impact of feature scaling and normalization on model performance. Furthermore, the authors discussed the role of hyperparameter tuning for both KNN and SVM. The paper highlighted that KNN's interpretability allows stakeholders to trace predictions back to similar historical cars, which can improve trust in automated systems. In addition to the original findings, the authors elaborated that used car pricing is inherently comparison-based, where buyers and sellers rely heavily on reference prices of similar vehicles, making instance-based learning approaches particularly relevant. The dataset was carefully preprocessed to remove inconsistencies, handle missing values, and standardize numerical attributes so that no single feature dominated the distance calculations. The authors further explained that improper scaling could significantly distort similarity measurements, leading to biased predictions, especially in distance-driven algorithms like KNN. The study also explored the influence of different distance metrics and neighborhood sizes, observing that prediction stability improves when the neighborhood size is neither too small nor excessively large. Moreover, the researchers analyzed SVM kernel selection and noted that while non-linear kernels can improve accuracy, they also introduce additional complexity and sensitivity to parameter selection. The study emphasized that hyperparameter tuning plays a decisive role in achieving optimal performance for both algorithms, requiring systematic experimentation. Additionally, the authors discussed scalability challenges in real-world deployment and suggested the use of optimized data structures to mitigate KNN's computational overhead. The research also highlighted practical implications for online car marketplaces, where transparent and explainable pricing models are increasingly demanded by users. The authors concluded that while SVM offers strong theoretical generalization capabilities, KNN provides a more intuitive and interpretable framework for used car price prediction, making it particularly suitable for applications where explainable, user trust, and market transparency are critical requirements. In addition to these findings, the study emphasized that used car pricing is inherently similarity-driven, making instance-based learning methods highly suitable for such applications. The authors also examined multiple distance metrics and found that Euclidean distance performed effectively after normalization, reinforcing the importance of preprocessing steps. The research further discussed scalability concerns and suggested optimization techniques for large datasets. The study concluded that while SVM remains a strong theoretical model, KNN offers practical advantages in transparency and real-world deployment for automotive price prediction systems.

2.2 Daniel Aprille Budiono et al., 2024 [5] This study proposed a machine learning approach using the K-Nearest Neighbors (KNN) regression model to predict used car prices, with a primary focus on achieving highly accurate and reliable valuation outcomes in real-world scenarios. The authors concentrated on optimizing the value of the parameter 'K' to minimize prediction error and enhance model stability across different subsets of data. Their experimental evaluation demonstrated an exceptionally high predictive performance, reporting an R-squared value of 98.8% along with a relatively low error rate of 8.3%, which clearly indicates the effectiveness of the proposed approach. The paper concluded that KNN is a highly dependable method for used car price estimation when it is trained on sufficiently large and representative datasets, emphasizing the algorithm's simplicity, transparency, and ease of implementation. The authors also stressed the importance of rigorous data preprocessing, particularly the treatment of missing values, outliers, and inconsistent records, as poor data quality can significantly distort distance-based predictions. Feature normalization was identified as a critical step to ensure that attributes measured on different scales, such as mileage and engine capacity, do not disproportionately influence distance calculations. In addition to the core findings, the study examined the influence of various vehicle-related features, including age, mileage, fuel type, transmission, brand reputation, and engine specifications, highlighting that careful feature selection plays a decisive role in improving predictive accuracy. The researchers observed that irrelevant or weakly correlated features introduce noise and reduce the discriminatory power of the KNN algorithm, thereby underscoring the need for feature relevance analysis prior to model training. The paper further analyzed the sensitivity of KNN performance to different values of K, explaining that very small K values can lead to overfitting and unstable predictions, whereas excessively large values may over smooth the results and fail to capture localized pricing patterns present in the market. Moreover, the authors discussed computational considerations associated with KNN, noting that although the algorithm does not require an explicit training phase, its prediction time grows linearly with the size of the dataset, which can present scalability challenges in large online marketplaces. To address this issue, the study suggested the use of efficient data structures, dimensionality reduction techniques, and weighted distance metrics to improve both accuracy and computational efficiency. The authors also explored the potential of combining KNN with ensemble or hybrid approaches to enhance robustness under fluctuating market conditions. Additionally, the research highlighted that the non-parametric nature of KNN allows it to adapt effectively to evolving market trends when retrained periodically with updated data. The study concluded that with meticulous preprocessing, optimal parameter tuning, and thoughtful feature selection, KNN can serve as a powerful, interpretable, and practically viable solution for used car price prediction systems deployed in real-world decision-support and e-commerce platforms. The study suggested that performance can be enhanced using weighted distance metrics, where closer neighbors have a greater influence on the predicted price, thereby improving accuracy in heterogeneous datasets. Additionally, the authors discussed the potential integration of KNN with ensemble or hybrid approaches to further stabilize predictions under fluctuating market conditions. The research highlighted that KNN's non-parametric nature allows it to adapt effectively to changing market trends when models are periodically retrained with recent data. The authors concluded that with careful preprocessing, parameter tuning, and feature selection, KNN serves as a powerful and interpretable tool for used car price prediction in online marketplaces and decision-support systems.

2.3 Amneh J. Kenana et al., 2024 [10] The researchers explored the application of multiple machine learning techniques, including Random Forest, Lasso Regression, Ridge Regression, and Linear Regression, to address the growing complexity associated with modern used car price prediction. The study was motivated by the observation that used car pricing is influenced by a wide range of interdependent variables such as brand reputation, model type, vehicle age, mileage, physical condition, fuel type, engine capacity, ownership history, and prevailing market demand, many of which demonstrate non-linear relationships that are difficult to model using traditional statistical approaches. To systematically evaluate the effectiveness of these algorithms, the authors conducted extensive experimental validation using real-world datasets and assessed model performance through established evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The experimental results clearly indicated that the Random Forest model achieved the lowest MAE and RMSE values among all evaluated techniques, demonstrating superior predictive accuracy and consistency in estimating used car prices. The authors attributed this strong performance to Random Forest's ensemble-based learning structure, which aggregates predictions from multiple decision trees and thereby captures complex interactions between features that linear regression-based models often fail to represent adequately. In contrast to Linear, Lasso, and Ridge Regression, Random Forest was shown to be less sensitive to multicollinearity and more capable of handling heterogeneous datasets containing both numerical and categorical attributes without relying on rigid assumptions about data distribution. The study further analysed the impact of hyperparameter tuning, emphasizing that parameters such as the number of trees, maximum tree depth, minimum samples required for node splitting, and feature subset selection significantly influence model accuracy, generalization capability, and resistance to overfitting. Additionally, the authors highlighted the critical role of data preprocessing techniques, including categorical feature encoding, normalization of numerical variables, outlier treatment, and noise reduction, in improving the overall reliability and stability of predictive models. Cross-validation strategies were employed extensively to ensure that performance gains were not limited to the training data and that the models generalized effectively to unseen vehicle listings. The research also emphasized that Random Forest demonstrated strong robustness when dealing with noisy, incomplete, or inconsistent data, which are common characteristics of real-world used car datasets collected from online marketplaces and classified advertisements. Furthermore, the authors discussed the interpretability advantages of Random Forest through feature importance analysis, enabling stakeholders to understand which vehicle attributes exert the greatest influence on pricing decisions and thereby improving transparency and trust in automated valuation systems. The study also noted that the ability of Random Forest to model non-linear depreciation patterns makes it particularly suitable for capturing real-world market behaviour. The researchers concluded that Random Forest offers an effective balance between predictive accuracy, robustness, and interpretability, making it a highly practical and reliable choice for real-world used car price prediction systems deployed in decision-support tools and online automotive platforms operating within dynamic and competitive market environments.

2.4 P. Bhatnagar, 2024 [4] The author conducted an in-depth investigation into the application of machine learning techniques for predicting used car prices, with the primary objective of improving valuation accuracy and ensuring fairness within second-hand automobile markets. The study was based on the understanding that used car pricing is affected by a broad and complex set of variables, including vehicle age, mileage, brand reputation, model popularity, engine specifications, fuel type, transmission system, ownership history, maintenance records, and prevailing market conditions, many of which interact in non-linear and unpredictable ways. To address this complexity, the author utilized a comprehensive dataset consisting of historically used car sales records and implemented multiple regression-based machine learning models, including Linear Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting techniques. Extensive data preprocessing was performed prior to model training, involving the treatment of missing values, removal of inconsistent records, encoding of categorical variables, normalization of numerical attributes, and outlier handling to enhance data quality and model reliability. The performance of each model was rigorously evaluated using standard error metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE), enabling a detailed and objective comparison of predictive accuracy. The experimental findings revealed that ensemble learning models, particularly Random Forest and Gradient Boosting, consistently outperformed traditional Linear Regression approaches by effectively capturing non-linear relationships and complex interactions among vehicle attributes. The study emphasized that these ensemble models demonstrated greater robustness against noise and data variability, which are common challenges in real-world used car datasets sourced from online platforms and classified advertisements. The author also examined the role of hyperparameter optimization, highlighting that careful tuning of parameters such as the number of trees, maximum tree depth, learning rate, and minimum samples per leaf significantly contributed to reducing prediction errors and improving generalization performance. Cross-validation techniques were applied extensively to ensure that the trained models did not suffer from overfitting and that their predictive capabilities remained stable when applied to unseen data. Furthermore, the study discussed the interpretability aspects of machine learning models, noting that feature importance analysis provided by ensemble methods helps identify key factors influencing car prices, thereby improving transparency and user trust in automated pricing systems. The research also highlighted the practical significance of deploying machine learning-based valuation models in online used car marketplaces, as such systems can assist buyers in making informed purchasing decisions and help sellers set competitive and realistic prices. The author concluded that machine learning-driven pricing frameworks, particularly those based on ensemble learning techniques, offer a scalable, accurate, and reliable solution for used car price prediction and play a crucial role in enhancing efficiency, fairness, and transparency in modern automotive markets.

2.5 S. Patel et al., 2023 [7] The authors presented a detailed study focused on the development of an accurate and reliable used car price prediction system by leveraging advanced machine learning techniques and comprehensive data analysis. The research was grounded in the recognition that used car pricing is a multifaceted problem influenced by numerous variables, including vehicle age, mileage, brand reputation, model demand, fuel type, transmission system, engine capacity, vehicle condition, ownership history, geographic location, and fluctuating market trends, all of which interact in complex and often non-linear ways. To effectively model these relationships, the authors utilized a large-scale dataset consisting of historically used car sales records collected from online automotive platforms. Prior to model development, extensive data preprocessing was performed, including the removal of duplicate records, treatment of missing values, normalization of numerical features, encoding of categorical variables, and outlier detection to ensure data consistency and quality. The study implemented multiple machine learning algorithms such as Linear Regression, Support Vector Regression, Decision Tree Regression, Random Forest Regression, and Gradient Boosting methods to compare their effectiveness in predicting vehicle prices. Model performance was evaluated using standard error metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE), enabling a robust comparison of prediction accuracy and stability across different techniques. The experimental results demonstrated that ensemble-based models, particularly Random Forest and Gradient Boosting, achieved superior predictive performance compared to traditional regression models by successfully capturing non-linear relationships and complex feature interactions inherent in used car pricing data. The authors emphasized that these models exhibited strong robustness to noise, missing information, and data variability, which are common challenges in real-world used car datasets. The study also highlighted the importance of feature selection and hyperparameter tuning, noting that optimizing parameters such as tree depth, number of estimators, learning rate, and minimum samples per split played a significant role in minimizing prediction errors and enhancing model generalization. Cross-validation techniques were extensively applied to validate model performance and prevent overfitting, ensuring that the predictive models remain reliable when applied to unseen data. Additionally, the research discussed the interpretability advantages of ensemble learning methods through feature importance analysis, allowing stakeholders to identify the most influential factors affecting vehicle prices and thereby improving transparency and trust in automated valuation systems. The authors further explored the practical implications of integrating machine learning-based price prediction models into online used car marketplaces, highlighting their potential to support fair pricing, reduce information asymmetry, and enhance decision-making for both buyers and sellers. The study concluded that machine learning-driven price prediction systems, particularly those based on ensemble learning techniques, provide an effective, scalable, and accurate solution for addressing the complexities of used car valuation in dynamic and competitive automotive markets.

2.6 F. Uysal, 2024 [19] This research focused on the comparative evaluation of classical machine learning algorithms and deep learning approaches for predicting used car resale prices within the Turkish automotive market, with the goal of identifying models that provide both high predictive accuracy and practical feasibility for deployment in real-world marketplaces. The study utilized a unique and comprehensive dataset named "Car Prices in the Wild," which encompassed historical sales records, vehicle specifications, maintenance history, ownership details, mileage, fuel type, engine capacity, and market location, allowing for a detailed exploration of the complex factors influencing used car pricing. Prior to analysis, extensive data preprocessing was undertaken, including handling of missing and inconsistent values, standardization of numerical features, encoding of categorical variables such as brand, model, and gearbox type, and outlier detection to ensure high-quality input for model training. The study implemented a variety of predictive models, including classical ensemble methods like Random Forest and XG Boost, as well as deep learning architectures such as feedforward neural networks and convolutional neural networks adapted for tabular data. Each model's performance was rigorously assessed using metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R-squared, and cross-validated predictive stability to ensure a comprehensive understanding of their relative effectiveness. The findings revealed that while deep learning models demonstrated promising results under controlled conditions, well-tuned classical ensemble methods frequently matched or exceeded their performance in terms of accuracy while requiring significantly less computational overhead and training time. The study emphasized the importance of region-specific data sets to accurately capture local market dynamics, noting that generic models often fail to account for specific trends in vehicle depreciation, demand variations, and brand popularity within a particular country. The research also highlighted the role of feature engineering, observing that incorporating variables such as recent service records, accident history, and fuel efficiency contributed to significant improvements in prediction reliability. Hyperparameter tuning was extensively explored for both ensemble and deep learning models, with careful adjustment of parameters such as tree depth, number of estimators, learning rate, batch size, and network architecture shown to have a substantial effect on predictive accuracy. The study additionally discussed interpretability considerations, noting that ensemble models offer the advantage of feature importance analysis, enabling stakeholders to identify key determinants of car value and enhance user trust in automated pricing platforms. Practical implications were further examined, demonstrating that integrating these models into online used car marketplaces could improve transparency, reduce pricing errors, and facilitate more informed decision-making for both buyers and sellers. The research concluded that hybrid approaches, combining classical ensemble methods with deep learning techniques, could be explored in future work to further enhance prediction accuracy in dynamic automotive markets, and emphasized that adopting machine learning-driven valuation systems can significantly contribute to fair, scalable, and transparent pricing practices in the modern used car industry.

2.7 S. Muti and K. Yildiz, 2023 [12] The authors conducted an in-depth investigation into the application of the Linear Regression model for predicting used car prices, aiming to understand both its practical effectiveness and limitations in real-world automotive markets. Recognizing that linear regression is often overlooked in favor of more complex models, the study sought to demonstrate that with meticulous data preprocessing, feature selection, and validation techniques, linear models can still provide robust and interpretable predictions. The dataset comprised a diverse set of historical used car sales, capturing a wide array of features including vehicle age, mileage, brand, model, fuel type, engine capacity, transmission type, vehicle condition, accident history, ownership duration, and regional market characteristics. The authors emphasized the importance of cleaning and preprocessing the dataset, which involved handling missing values through imputation strategies, removing duplicate and inconsistent entries, detecting and correcting outliers, normalizing numerical features, and encoding categorical variables to ensure the integrity and consistency of the input data. Special attention was given to feature analysis and multicollinearity assessment, as correlated predictors can significantly distort model coefficients and reduce interpretability. Linear Regression was trained and validated using standard regression techniques, and performance metrics including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared were carefully analyzed to quantify predictive accuracy. The model achieved an R-squared value of 73%, indicating a moderate but reliable level of accuracy while preserving the advantages of interpretability. The study further explored the strengths of Linear Regression in highlighting the contribution of individual features to pricing decisions, allowing stakeholders such as dealers, buyers, and online marketplaces to understand the rationale behind predicted prices. Additionally, the authors addressed the inherent limitations of linear models in capturing non-linear interactions and complex dependencies, discussing strategies such as polynomial feature expansion, interaction terms, and hybrid approaches combining linear regression with ensemble methods to enhance predictive power without sacrificing interpretability. Cross-validation techniques were extensively applied to ensure the robustness of the model across various subsets of data, and sensitivity analyses were conducted to evaluate the influence of different features on pricing predictions. The authors highlighted practical deployment considerations, noting that linear regression models are computationally efficient, easy to implement in real-time pricing platforms, and provide a transparent framework for decision-making. Moreover, the study discussed the relevance of continuous monitoring and periodic retraining to adapt to market fluctuations, seasonal trends, and evolving buyer preferences, ensuring sustained accuracy over time. The research concluded that, despite being a simpler approach compared to ensemble or deep learning models, linear regression remains a valuable and interpretable tool for used price prediction, particularly for applications where transparency, simplicity, and stakeholder trust are critical. The authors emphasized that linear models could serve both as reliable standalone solutions in moderately complex markets and as benchmarks for assessing the performance of more sophisticated machine learning algorithms, ultimately contributing to fair, explainable, and data-driven pricing strategies in the automotive domain.

2.8 Aarone Steve J. Alex stan et al., 2023 [2] This study focused on developing a robust predictive model for estimating continuous price values of used cars by applying a combination of regression algorithms and evaluating their performance across multiple metrics to determine the most suitable approach for practical deployment in online automotive marketplaces. The researchers collected a comprehensive dataset encompassing a wide range of vehicle attributes, including age, mileage, brand, model, fuel type, engine size, transmission type, accident history, service records, ownership details, and regional market variations, ensuring that the model accounted for the multifaceted factors influencing car prices. Data preprocessing played a critical role, with the authors undertaking missing value imputation, outlier detection and removal, standardization of numerical variables, and encoding of categorical attributes, thereby enhancing the quality and consistency of inputs and preventing biases in regression calculations. The study implemented and compared multiple algorithms including Linear Regression, Decision Tree, and Random Forest regressors, with each model rigorously evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R-squared values, and cross-validation techniques to ensure generalizability and stability of predictions. Results demonstrated that the Random Forest Regressor consistently outperformed other models due to its ability to handle high-dimensional data, mitigate overfitting, and capture non-linear interactions between variables, which are prevalent in used car datasets. The research further analysed the impact of hyperparameter tuning, such as adjusting the number of trees, maximum depth, minimum samples per leaf, and splitting criteria, observing that optimal configurations significantly enhanced model accuracy and stability. Feature importance analysis was conducted to identify the most influential attributes affecting price prediction, revealing that mileage, age, brand, and engine size were critical determinants, while less prominent features such as optional packages or minor service histories also contributed subtly to pricing variations. The authors emphasized the importance of interpretability in predictive modelling, noting that Random Forest models, despite their complexity, allow stakeholders to understand the relative influence of each feature, enhancing transparency and user trust in automated pricing systems. Moreover, practical deployment considerations were discussed, including computational efficiency, scalability for large online marketplaces, integration with real-time platforms, and adaptability to market trends and seasonal fluctuations. The study highlighted the necessity of continuous model retraining and monitoring to ensure performance under changing market dynamics and emerging vehicle features. The research concluded that Random Forest regression provides a powerful, reliable, and interpretable tool for used car price prediction, capable of supporting buyers, sellers, and online platforms in making informed pricing decisions, while suggesting that hybrid approaches combining Random Forest with other models could further improve robustness and accuracy in dynamic market environments.

2.9 Shashwat Singh Raghav, 2023 [14] The author conducted a comprehensive comparative study between Random Forest and Linear Regression models for predicting used car prices, with the primary objective of evaluating their effectiveness in capturing complex pricing patterns in heterogeneous automotive datasets. The research utilized a large and diverse dataset consisting of historical sales records, encompassing detailed features such as vehicle age, mileage, brand, model, fuel type, engine capacity, transmission type, accident and service history, ownership duration, optional packages, and regional market variations, ensuring a holistic representation of factors influencing vehicle valuation. Prior to model development, extensive data preprocessing was carried out, including handling missing values, removing duplicate and inconsistent entries, outlier detection and mitigation, normalization of numerical features, and encoding of categorical variables, which collectively ensured data integrity and consistency. The Linear Regression model served as a baseline due to its interpretability and simplicity, while the Random Forest model was implemented to leverage its ability to capture non-linear interactions and complex feature dependencies. Both models were trained and evaluated using robust cross-validation techniques, and their performance was assessed using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R-squared, providing a comprehensive assessment of predictive accuracy. Results indicated that the Random Forest model consistently outperformed Linear Regression, effectively capturing subtle and non-linear patterns that Linear Regression missed, particularly in cases where feature interactions and market fluctuations played a significant role in pricing. The study emphasized hyperparameter tuning for Random Forest, including the number of trees, maximum depth, minimum samples per split, and feature selection criteria, which significantly influenced model performance and stability. Feature importance analysis highlighted critical attributes such as mileage, vehicle age, brand reputation, engine capacity, and optional packages, while also identifying the contribution of less prominent variables in influencing price predictions. The author discussed the practical implications of these findings for online automotive marketplaces, dealerships, and valuation platforms, noting that Random Forest provides both accurate and interpretable outputs that can improve transparency, stakeholder trust, and decision-making in used car transactions. Additionally, the research explored computational considerations, including the efficiency of prediction, scalability to large datasets, and integration with real-time platforms, ensuring practical applicability in dynamic market environments. Sensitivity analysis and scenario testing were conducted to evaluate the robustness of predictions under varying data conditions, further validating the reliability of Random Forest in real-world deployment. The study concluded that, while Linear Regression offers interpretability and baseline predictive power, Random Forest is a superior choice for comprehensive, accurate, and practical used car price prediction, especially in heterogeneous datasets with complex feature interactions, making it highly suitable for modern automotive pricing systems and dynamic online marketplaces.

2.10 N. A. Deepak et al., 2023 [7] The authors conducted a comprehensive study on the application of machine learning algorithms for automobile valuation prediction, focusing on automating the traditional manual and subjective process of used car appraisal to enhance accuracy, efficiency, and transparency. The research utilized an extensive dataset containing historical sales records from diverse sources, capturing a wide spectrum of vehicle attributes including age, mileage, brand, model, engine capacity, fuel type, transmission, accident and service history, ownership patterns, and regional market influences, ensuring that the model accounted for all critical factors impacting pricing. Data preprocessing was a central aspect of the study, encompassing missing value imputation, outlier detection and correction, removal of duplicate entries, normalization of numerical features, and encoding of categorical variables, which collectively enhanced data consistency and prevented biases in predictive modeling. The study implemented multiple algorithms, including Linear Regression, Decision Trees, Random Forest, and gradient boosting techniques, each rigorously evaluated to determine their suitability for accurate price prediction. Performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R-squared were carefully analyzed, complemented by cross-validation procedures to ensure the reliability and generalizability of the models. The authors observed that while simpler models like Linear Regression provided interpretability, more sophisticated ensemble methods, particularly Random Forest and gradient boosting models, exhibited superior predictive accuracy due to their ability to capture non-linear interactions among features, handle complex datasets with high dimensionality, and mitigate overfitting through averaging and boosting mechanisms. Hyperparameter tuning was explored extensively, optimizing parameters such as the number of trees, maximum depth, learning rate, and splitting criteria to achieve optimal performance. Feature importance analysis revealed that attributes such as mileage, age, brand reputation, vehicle condition, and optional features played pivotal roles in determining car prices, while additional contextual factors such as seasonal trends, regional demand, and market fluctuations were also found to influence valuation outcomes. The study further examined practical considerations for integrating predictive models into real-world applications, including computational efficiency, scalability for large online marketplaces, and adaptability to evolving market dynamics. The authors highlighted the importance of combining multiple predictive models to enhance robustness and reduce prediction variance, especially in heterogeneous datasets where individual algorithms might exhibit varying performance. Additionally, strategies for monitoring model performance over time, updating models with new data, and incorporating macroeconomic and regional market indicators were discussed to ensure sustained accuracy in dynamic environments. The research concluded that machine learning-driven valuation systems provide significant improvements over traditional manual appraisal methods, enabling transparent, reliable, and data-driven pricing for used cars, while supporting decision-making for buyers, sellers, dealerships, and online platforms in increasingly competitive and complex automotive markets.

2.11 F. Abdullah et al., 2023 [1] This study focused on predicting the selling price of reconditioned cars, a specialized and complex segment of the used car market where vehicles undergo refurbishment, parts replacement, and quality improvements that significantly influence market valuation. The authors compiled a comprehensive dataset from multiple sources, including online marketplaces, dealership records, and auction data, encompassing a wide range of vehicle attributes such as age, mileage, brand, model, engine capacity, fuel type, transmission type, accident history, service and maintenance records, reconditioning details, ownership patterns, optional features, and regional market factors. This extensive dataset allowed the researchers to capture the nuances of reconditioned car pricing and ensure that the predictive models could generalize effectively across diverse scenarios. The study placed a strong emphasis on data preprocessing, including handling missing or inconsistent data, correcting errors in refurbishment information, outlier detection and removal, normalization of numerical features, and encoding of categorical variables, thereby maintaining high data integrity and consistency. Multiple regression techniques were applied, including Linear Regression, Random Forest, and gradient boosting algorithms, and were evaluated using standard metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R-squared, alongside k-fold cross-validation to assess stability and generalizability. The findings demonstrated that models incorporating specific reconditioning-related features, such as the quality of replaced parts, refurbishment cost, and service history, achieved significantly higher predictive accuracy compared to models that treated reconditioned cars the same as standard used vehicles. Hyperparameter optimization, including tuning the number of trees, maximum depth, learning rate, and minimum samples per split for ensemble models, was performed to enhance predictive performance. Feature importance analysis highlighted that key attributes such as refurbishment quality, vehicle age, mileage, brand reputation, engine capacity, accident history, and optional features were critical in influencing price predictions, while additional factors like regional demand, seasonal trends, and market fluctuations also played a meaningful role. The research further addressed practical deployment considerations, including computational efficiency for large datasets, prediction speed for real-time applications, model scalability, and retraining strategies to accommodate evolving market dynamics. The authors discussed the potential of integrating hybrid approaches combining ensemble and regression models to improve robustness and stability under fluctuating conditions. The study emphasized interpretability, enabling users to trace predictions back to influential features, which fosters transparency and trust among buyers, sellers, and online marketplace operators. Additionally, the research examined scenario analysis to test model performance under varying market conditions, such as changes in refurbishment standards, shifts in consumer preferences, and economic fluctuations, demonstrating the resilience and adaptability of the models. The authors highlighted that careful feature selection, domain-specific data engineering, and robust model validation are essential for achieving high accuracy in specialized automotive segments, making these approaches critical tools for stakeholders operating in the reconditioned vehicle market.

2.12 Yian Zhu, 2023 [23] This study focused on predicting used car prices by leveraging a comprehensive suite of machine learning algorithms, with the goal of identifying the key determinants of vehicle value and providing reliable pricing models for online marketplaces and dealership platforms. The researcher compiled an extensive dataset including vehicle attributes such as manufacturing year, mileage, engine power, fuel type, transmission, brand, model, trim level, optional packages, accident history, maintenance records, and regional market factors. Data preprocessing was a critical aspect of the study, encompassing handling missing values, correcting inconsistencies, removing outliers, standardizing numerical features, and encoding categorical variables to ensure consistency and improve model performance. The study applied several machine learning algorithms, including XG Boost, Random Forest, Gradient Boosting, and Support Vector Regression, and rigorously evaluated their performance using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R-squared values. Feature importance analysis revealed that attributes such as vehicle age, engine power, mileage, and optional packages were among the most influential in determining car prices, while less common factors such as trim level, specific features, and regional market conditions also contributed significantly to price variability. The researcher explored interactions between features, observing that non-linear relationships between variables like mileage, engine power, and vehicle age could significantly impact predicted prices, necessitating algorithms capable of capturing these complexities. Hyperparameter tuning was performed to optimize model performance, including adjusting parameters like tree depth, learning rate, number of estimators, and regularization factors in ensemble models. The study also examined the computational efficiency of each algorithm, considering scalability for large datasets typical of online marketplaces and real-time applications. The research emphasized interpretability, particularly for tree-based models like XG Boost and Random Forest, enabling stakeholders to trace predictions back to the most influential features, thereby increasing trust in automated pricing systems. Additionally, the study discussed the practical implications of deploying such models in dynamic markets, highlighting the importance of retraining models regularly to account for seasonal trends, changes in market demand, macroeconomic fluctuations, and new vehicle features. The research also proposed potential integration of hybrid approaches combining multiple algorithms to further enhance predictive accuracy and robustness under variable market conditions. The author concluded that comprehensive machine learning models, supported by careful feature selection, thorough preprocessing, and regular model updates, provide highly reliable, interpretable, and actionable insights for pricing used cars, supporting fairness, transparency, and data-driven decision-making for buyers, sellers, and marketplace operators. This work highlighted the critical role of domain-specific data engineering and advanced modeling techniques in achieving accurate valuations and facilitating trust in digital automotive marketplaces.

2.13 Prof. Dipti A. Gaikwad et al., 2023 [8] This comparative study evaluated multiple machine learning algorithms, including Linear Regression, Lasso Regression, Support Vector Machine (SVM), and Random Forest, for predicting used car prices, with the aim of identifying an optimal balance between prediction accuracy, model complexity, and practical applicability in real-world automotive marketplaces. The authors compiled a large and diverse dataset encompassing attributes such as vehicle age, mileage, brand, model, engine capacity, fuel type, transmission, accident history, optional features, service and maintenance records, and regional demand factors. Comprehensive data preprocessing was performed to ensure the quality and consistency of the dataset, including handling missing values, outlier detection and removal, normalization of numerical features, and encoding of categorical variables. The study applied a rigorous evaluation framework, employing metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R-squared to measure model performance, alongside k-fold cross-validation to assess generalizability and reduce overfitting risks. Hyperparameter tuning was conducted for all algorithms, including adjusting regularization parameters for Lasso, kernel selection and gamma tuning for SVM, and tree number and maximum depth optimization for Random Forest, to achieve the most robust and accurate predictions possible. The findings revealed that Random Forest provided the best trade-off between interpretability, computational efficiency, and predictive accuracy, delivering high performance without the extensive tuning required for SVM and outperforming simpler linear models in capturing non-linear relationships and complex interactions between car features. The authors also emphasized the importance of feature selection and importance analysis, noting that variables such as vehicle age, mileage, brand, engine capacity, and optional features had the most significant impact on predicted prices, while less frequent or weakly correlated attributes contributed minimal information. The study explored practical deployment considerations, including model scalability for large online marketplaces, prediction latency for real-time valuation, retraining strategies to adapt to market changes, and integration into decision-support platforms for dealers and consumers. Additionally, the authors discussed the interpretability of Random Forest, which allows stakeholders to trace price predictions back to influential features, thereby improving transparency, user trust, and confidence in automated pricing systems. The research further analyzed how ensemble approaches, like Random Forest, mitigate bias and variance issues present in individual models, enhancing prediction stability in volatile market conditions. Scenario analysis was conducted to evaluate model robustness under varying economic and regional conditions, revealing that Random Forest consistently maintained high accuracy and reliability across different contexts. The study concluded that careful data preprocessing, rigorous feature engineering, hyperparameter optimization, and the use of ensemble-based methods are essential for developing effective, interpretable, and practically deployable used car price prediction systems, enabling informed, fair, and transparent pricing decisions in both online and offline automotive markets.

2.14 B. V. Raghu rami Reddy and Dr. K. Santhi Shree, 2022 [15] The authors presented a comprehensive analysis of various machine learning algorithms for used car price prediction, aiming to provide insights into the comparative strengths, weaknesses, and practical applications of different approaches within dynamic automotive markets. The study reviewed a broad spectrum of models ranging from simple regression techniques such as Linear Regression and Multiple Linear Regression to complex ensemble methods including Random Forest, Gradient Boosting, and XG Boost, examining their effectiveness across diverse datasets containing attributes like vehicle age, mileage, brand, model, engine specifications, fuel type, transmission, optional features, service history, accident records, and regional market trends. The research emphasized the necessity of thorough data preprocessing to ensure accurate predictions, including handling missing values, outlier removal, feature scaling, and categorical encoding, noting that poor data quality can significantly impair model performance, particularly in algorithms sensitive to feature distributions such as distance-based models or gradient boosting frameworks. The study conducted rigorous performance evaluation using metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and R-squared values, coupled with cross-validation to assess generalization capabilities and reduce overfitting risks, ensuring reliable model comparison under practical scenarios. Hyperparameter tuning played a pivotal role in optimizing model outcomes, with adjustments such as the number of estimators and maximum depth in ensemble models, regularization coefficients in linear models, and learning rates in boosting techniques being carefully calibrated to achieve high predictive accuracy. The authors found that while deep learning models have potential, ensemble methods such as Random Forest and Gradient Boosting currently deliver superior performance for tabular data typical of used car datasets, offering robustness to noise, the ability to capture non-linear interactions, and interpretability through feature importance analysis. The study also highlighted practical implications for deployment, including the integration of predictive models into online marketplaces, dealership valuation systems, and mobile applications, emphasizing scalability, computational efficiency, and the need for regular retraining to reflect seasonal trends, market fluctuations, and emerging vehicle features. The research concluded that the combination of robust preprocessing, appropriate feature selection, ensemble-based modeling, and systematic hyperparameter optimization provides a reliable framework for accurate, interpretable, and deployable used car price prediction, supporting transparency, fairness, and informed decision-making in automotive transactions while also offering guidance for practitioners in algorithm selection based on dataset characteristics, computational resources, and application requirements.

2.15 N. Radojevic et al., 2022 [13] This analysis focused on predicting used car prices using advanced machine learning methodologies, with particular emphasis on understanding the relationship between vehicle features, depreciation patterns, and market dynamics, providing a practical framework for accurate residual value estimation in automotive markets. The researchers compiled a large dataset spanning multiple car brands, models, manufacturing years, mileage, engine specifications, fuel types, transmission systems, trim levels, optional features, accident histories, maintenance records, and regional demand factors, ensuring that the dataset represented both typical and edge-case vehicles. The study placed significant emphasis on data preprocessing to enhance model reliability and accuracy, including strategies for handling missing data, outlier detection and correction, normalization of numerical features, encoding of categorical variables, and feature scaling to ensure consistency across diverse attributes. Multiple machine learning algorithms were applied, including Random Forest, Gradient Boosting, XG Boost, Lasso Regression, Ridge Regression, and Artificial Neural Networks (ANN), and their predictive performance was evaluated using standard metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and R-squared, along with k-fold cross-validation to ensure generalizability and reduce the likelihood of overfitting. Feature importance analysis highlighted the significant impact of variables such as mileage, age, brand, condition, and maintenance history on vehicle depreciation, while the study also considered less common factors such as trim packages, optional features, and regional economic indicators, demonstrating that combining multiple relevant features can substantially improve model accuracy. The authors explored non-linear models for capturing complex interactions among vehicle features, noting that such models outperform traditional linear methods in predicting varying depreciation curves across different brands and models. The research also analyzed practical deployment considerations, including computational efficiency, model scalability for large online datasets, interpretability for stakeholders, and integration into decision-support systems for dealers, fleet managers, leasing companies, and insurance providers. Additionally, the study emphasized the importance of scenario analysis to evaluate model robustness under fluctuating market conditions, seasonal demand changes, and macroeconomic influences, demonstrating that hybrid or ensemble approaches combining multiple algorithms further enhance prediction stability and accuracy. The authors concluded that well-designed machine learning models, supported by comprehensive preprocessing, thoughtful feature selection, hyperparameter optimization, and model validation, are capable of providing highly accurate, interpretable, and actionable estimates of used car residual values, supporting transparent pricing, informed decision-making, and trust in automotive markets, while also providing a roadmap for integrating data-driven valuation systems into commercial platforms, fleet management tools, leasing operations, and insurance assessments, ultimately contributing to greater market efficiency and fairness.

2.16 L. Bukvić et al., 2022 [6] The researchers conducted an extensive study on predicting used vehicle prices in the Croatian market using data-driven models, with a particular focus on capturing the nuances of local market dynamics that influence pricing decisions. The study leveraged data mining techniques to extract structured and semi-structured information from online advertisements, dealership listings, and transactional records, encompassing a wide range of vehicle attributes such as brand, model, year of manufacture, mileage, engine size, fuel type, transmission, trim levels, accident history, service records, optional features, and geographic location. A central theme of the research was the importance of incorporating region-specific factors, noting that localized data significantly improves prediction accuracy compared to generic, global datasets that may fail to reflect unique consumer preferences, supply-demand variations, and economic conditions in a specific market. The authors performed comprehensive preprocessing, including handling missing values, outlier detection and removal, normalization of numerical features, encoding of categorical variables, and feature scaling, to ensure consistency and reliability of the dataset for model training. Multiple predictive models were tested, including Linear Regression, Decision Trees, Random Forest, Gradient Boosting, and ensemble approaches, with performance evaluated using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), Mean Absolute Percentage Error (MAPE), and R-squared values, coupled with cross-validation techniques to assess generalization and mitigate overfitting. The study highlighted that ensemble methods, particularly Random Forest and Gradient Boosting, were highly effective in capturing non-linear interactions among features, managing noisy and inconsistent data, and providing robust, accurate predictions. Additionally, the research emphasized the critical role of feature engineering, noting that the inclusion of macroeconomic indicators such as inflation, fuel prices, regional demand fluctuations, and seasonal trends further enhanced model accuracy and predictive reliability. The authors also analysed the interpretability of different models, highlighting that feature importance analysis allows stakeholders to understand the key drivers of vehicle prices, improving trust, transparency, and decision-making for buyers, sellers, dealerships, and insurance companies. The study explored practical deployment considerations, including computational efficiency, scalability for real-time applications, retraining strategies to adapt to dynamic market conditions, and integration into online marketplaces and valuation tools. Scenario analysis demonstrated that localized models significantly outperformed global models, particularly in predicting price growth trends and residual values for specific vehicle categories, showcasing the necessity of region-specific data in developing accurate, actionable, and contextually relevant predictive systems. The authors concluded that combining comprehensive preprocessing, ensemble modelling, region-specific data integration, hyperparameter tuning, and continuous model updates is essential to building robust, interpretable, and operationally viable used car price prediction systems capable of supporting informed decision-making, fair pricing, and efficient automotive market operations across both consumer and commercial sectors.

2.17 Satapathy et al., 2022 [17] This study focused on accurately estimating used car prices by conducting a comprehensive comparison of multiple machine learning algorithms, including Multiple Linear Regression, Lasso Regression, Ridge Regression, XG Boost, and Random Forest, with the objective of identifying the most effective predictive model for practical deployment in automotive markets. The authors collected and analyzed a rich dataset containing detailed information about vehicles, such as make, model, manufacturing year, mileage, engine size, fuel type, transmission, trim level, optional features, accident history, maintenance records, and regional market variations, ensuring representation across a wide range of pricing and vehicle conditions. The study placed strong emphasis on preprocessing and feature engineering, addressing challenges such as missing data, outlier removal, normalization of numerical features, categorical encoding, and multicollinearity, as these steps were critical for improving model stability, reducing noise, and enhancing prediction accuracy. Each algorithm underwent hyperparameter tuning to optimize performance, including adjustment of tree depth, number of estimators, learning rate, regularization parameters, and L1/L2 coefficients for linear models, ensuring that models were fine-tuned for both accuracy and generalization. Through rigorous evaluation using metrics such as R-squared, Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE), the researchers found that XG Boost outperformed other models, achieving an R-squared value of 0.92, effectively capturing complex non-linear interactions among vehicle features and market trends that linear models were unable to fully account for. The study highlighted the superior capability of gradient boosting techniques in handling missing values, modeling feature interactions, and providing stable predictions even in heterogeneous datasets with diverse feature types. The authors also conducted sensitivity analysis to assess the influence of individual features on model performance, revealing that attributes such as mileage, vehicle age, brand reputation, engine specifications, and optional features played a crucial role in driving predictive accuracy. Additionally, the research explored practical deployment considerations, including computational efficiency, scalability for large online marketplaces, real-time prediction feasibility, retraining strategies to account for evolving market trends, and integration into dealership pricing tools, fleet management systems, and insurance valuation platforms. Scenario analysis was performed to test model robustness under fluctuating market conditions, seasonal variations, and regional demand shifts, demonstrating that XG Boost provides reliable, adaptable, and interpretable predictions suitable for operational use. The authors concluded that the combination of comprehensive preprocessing, feature selection, hyperparameter optimization, and the adoption of advanced gradient boosting algorithms creates a highly effective framework for used car price prediction, supporting transparent pricing, informed decision-making, and enhanced trust for stakeholders in dynamic automotive markets. Furthermore, the study emphasized that such predictive models could be continuously updated with new data to maintain accuracy and relevance, providing a sustainable solution for online marketplaces, dealerships, and policy makers seeking data-driven insights for vehicle valuation.

2.18 Zhang, 2022 [24] In this study, the author developed a highly comprehensive price evaluation model for used cars using the Light GBM algorithm, aiming to address the multifaceted challenges inherent in automotive pricing, such as large-scale data processing, high-dimensional feature spaces, dynamic market conditions, and heterogeneity in vehicle characteristics. The research involved compiling an extensive dataset with detailed attributes for each vehicle, including make, model, manufacturing year, mileage, engine capacity, fuel type, transmission, trim level, accident and repair history, service and maintenance records, optional packages, geographic location, and even regional economic indicators, ensuring representation across a broad spectrum of vehicle types, price ranges, and market segments. The dataset underwent rigorous preprocessing, including missing value imputation, outlier detection and removal, standardization of numerical features, and encoding of categorical variables to improve model stability and predictive accuracy. The study placed significant emphasis on feature engineering, highlighting that certain feature, such as mileage, vehicle age, brand reputation, and engine specifications, had a disproportionate impact on price prediction, while also considering interactions between features and regional market variations. The Light GBM model was benchmarked against other gradient boosting frameworks such as XG Boost and Cat Boost, as well as traditional ensemble methods, demonstrating that Light GBM achieved not only faster training times but also comparable or superior predictive accuracy, particularly for large-scale datasets typical of online marketplaces. Extensive hyperparameter tuning was performed, adjusting parameters such as the number of leaves, learning rate, maximum depth, regularization coefficients, and boosting iterations to optimize model performance while mitigating overfitting and improving generalization to unseen data. The research also explored model interpretability by analyzing feature importance and SHAP values, providing stakeholders with actionable insights into the key determinants of vehicle prices, including optional features, maintenance history, fuel type, and trim level, which could guide dealers, buyers, and online platforms in making informed pricing and purchasing decisions. The study further examined computational efficiency and scalability, demonstrating that Light GBM could process millions of listings in near real-time without compromising accuracy, making it particularly suitable for integration into automated pricing platforms, e-commerce marketplaces, dealership inventory management systems, and fleet valuation tools. Additionally, the model was tested for robustness under varying market conditions, including seasonal fluctuations, changes in regional demand, economic shocks, and the introduction of new vehicle models, confirming its stability and reliability for practical applications. The author concluded that combining thorough data preprocessing, advanced gradient boosting techniques, comprehensive feature engineering, and continuous model retraining creates an exceptionally effective and practical framework for used car price prediction. This approach not only ensures high accuracy but also supports transparency, fairness, and adaptability in dynamic automotive markets. Furthermore, the study highlighted the potential of integrating Light GBM-based predictive systems into decision-support tools for dealerships, online platforms, fleet management, and insurance companies, ultimately enhancing operational efficiency, reducing human bias, increasing trust among buyers and sellers, and enabling real-time dynamic pricing that reflects both vehicle-specific attributes and market trends comprehensively.

2.19 Liu et al., 2022 [11] This study proposed a sophisticated hybrid model, combining Particle Swarm Optimization (PSO), Grey Relational Analysis (GRA), and a Backpropagation Neural Network (BPNN), to accurately forecast used car prices. The researchers aimed to address the limitations of traditional machine learning and neural network models, particularly their susceptibility to local minimum, overfitting, and sensitivity to feature selection in high-dimensional datasets. The dataset compiled for this research was extensive, including attributes such as make, model, year of manufacture, mileage, engine capacity, fuel type, transmission, vehicle condition, accident and service history, trim level, optional features, and regional market variables, ensuring comprehensive coverage of both high- and low-end vehicles in the market. Rigorous preprocessing was performed to handle missing values, outliers, inconsistent records, and scaling differences between numerical attributes, which is critical for hybrid optimization and neural network models that are sensitive to data quality and feature distribution. The PSO component was used to optimize the initial weights and biases of the BPNN, enhancing convergence speed and avoiding local minima, while GRA was employed to determine the relative importance and weighting of each input feature, allowing the model to prioritize influential variables and reduce noise from less significant attributes. The hybrid PSO-GRA-BPNN approach demonstrated significant improvements in predictive accuracy, achieving a Mean Absolute Percentage Error (MAPE) of just 3.936%, which reflects a highly reliable estimation of vehicle prices even under complex, heterogeneous, and non-linear relationships between features. The study also examined the effects of varying network architectures, including the number of hidden layers, neurons per layer, activation functions, and learning rates, highlighting the sensitivity of neural networks to structural configuration and hyperparameter selection. Additionally, the authors performed robust testing by introducing synthetic perturbations, simulating market fluctuations, seasonal price variations, and changes in regional demand, confirming that the hybrid model-maintained stability and accuracy under realistic dynamic market conditions. Feature importance analysis was conducted using the GRA component, providing insights into which vehicle characteristics strongly influenced price predictions, such as mileage, age, brand reputation, fuel efficiency, engine capacity, accident history, and optional packages, which can aid dealers, buyers, and online platforms in making informed pricing decisions. The study emphasized that integrating optimization algorithms with neural networks not only enhances model generalization and reduces prediction error but also provides a flexible framework that can adapt to evolving market trends when retrained periodically with new data. Furthermore, the research highlighted practical applications of the hybrid model in real-time online car marketplaces, automated dealership pricing systems, fleet valuation tools, and insurance assessment platforms, demonstrating its potential to reduce human bias, improve transparency, and enable dynamic pricing that reflects both intrinsic vehicle attributes and prevailing market conditions. The authors concluded that the PSO-GRA-BPNN hybrid model offers a highly accurate, robust, and interpretable solution for used car price prediction, capable of supporting decision-making in a variety of real-world automotive contexts and providing a scalable framework for future extensions incorporating additional optimization algorithms, macroeconomic indicators, and larger datasets for even more precise valuations.

2.20 Varshitha et al., 2022 [20] In this study, the authors investigated the effectiveness of advanced machine learning models, specifically Artificial Neural Networks (ANN) and Random Forest (RF), for predicting used car prices, aiming to capture complex, non-linear relationships between vehicle features and market valuation. The research compiled a comprehensive dataset comprising attributes such as make, model, manufacturing year, mileage, engine specifications, fuel type, transmission, trim levels, optional features, accident and service history, regional market indicators, and macroeconomic factors, ensuring broad representation across different market segments and vehicle types. Rigorous preprocessing steps were applied, including handling missing values, outlier detection, standardization of numerical variables, encoding of categorical features, and feature scaling, all critical for enhancing the predictive performance of both ANN and RF models. The study emphasized the importance of feature engineering and selection, identifying key variables such as mileage, vehicle age, brand prestige, fuel economy, engine capacity, and optional packages that strongly influence resale value, while also examining interaction effects among features to improve model robustness. The authors compared the performance of ANN and RF with simpler linear models, including Multiple Linear Regression, demonstrating that non-linear models significantly outperform linear approaches, particularly in datasets with heterogeneous and complex feature interactions. Hyperparameter tuning was extensively conducted for both models, including the number of hidden layers, neurons per layer, activation functions, learning rate, and regularization parameters for ANN, and the number of trees, maximum depth, minimum samples per split, and feature selection strategy for RF, to optimize model accuracy and prevent overfitting. The study further analyzed the models' robustness by testing against fluctuating market conditions, including seasonal variations, regional demand shifts, and changes in macroeconomic indicators, confirming the stability and reliability of ANN and RF under dynamic conditions. Additionally, feature importance analysis in the RF model and sensitivity analysis in the ANN model provided interpretable insights into how different vehicle characteristics affect price predictions, supporting data-driven decision-making for dealers, buyers, fleet managers, and online marketplaces. The research also highlighted the practical application of these models in real-time online pricing platforms, automated dealership valuation systems, insurance assessment, and fleet management tools, emphasizing their ability to reduce human bias, enhance transparency, and provide accurate dynamic pricing. The authors concluded that as dataset complexity increases and market conditions become more volatile, the adoption of non-linear models such as ANN and RF is essential for reliable used car price prediction, and that integrating these models with feature importance insights, continuous retraining, and regionalized market data can further improve fairness, accuracy, and trustworthiness in automotive valuation systems. Moreover, the study suggested that combining ANN and RF in hybrid or ensemble frameworks could potentially enhance predictive performance even further, enabling robust, scalable, and adaptive solutions for diverse automotive marketplaces and decision-support applications.

2.21 K. Samruddhi and Kumar, 2020 [16] This study focused on the application of a Supervised Machine Learning model using the K-Nearest Neighbor (KNN) algorithm for predicting used car prices, aiming to provide an accurate, interpretable, and practically deployable pricing system for online marketplaces and dealership platforms. The authors compiled a detailed dataset that included features such as make, model, year of manufacture, mileage, engine specifications, fuel type, transmission type, vehicle condition, accident history, optional packages, trim levels, and regional market factors. Rigorous preprocessing was carried out to address missing values, standardize numerical attributes, handle outliers, and encode categorical variables, ensuring that the distance-based KNN model would generate reliable and unbiased predictions. The study emphasized the significance of feature scaling, particularly for distance-sensitive algorithms like KNN, where unscaled attributes can dominate Euclidean or Manhattan distance calculations, leading to skewed predictions. Extensive experimentation was conducted to determine the optimal value of K, balancing model bias and variance to avoid overfitting and underfitting, while the researchers also examined the effects of different distance metrics, such as Euclidean, Manhattan, and Minkowski distances, to assess their influence on predictive accuracy. Cross-validation techniques, including 5-fold and 10-fold validation, were applied to evaluate model stability and generalization across diverse subsets of the dataset. The study found that KNN achieved a prediction accuracy of up to 85%, demonstrating its effectiveness as a simple yet powerful non-parametric method for used car price estimation. In addition, the authors explored combining KNN with ensemble techniques and weighted distance metrics, where closer neighbors exert greater influence on the predicted price, further enhancing the model's robustness and reliability. The research also highlighted the importance of careful feature selection, noting that attributes such as mileage, car age, brand reputation, engine capacity, and accident history had a particularly strong impact on prediction performance, whereas irrelevant or weakly correlated features could introduce noise and reduce overall model accuracy. Sensitivity analysis was conducted to evaluate the model's responsiveness to variations in input features and neighborhood size, confirming the necessity of periodic retraining to adapt to evolving market trends and seasonal fluctuations. Furthermore, the study examined the computational challenges associated with KNN in large-scale datasets, suggesting the use of optimized data structures and dimensionality reduction techniques to improve prediction efficiency and reduce processing time. The authors concluded that, with meticulous preprocessing, careful feature selection, optimal K tuning, and appropriate handling of distance metrics, KNN offers a practical, interpretable, and high-performing solution for used car price prediction, capable of supporting real-world decision-making, dynamic online pricing systems, and fair market valuation. Moreover, the study emphasized that KNN's simplicity, transparency, and adaptability make it particularly suitable for integration into dealership software, online marketplaces, fleet management platforms, and insurance assessment tools, ensuring accurate, explainable, and trustworthy pricing across diverse automotive segments and regional markets.

2.22 F. R. Amik et al., 2021 [3] This study explored the use of machine learning techniques for predicting the price of pre-owned cars in Bangladesh, with a primary objective of creating accurate, localized pricing tools that reflect the unique characteristics of the regional automotive market. The authors highlighted the prevalence of reconditioned Japanese vehicles in Bangladesh, emphasizing that traditional global models often fail to capture the specific market dynamics and feature distributions of this region. To address these challenges, the research compiled a detailed dataset that included vehicle attributes such as make, model, year of manufacture, mileage, engine capacity, fuel type, transmission, accident and service history, trim levels, optional packages, and local market indicators. Rigorous preprocessing steps were applied, including handling missing data, encoding categorical features, standardizing numerical variables, and detecting and correcting inconsistencies, ensuring that the machine learning models could learn accurate patterns from high-quality input. The study evaluated several algorithms, including K-Nearest Neighbors (KNN), Random Forest (RF), Gradient Boosting Machines (GBM), and Linear Regression, assessing their predictive performance in terms of accuracy, R-squared values, mean absolute error, and mean squared error. The authors found that training models on localized data significantly improved predictive accuracy compared to applying global models, highlighting the importance of region-specific training in markets with unique vehicle compositions and pricing structures. Feature selection and engineering were emphasized as critical steps, identifying mileage, vehicle age, brand reputation, engine size, fuel type, and reconditioning features as the most influential factors in determining resale value. The study also explored the integration of macroeconomic indicators, such as inflation, fuel prices, and regional demand trends, to further enhance predictive reliability. Computational considerations were addressed, including the implementation of efficient data structures, hyperparameter tuning, and cross-validation techniques to ensure robust and scalable model deployment. Additionally, the research discussed the practical application of the models for online marketplaces, dealerships, and fleet management, emphasizing their potential to reduce information asymmetry between buyers and sellers, improving market transparency, and foster trust. The authors concluded that with meticulous preprocessing, feature selection, and region-specific modeling, machine learning can provide highly accurate, interpretable, and adaptable solutions for used car price prediction in emerging markets like Bangladesh. Furthermore, the study suggested that combining models in ensemble or hybrid frameworks could enhance robustness under changing market conditions, while continuous retraining with updated data ensures that predictions remain relevant and accurate, supporting fair and transparent pricing across the automotive sector.

2.23 Yadav et al., 2021 [22] The researchers conducted a comprehensive study to determine the most effective machine learning models for predicting used car prices across various Indian cities, using a rich dataset obtained from Kaggle. The dataset included extensive features such as car make, model, year of manufacture, mileage, engine capacity, fuel type, transmission type, vehicle condition, trim levels, optional packages, regional economic factors, city-specific demand, and brand reputation. The primary goal was to identify models that could accurately capture the complex relationships between these diverse variables and provide reliable price predictions. The study performed extensive preprocessing, including handling missing values, encoding categorical features, standardizing numerical attributes, detecting and managing outliers, and ensuring data consistency, which allowed the models to learn meaningful patterns without distortion. Feature importance analysis was emphasized to identify the most influential variables, highlighting mileage, vehicle age, brand value, fuel type, engine capacity, city-specific demand, and optional features as critical determinants of car prices. The authors employed both Multiple Linear Regression and Random Forest models, rigorously comparing their predictive performance using metrics such as R-squared, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). Cross-validation strategies, including k-fold validation, were applied to ensure model stability, reduce variance, and assess generalization capabilities across different subsets of the dataset. The study found that Random Forest consistently outperformed Linear Regression by effectively capturing non-linear dependencies and interactions among features, while also demonstrating robustness to overfitting and scalability for large heterogeneous datasets. Hyperparameter tuning was applied to optimize Random Forest performance, considering parameters such as the number of trees, maximum depth, minimum samples per leaf, and feature selection methods. The research further explored practical applications of these models in online car marketplaces, dealerships, and fleet management platforms, emphasizing their potential to improve price transparency, reduce information asymmetry between buyers and sellers, and enable automated valuation systems for diverse urban markets. Additionally, the study highlighted the importance of including city-specific and regional factors, such as local economic conditions, traffic patterns, and population density, demonstrating that such localized modeling significantly enhances predictive accuracy compared to generic models. The authors also suggested that combining Random Forest with ensemble or hybrid techniques could further improve robustness and stability under fluctuating market conditions, while continuous retraining with updated transactional data ensures predictions remain accurate and relevant. Overall, the study concluded that Random Forest provides a scalable, interpretable, and highly reliable solution for used car price prediction in Indian cities, supporting informed decision-making for buyers, sellers, dealers, online marketplaces, and regulatory authorities, while also emphasizing the critical role of preprocessing, feature engineering, parameter tuning, and localized modeling in achieving accurate and consistent predictions in real-world deployments.

2.24 Shanti et al., 2021 [18] This study focused on automating the process of buying and selling used cars in Palestine using advanced machine learning techniques, with the goal of creating a transparent, accurate, and reliable pricing system that could assist buyers, sellers, and dealerships in making informed decisions. The authors collected a comprehensive dataset from multiple commercial websites, which included detailed features such as car make, model, year of manufacture, mileage, engine capacity, fuel type, transmission, trim levels, optional packages, vehicle condition, and region-specific demand indicators. Data preprocessing was carried out meticulously, including handling missing values, standardizing numerical variables, encoding categorical features, detecting and managing outliers, and ensuring consistency across various sources to create a robust dataset suitable for machine learning applications. The study evaluated multiple regression-based and ensemble machine learning models, including Linear Regression, Decision Trees, Random Forest, Gradient Boosting, and XG Boost, to determine which algorithms could most accurately predict used car prices in a real-world, unregulated market. Extensive hyperparameter tuning and cross-validation techniques were applied to optimize model performance, reduce overfitting, and assess generalization across different subsets of the dataset. The authors found that the Random Forest model, with its ability to handle non-linear dependencies, complex interactions, and heterogeneous feature types, consistently outperformed other models, achieving a high R-squared value of 0.90, which indicates strong predictive accuracy. The research further emphasized the practical significance of incorporating region-specific variables, such as local market demand, seasonal trends, economic indicators, traffic conditions, and city-specific vehicle popularity, which significantly improved model accuracy compared to generic models. Additionally, the study explored the integration of predictive models into mobile and web-based platforms, allowing users to instantly estimate car prices, ensuring fairness and transparency in markets where pricing information is often inconsistent or opaque. The authors highlighted the importance of continuous model retraining using updated data to reflect changing market conditions, new car features, and seasonal variations in demand, ensuring that predictions remain accurate over time. Furthermore, the study analyzed feature importance, showing that mileage, vehicle age, fuel type, engine capacity, and brand reputation were critical determinants of pricing, while optional features, trim levels, and local market factors contributed significantly to prediction refinement. The authors also discussed potential extensions, such as combining ensemble models with hybrid frameworks, incorporating macroeconomic indicators, and leveraging real-time data streams for automated dynamic pricing. Overall, the study concluded that advanced machine learning techniques, particularly Random Forest, provide a scalable, interpretable, and reliable solution for used car price prediction in Palestine, capable of enhancing trust, reducing information asymmetry, supporting dealers and online marketplaces, and improving overall market efficiency, while emphasizing the importance of rigorous preprocessing, feature engineering, hyperparameter tuning, and continuous model updates to achieve robust and consistent predictions in real-world applications.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Research Design

The research design functions as the structural blueprint that governs how a scientific investigation is conducted. It defines the systematic process through which data is collected, processed, analysed, and interpreted to derive meaningful conclusions. In data science–driven systems, research design ensures that the analytical workflow follows a logical, structured, and reproducible methodology that allows reliable results to be obtained from complex datasets.

For the Fair Car – AI Based Used Car Price Prediction project, the research design provides the framework through which machine learning techniques are applied to analyse historical used-car data and predict realistic vehicle prices. The used car market is inherently complex, as vehicle pricing is influenced by numerous variables including manufacturing year, total distance driven, ownership history, fuel type, drivetrain configuration, vehicle category, and brand reputation. These variables interact with each other in intricate ways, making manual price estimation highly subjective and often inaccurate.

To address this challenge, the present research adopts a structured computational approach where historical vehicle data is processed using machine learning algorithms to discover patterns that influence pricing behaviour. The dataset used for this research consists of thousands of used-car listings obtained from the Cars24 dataset, which contains detailed attributes describing each vehicle along with its actual selling price.
The research design implemented in this study follows a systematic pipeline that includes data preprocessing, feature engineering, model training, and predictive evaluation. During preprocessing, the dataset is cleaned by removing incomplete records and standardizing column names. Feature engineering techniques are then applied to extract useful attributes such as vehicle brand and model from the car name field. These features are subsequently transformed into numerical representations using encoding techniques so that they can be processed by machine learning algorithms.

The predictive model is then developed using a Random Forest Regression algorithm implemented through a machine learning pipeline. The pipeline integrates preprocessing steps such as categorical encoding and numerical feature handling with the predictive algorithm, ensuring that the entire modelling workflow operates in a unified and automated manner. The trained model is evaluated using statistical performance metrics to determine its accuracy and reliability in predicting used car prices.

By combining structured data processing, statistical modelling, and automated prediction systems, the research design ensures that the developed solution provides a scientifically valid and practically useful framework for estimating used vehicle prices based on historical data.

3.1.1 Quantitative Research Approach

The study adopts a Quantitative Research Approach, which relies on numerical data, statistical computation, and mathematical modelling to analyse relationships between variables. Quantitative research is particularly suitable for data-driven predictive systems because it enables objective analysis based on measurable values rather than subjective judgement.

In the context of used car price prediction, the dependent variable being analysed—vehicle price—is inherently numerical and continuous. Predicting the selling price of a vehicle requires analysing several measurable attributes such as manufacturing year, kilometres driven, and ownership status. These attributes can be mathematically represented and processed using machine learning algorithms to uncover underlying patterns within the dataset.

The dataset used in this study contains structured records describing various vehicle attributes along with their corresponding prices. Each record represents a real-world transaction or listing in the used car market. By analysing these historical examples, the machine learning model learns how different vehicle characteristics influence market valuation.

Unlike qualitative approaches that depend on interviews, surveys, or subjective opinions of car dealers and buyers, the present study relies entirely on objective numerical data. The system does not rely on human assumptions about vehicle value; instead, it extracts knowledge directly from historical sales patterns contained within the dataset.

To ensure that machine learning algorithms can process the information effectively, categorical variables such as fuel type, vehicle category, ownership status, drivetrain configuration, and brand identity must be converted into numerical representations. This is accomplished through One-Hot Encoding, a preprocessing technique implemented using the Scikit-Learn library. One-Hot Encoding transforms categorical variables into binary vectors, allowing the algorithm to interpret each category as a separate feature.

The preprocessing stage is implemented using a Column Transformer, which systematically applies different transformations to categorical and numerical features. Numerical attributes such as year and kilometres driven are passed directly to the model, while categorical attributes are encoded into machine-readable form.

Through this transformation process, all relevant vehicle attributes are converted into a structured numerical format suitable for computational analysis. The machine learning model then processes these numerical representations to learn complex relationships between vehicle characteristics and their corresponding prices.
The quantitative approach therefore provides a rigorous framework for analysing the dataset using mathematical models, ensuring that predictions are data-driven, objective, and reproducible.

3.1.2 Correlational Framework

Within the broader quantitative paradigm, this research adopts a Correlational Research Framework. Correlational research focuses on identifying relationships between variables and understanding how changes in one variable are associated with changes in another.

In the used car market, the selling price of a vehicle is not determined by a single factor. Instead, it emerges from the interaction of multiple variables such as vehicle age, mileage, fuel type, ownership status, and brand reputation. A correlational framework allows the study to analyse these relationships and understand how each attribute contributes to the overall price determination process.

For instance, certain variables are expected to demonstrate negative correlations with price. As the kilometres driven increases, the market value of a vehicle typically decreases due to increased wear and tear. Similarly, older vehicles generally have lower resale values because depreciation reduces their market desirability over time.

Conversely, some variables demonstrate positive correlations with price. Vehicles belonging to premium brands or higher vehicle categories, such as SUVs, often command higher resale values due to greater demand and perceived quality. Similarly, newer vehicles with lower mileage typically maintain higher market prices.

The correlational framework in this study is further enhanced through feature engineering techniques applied during data preprocessing. One important transformation implemented in the code involves splitting the car name column into separate brand and model attributes. This allows the model to independently analyse how different manufacturers and specific vehicle models influence resale value.

The relationships between these attributes are not analysed individually but rather collectively through the machine learning model. The Random Forest algorithm used in this research can capture complex nonlinear relationships between variables. Instead of relying on simple linear correlations, the algorithm constructs multiple decision trees that analyse different combinations of features to determine how they influence vehicle price.

This multivariate correlational analysis allows the model to understand intricate patterns within the dataset and generate predictions that reflect real-world pricing behaviour in the used car market.

3.1.3 Experimental Modelling Design

The third component of the research design is the Experimental Modelling Approach, which involves systematically training and evaluating machine learning models to determine their predictive performance. In the context of machine learning research, experimentation refers to the process of training algorithms on historical data and assessing how accurately they can predict unseen data.

The modelling process implemented in this study follows a structured pipeline architecture developed using the Scikit-Learn machine learning framework. The pipeline integrates data preprocessing, feature transformation, and predictive modelling into a single automated workflow. This design ensures that all data transformations are consistently applied during both training and prediction stages.

The experimental framework consists of three primary components: control variables, independent variables, and dependent variables.

The control variables represent the elements that remain constant during the experiment. In this research, the dataset acts as the primary control variable. The Cars24 used-car dataset containing thousands of vehicle records is used consistently throughout the modelling process. Maintaining a fixed dataset ensures that the evaluation results are reliable and that the model performance can be accurately measured.

The independent variables represent the elements that are manipulated during the experiment. In this study, the independent variable is the machine learning algorithm used for prediction. The predictive system employs a Random Forest Regression model, which is an ensemble learning algorithm that constructs multiple decision trees and aggregates their predictions to produce a final output. The model is configured with 300 decision trees, allowing it to capture complex relationships between vehicle attributes and price.

The dependent variables represent the outcomes measured during the experiment. In this study, the dependent variables are the model performance metrics that quantify how accurately the algorithm predicts vehicle prices. To evaluate predictive accuracy, the dataset is divided into training and testing subsets using an 80–20 train-test split. The model is trained on 80% of the data and evaluated on the remaining 20% to assess its ability to generalize to unseen examples.

Two key evaluation metrics are used to measure model performance: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics quantify the average difference between the predicted prices and the actual prices in the dataset. Lower values indicate that the model produces predictions that are closer to real market values.

Once the model achieves satisfactory performance, it is saved using the Joblib serialization library. Saving the trained model enables it to be reused for real-time predictions without retraining. This functionality allows the system to predict prices for new vehicles by simply providing their attributes such as brand, model, year, mileage, and fuel type.

Through this experimental modelling design, the research moves beyond theoretical analysis and implements a fully functional predictive system capable of estimating used car prices based on historical market data. The structured experimentation process ensures that the final model is both accurate and practically applicable in real-world scenarios.

3.2 Hypothesis Formulation

This section outlines the key assumptions that guide the development and evaluation of the Fair Car – AI-Based Used Car Price Prediction system. The hypotheses are framed to test whether the machine learning–based price prediction system, which uses structured vehicle attributes and a Random Forest Regression model,

meaningfully improves the accuracy and reliability of used car price estimation compared to traditional manual valuation or rule-based pricing methods.
The system analyses important vehicle features such as brand, manufacturing year, kilometres driven, fuel type, ownership history, and vehicle category to estimate the fair resale price of used cars. By learning patterns from historical vehicle datasets, the model aims to generate realistic and unbiased price predictions for buyers and sellers in the used car market.
These testable assumptions provide a measurable foundation for assessing the effectiveness of the system in predicting fair car prices.

Null Hypothesis (H_0):
An AI-based used car price prediction system like Fair Car does not significantly enhance the accuracy, consistency, or reliability of estimating used car prices when compared to traditional pricing methods or simple rule-based valuation approaches.

Alternative Hypothesis (H_1):
An AI-based used car price prediction system like Fair Car significantly improves the accuracy, consistency, and reliability of used car price estimation by systematically analysing structured vehicle attributes and applying a Random Forest Regression model to learn patterns from historical data.

These hypotheses are evaluated through practical metrics such as:

- Prediction Accuracy: Evaluation of how close the predicted car price is to the actual market price using performance metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
- Feature Influence: Analysis of how different vehicle attributes—such as brand, manufacturing year, and kilometres driven—affect the predicted resale price of used cars.
- Model Reliability: Assessment of the model's ability to consistently produce stable price predictions across different vehicle types and datasets.
- System Usability: Evaluation of how easily users can input vehicle details and obtain an estimated fair price through the prediction system interface.

3.3 Objectives Aligned with Hypothesis

The objectives of the Fair Car – AI-Based Used Car Price Prediction system are designed to directly test and validate the hypotheses defined in Section 3.2 Hypothesis Formulation. These objectives guide the research and development process by ensuring that the machine learning model is evaluated based on measurable performance indicators such as prediction accuracy, feature influence, reliability, and usability.

To develop an AI-based used car price prediction system using machine learning techniques. The primary objective of this research is to design and implement a data-driven price prediction model using Random Forest Regression that can estimate the fair resale price of used cars based on structured vehicle attributes such as brand, manufacturing year, kilometres driven, fuel type, ownership history, and vehicle category.

To evaluate the prediction accuracy of the machine learning model.
This objective aims to assess how closely the predicted prices match the actual market prices of used cars. The performance of the system will be evaluated using statistical metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to determine whether the AI-based system provides more accurate estimations compared to traditional valuation methods.

To analyse the influence of vehicle attributes on resale price prediction.
Another important objective is to examine how different vehicle characteristics—such as brand reputation, manufacturing year, kilometres driven, and fuel type—affect the predicted price of a car. This analysis helps identify the most influential features used by the model while making predictions.

To assess the reliability and consistency of the prediction model.
This objective focuses on determining whether the trained model produces stable and consistent price predictions across different vehicle datasets and categories. Reliability is essential for ensuring that the prediction system can be trusted by users in real-world applications.

To develop a user-friendly system interface for price estimation.
The final objective is to design an accessible prediction interface where users can easily input vehicle details and obtain an estimated fair price. This ensures that the system is not only technically accurate but also practical and usable for buyers and sellers in the used car market.

Together, these objectives provide a structured framework for validating the null hypothesis (H_0) and the alternative hypothesis (H_1) by systematically evaluating the effectiveness, accuracy, and usability of the AI-based used car price prediction system.

Hypothesis Testing Result
The experimental evaluation of the used car price prediction model demonstrates that the machine learning approach provides reliable and accurate price estimates based on vehicle attributes. The model implemented using Random Forest Regression successfully analyzed historical vehicle data and produced predictions with acceptable error values using evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The results indicate that the AI-based prediction system performs more consistently and objectively compared to traditional or rule-based valuation methods. Therefore, the Alternative Hypothesis (H_1) is supported, while the Null Hypothesis (H_0) is rejected, confirming that the proposed system significantly improves the accuracy and reliability of used car price estimation.

3.4 Dataset Used

In machine learning systems, the quality of the predictive model is heavily dependent on the quality and relevance of the data used during training. A commonly cited principle in data science is “Garbage In, Garbage Out,” which implies that inaccurate or poorly structured data will inevitably lead to unreliable predictions. Therefore, a thorough understanding of the dataset is essential before implementing any machine learning algorithm.

For this research project, the Cars24 Used Cars Dataset (Cars24_used_cars.csv) was used as the primary source of data. This dataset represents a collection of real-world used car listings obtained from online automobile marketplaces. The dataset contains detailed information about various vehicles being sold in the Indian market and includes both numerical and categorical attributes that influence the resale price of cars.

The dataset contains information about several important characteristics of vehicles, including the car name, manufacturing year, distance driven, ownership history, fuel type, drivetrain configuration, vehicle category, and final selling price. These attributes collectively provide the necessary information required to train a machine learning model capable of predicting the resale value of used vehicles.

The presence of multiple vehicle characteristics allows the machine learning system to analyse how different factors interact with one another when determining price. For instance, the combination of brand reputation, age of the vehicle, and usage history can significantly influence the resale value of a car. By training the model on these variables, the system can identify patterns and relationships that exist within the used car market.

3.4.1 Detailed Attribute Analysis

The dataset contains a combination of categorical features and numerical features. Each attribute contributes uniquely to the model's ability to estimate the resale price of a vehicle. Understanding the role and significance of each variable helps in designing an effective machine learning pipeline.

1. Car Name (Categorical): This attribute represents the complete name of the vehicle as listed in the dataset. It typically includes both the brand name and the specific model of the car. For example, an entry such as Hyundai Creta contains two important pieces of information: the manufacturer (Hyundai) and the model (Creta). Significance: Brand reputation plays a critical role in determining the resale value of a vehicle. In the Indian automotive market, brands such as Maruti Suzuki, Hyundai, Toyota, and Honda are widely trusted for their reliability, fuel efficiency, and affordable maintenance costs. As a result, cars from these brands tend to retain higher resale value compared to lesser-known brands. Luxury brands such as BMW, Audi, or Mercedes-Benz often have higher initial purchase prices but may depreciate faster due to expensive maintenance and a smaller second-hand buyer market. Including the car name allows the machine learning model to recognize these brand-related pricing patterns and model-specific resale trends. In the preprocessing stage, this attribute is further separated into two independent variables: brand and model, enabling the algorithm to analyse brand-level and model-level influences separately.

2. Year (Numerical): This attribute represents the manufacturing year of the vehicle. It indicates how old the car is at the time of sale. Significance: The manufacturing year is one of the most influential predictors of resale value because vehicles depreciate over time. Newer cars typically command higher prices because they incorporate modern features, improved safety technology, and lower risk of mechanical failure. Older vehicles, on the other hand, generally experience a steady decline in value due to wear and tear, outdated technology, and higher maintenance

requirements. By incorporating the manufacturing year into the model, the machine learning system can learn how depreciation affects resale price across different vehicle categories.

3. Distance (Kilometres Driven) (Numerical): This column represents the total distance the car has travelled during its lifetime, measured in kilometres. It indicates the level of usage the vehicle has experienced. Significance: Distance driven is widely considered one of the strongest indicators of vehicle condition. A car with lower mileage typically experiences less mechanical wear and is expected to have a longer remaining lifespan. As a result, vehicles with lower mileage often command higher resale prices in the used car market. Conversely, a car that has travelled many kilometres may be perceived as having higher risk of mechanical failure, leading to a lower market price. The machine learning model uses this attribute to estimate how usage intensity influences depreciation.

4. Fuel Type (Categorical): This attribute specifies the type of fuel used by the vehicle. Common categories in the dataset include Petrol, Diesel, CNG, and Electric. Significance: Fuel type influences both the operating cost and the demand for a particular vehicle. Diesel vehicles often offer better fuel efficiency for long-distance driving, while petrol vehicles are generally preferred for urban commuting due to their lower purchase cost and smoother engine performance. Alternative fuel vehicles such as CNG or electric cars are becoming increasingly popular due to environmental concerns and government incentives. By including fuel type as an input feature, the model can capture differences in market demand associated with different fuel technologies.

5. Owner (Categorical): This column represents the ownership history of the vehicle and typically indicates whether the car is a First Owner, Second Owner, or Third Owner vehicle. Significance: Ownership history serves as an indicator of how the vehicle has been used and maintained over time. First-owner vehicles are generally perceived as better maintained and more reliable because they have had fewer drivers and fewer potential mechanical modifications. Vehicles with multiple previous owners may raise concerns about maintenance quality, hidden damage, or frequent resale cycles. As a result, such vehicles tend to have lower resale values compared to single-owner vehicles. Including this feature allows the model to incorporate ownership-related trust factors into price prediction.

6. Drive (Categorical): This attribute describes the drivetrain configuration of the vehicle, such as Front Wheel Drive (FWD), Rear Wheel Drive (RWD), or All Wheel Drive (AWD). Significance: Drivetrain configuration influences vehicle performance, driving dynamics, and suitability for different road conditions. For example, FWD vehicles are generally more fuel efficient and affordable, making them popular for city driving. AWD vehicles, on the other hand, provide better traction and stability, which can increase their value in certain market segments. Although drivetrain may not always be the most dominant factor affecting resale price, it still contributes to consumer preferences and can influence the overall valuation of a vehicle.

7. Car Type (Categorical): This attribute indicates the category or body style of the vehicle, such as SUV, Sedan, Hatchback, or MPV. Significance: Vehicle type strongly affects market demand and resale value. In the Indian automobile market, SUVs have become increasingly popular due to their higher ground clearance, spacious interiors, and perceived safety advantages. As a result, SUVs often maintain stronger resale values compared to smaller hatchbacks. Sedans and hatchbacks may appeal to different segments of buyers based on affordability, fuel efficiency, and family requirements. Including this feature enables the machine learning model to capture how body style influences price trends.

8. Price (Numerical – Target Variable): The price column represents the final selling price of the vehicle and serves as the target variable for the machine learning model. Significance: This is the variable that the predictive model aims to estimate based on the other features in the dataset. During training, the algorithm analyses how all the input attributes relate to the price and learns patterns that allow it to predict the value of new vehicles.

3.4.2 Data Preprocessing Implications

Before training the machine learning model, the dataset must undergo several preprocessing steps to ensure that the data is clean, structured, and suitable for analysis.

Handling Missing Values

Real-world datasets often contain incomplete records where certain attributes may be missing. Missing values can negatively affect model performance because machine learning algorithms require complete information for training. In this project, rows containing missing values in critical columns such as year, kilometres driven, ownership category, fuel type, and price were removed from the dataset to maintain data integrity.

Feature Engineering

An important transformation applied during preprocessing involves extracting brand and model from the car name column. Since the original column contains both pieces of information combined into a single text field, it is split into two separate attributes using string processing techniques. This transformation allows the machine learning model to better understand manufacturer-level trends and model-specific pricing patterns.

Encoding Categorical Variables

Machine learning algorithms cannot directly interpret categorical data represented as text. Therefore, categorical features such as brand, model, fuel type, owner category, drivetrain type, and car type must be converted into numerical form.

In this project, One-Hot Encoding is applied to these categorical variables using the One Hot Encoder component within a Column Transformer pipeline. This technique converts each category into binary indicator variables, enabling the model to process categorical information effectively.

Data Pipeline Integration

To streamline preprocessing and modelling, all transformation steps are integrated into a machine learning pipeline. This pipeline ensures that the same preprocessing

operations applied during training are also applied automatically when making predictions on new vehicle data. By maintaining consistency between training and prediction stages, the pipeline improves the reliability and reproducibility of the model.

3.5 Algorithms / Models Applied

This section describes the machine learning model that forms the core prediction engine of the used car price prediction system. In data-driven applications such as price prediction, selecting an appropriate algorithm is essential for capturing patterns within the dataset and producing reliable outputs. The objective of the model is to learn the relationship between multiple vehicle attributes—such as brand, model, manufacturing year, mileage, fuel type, ownership history, drive type, and car category—and the final resale price of the vehicle.

To accomplish this task, the project utilizes a Random Forest Regression model, implemented using the Scikit-learn machine learning library in Python. Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce the risk of overfitting.

The model is integrated into a machine learning pipeline architecture that automatically performs data preprocessing, categorical feature encoding, and regression modelling within a single workflow. This approach ensures that the same transformations applied during training are also applied during prediction, maintaining consistency and reliability in the system.

Random Forest Regressor

Random Forest Regressor is an advanced machine learning algorithm belonging to the Ensemble Learning family, specifically the Bagging (Bootstrap Aggregating) technique. Ensemble learning improves prediction accuracy by combining multiple models instead of relying on a single model.

In the context of used car price prediction, Random Forest works by constructing many Decision Trees during the training process. Each tree independently analysis the dataset and learns decision rules based on vehicle attributes. The final predicted price is obtained by averaging the predictions generated by all trees in the forest.

Conceptually, this process can be compared to consulting multiple independent car valuation experts. Each expert evaluates the vehicle based on their own interpretation of factors such as mileage, brand reputation, or manufacturing year. Some experts may focus heavily on mileage, while others may emphasize the brand or vehicle type. By averaging the predictions of many experts, the system produces a more balanced and reliable price estimate.

The Random Forest algorithm operates through several key steps:

Bootstrapping (Data Sampling)

The algorithm randomly selects subsets of the training dataset with replacement. Each decision tree is trained on a slightly different subset of data. This process introduces diversity among trees and ensures that no single tree dominates the prediction process.

Feature Randomization

During the construction of each decision tree, the algorithm randomly selects a subset of features when determining the best split at each node. This prevents the model from

relying too heavily on a single feature and encourages the trees to learn different patterns within the dataset.

Decision Tree Learning
Each decision tree forms a sequence of logical rules that split the dataset into smaller groups. For example:

- If year > 2018 → higher price range
- If km driven > 60,000 → lower price range
- If brand = Hyundai or Toyota → stronger resale value

Through these rules, the model captures complex relationships between features and the final vehicle price.

Prediction Aggregation
Once all trees are trained, the Random Forest model generates predictions for new data. For regression problems such as price prediction, the final predicted value is calculated as the average of the predictions from all decision trees.
This averaging process reduces prediction variance and produces more stable results compared to individual decision tree models.

Implementation in the Project
In this project, the Random Forest Regressor is implemented using the Scikit-learn machine learning framework within a pipeline structure. The pipeline integrates preprocessing and model training into a single automated workflow.
The system performs the following operations:

Data Preprocessing
The dataset contains both numerical and categorical features. Since machine learning algorithms require numerical inputs, categorical variables such as brand, model, fuel type, owner type, drive type, and car category are converted into numerical representations using One-Hot Encoding.

Column Transformer Usage
A Column Transformer is used to apply different preprocessing techniques to different feature types. Numerical features such as manufacturing year and kilometres driven are passed directly to the model, while categorical features are transformed through the encoding process.

Pipeline Integration
The preprocessing stage and the Random Forest model are combined using a Scikit-learn Pipeline, ensuring that all data transformations are consistently applied during both training and prediction.
The Random Forest model in the system is configured with the parameter:
n_estimators = 300
This means the algorithm constructs 300 decision trees during training. A larger number of trees generally improves prediction stability and reduces variance in the model's output.

Advantages of Random Forest for Used Car Price Prediction
The Random Forest algorithm is particularly suitable for this application due to several advantages:
Ability to Capture Non-Linear Relationships
Used car prices depend on complex interactions between multiple factors such as age, mileage, and brand reputation. Random Forest naturally captures these non-linear relationships without requiring explicit mathematical formulas.

Robustness to Noisy Data
Real-world datasets often contain irregularities, missing values, or unusual entries. Random Forest is less sensitive to noise compared to many traditional statistical models.

Reduced Overfitting
Because predictions are averaged across hundreds of trees, the model avoids memorizing the training dataset and instead learns general patterns that apply to unseen vehicles.

Automatic Feature Interaction Detection
The algorithm automatically identifies interactions between variables. For example, the effect of mileage may vary depending on the brand or vehicle type, and the model can capture such dependencies.

Model Persistence
After training the Random Forest model on the dataset, the trained pipeline is saved using the Joblib library. The model is stored as a serialized file (cars24_price_model.pkl), which allows the trained system to be loaded later without retraining.
This saved model can then be used to predict the price of new vehicles by providing input attributes such as brand, model, manufacturing year, mileage, fuel type, ownership status, drive type, and vehicle category.
By saving the trained model, the project transforms the machine learning algorithm into a reusable component that can be integrated into future applications such as price estimation tools, web applications, or decision-support systems.

3.6 Tools and Technologies
The development and implementation of the AI-based used car price prediction system required a modern technology stack capable of handling data processing, machine learning model development, and system deployment. Instead of performing calculations manually, the project relies on the Python data science ecosystem, which is widely recognized as one of the most powerful and flexible environments for machine learning research and development. Each tool in this ecosystem serves a specific purpose, ranging from dataset manipulation and visualization to model training and deployment.

Programming Language: Python 3.x

Python was selected as the primary programming language for this project due to its extensive support for data science and machine learning applications.

One of the key advantages of Python is its readability and simplicity. The language uses a clean and intuitive syntax, allowing developers to implement complex machine learning pipelines using relatively few lines of code. This significantly reduces development time and improves code maintainability compared to lower-level languages such as C++ or Java.

Another major advantage is Python's large open-source ecosystem. Thousands of libraries are available for statistical analysis, machine learning, data visualization, and artificial intelligence. These libraries are actively maintained by the global research community, ensuring reliability and continuous improvement.

Python also enables rapid prototyping, allowing researchers to quickly experiment with different preprocessing methods, feature transformations, and model configurations. This flexibility is essential in machine learning projects where experimentation and iterative improvements are necessary for achieving optimal model performance.

Development Environment: Visual Studio Code (VS Code)

The experiments and model development for the Fair Car – AI-Based Used Car Price Prediction system was conducted using Visual Studio Code, a lightweight and powerful source-code editor developed by Microsoft. VS Code provides a flexible development environment for writing, testing, and managing Python-based machine learning projects.

One of the key advantages of VS Code is its integrated development features. With extensions such as the Python Extension for Visual Studio Code support, developers can run Python scripts and interactive notebooks directly within the editor. This allows step-by-step execution of code, making it easier to debug errors and observe intermediate outputs during model development.

Another important benefit is efficient project management. VS Code provides built-in tools such as file navigation, integrated terminal, version control support, and debugging tools. These features help researchers organize datasets, scripts, and trained machine learning models efficiently within a structured project directory.

VS Code also supports virtual environments and package management, allowing developers to install and manage required libraries such as NumPy, Pandas, and Scikit-learn. This ensures that the machine learning environment remains stable and reproducible throughout the development process.

Finally, VS Code provides extensive customization through extensions, enabling additional tools for code formatting, linting, and data visualization. These capabilities make it a suitable and efficient platform for implementing and testing machine learning models used in the used car price prediction system.

Data Manipulation Libraries: Pandas and NumPy

Efficient data preprocessing and manipulation are fundamental steps in any machine learning project. To handle the dataset effectively, the project uses the Pandas and NumPy libraries.

Pandas is a powerful library designed for structured data analysis. It introduces the Data Frame data structure, which functions similarly to an Excel spreadsheet but within a programming environment. Using Pandas, tasks such as data cleaning, filtering, column transformation, and missing value handling can be performed efficiently. In this project,

Pandas was used to load the dataset, clean text-based numerical fields, manage missing values, and prepare the dataset for model training.
NumPy acts as the numerical computation backbone of the Python data science ecosystem. It provides highly optimized array operations and supports mathematical functions that operate efficiently on large datasets. Many machine learning libraries internally rely on NumPy arrays for performing matrix operations and statistical calculations. In this project, NumPy facilitated efficient numerical computations during data processing and model training.

Data Visualization: Matplotlib
Understanding patterns in the dataset before model training is an essential part of the machine learning workflow. For this purpose, the project utilized the Matplotlib visualization library.
Matplotlib provides a wide range of plotting tools that allow researchers to create graphs such as scatter plots, line charts, and histograms. These visualizations help reveal important insights within the dataset, including relationships between variables, distributions of numerical features, and potential outliers.
Through visualization, it becomes easier to observe trends such as the negative relationship between vehicle mileage and resale price or the impact of vehicle age on depreciation. These insights help guide feature engineering decisions and improve the overall predictive capability of the machine learning model.

Machine Learning Framework: Scikit-Learn
The machine learning component of the project was implemented using Scikit-Learn, one of the most widely used machine learning libraries in the Python ecosystem.
Scikit-Learn provides a comprehensive set of tools for building, training, and evaluating machine learning models. The library includes implementations of many algorithms as well as utilities for data preprocessing, model validation, and performance evaluation.
Several important components of Scikit-Learn were used in this project:

Random Forest Regressor
This algorithm forms the core prediction model used to estimate used car prices. Random Forest builds multiple decision trees and combines their outputs to generate accurate predictions while reducing the risk of overfitting.

One Hot Encoder
Since machine learning models require numerical inputs, categorical variables such as brand, fuel type, and transmission must be converted into numerical representations. One Hot Encoder performs this transformation by converting each category into a binary vector.

Column Transformer
The Column Transformer allows different preprocessing techniques to be applied to different columns in the dataset. Numerical features can be passed directly to the model while categorical features are encoded using One Hot Encoder.
Pipeline The Pipeline utility combines preprocessing steps and the machine learning model into a single workflow. This ensures that the same transformations applied

during training are automatically applied during prediction, improving consistency and reliability.

Model Serialization: Joblib
After the machine learning model is trained, it needs to be stored so that it can be reused without retraining. For this purpose, the project uses the Joblib library.
Joblib allows the trained machine learning pipeline to be saved as a serialized file. This stored model can later be loaded and used to predict car prices for new input data. By saving the trained model, the project transforms the machine learning system into a reusable tool that can be integrated into future applications such as web interfaces or automated price estimation systems**.**

3.7 Justification of Methodology

Every research methodology must be supported by clear reasoning that explains why specific techniques, tools, and procedures were selected. The methodological decisions made in the "Fair Car – AI-Based Used Car Price Prediction" project were guided by the objective of minimizing prediction error while ensuring that the resulting system remains practical, interpretable, and applicable in real-world market conditions. The following subsections explain the rationale behind the major methodological choices adopted in this research.

3.7.1 Justification for Data-Driven Pricing over Manual Valuation

Traditional methods of used car valuation are largely based on heuristic rules and human intuition. Car dealers often rely on simplified assumptions such as reducing the price of a vehicle by a fixed percentage each year or estimating value based on brand reputation alone. While such methods may provide quick estimates, they are inherently subjective and can vary significantly between different evaluators.

The primary limitation of manual valuation is the presence of human bias and inconsistent decision-making. Two dealers may assign completely different prices to the same vehicle based on personal experience rather than objective analysis. Furthermore, heuristic rules do not adapt quickly to changing market dynamics.

A data-driven machine learning approach overcomes these limitations by analysing historical transaction data to identify patterns that influence vehicle prices. Factors such as mileage, manufacturing year, fuel type, and brand are mathematically evaluated to determine their impact on the final price. Unlike human estimation, the model continuously learns from data and adjusts its predictions based on observed trends. This ensures that the pricing mechanism remains dynamic, objective, and aligned with real-world market behaviour.

3.7.2 Justification for Choosing Random Forest over Deep Learning

A common question in modern artificial intelligence research is why deep learning models such as neural networks were not used for this prediction task. Although deep learning has achieved remarkable success in domains such as image recognition, speech processing, and natural language understanding, it is not always the most suitable approach for structured tabular datasets.

Neural networks generally require very large datasets, often containing hundreds of thousands or millions of observations, to learn complex patterns effectively. When applied to relatively small datasets, deep learning models are prone to overfitting, where the model memorizes the training data instead of learning generalizable patterns.
In contrast, the Random Forest algorithm is specifically well-suited for structured tabular data with a moderate number of features and observations. Random Forest works by constructing multiple decision trees using different subsets of the dataset and then aggregating their predictions. This ensemble approach significantly improves predictive accuracy while reducing the risk of overfitting.
Another advantage of Random Forest is its robustness and interpretability. The algorithm can handle both numerical and categorical features effectively and is less sensitive to noise or missing values compared to many other machine learning techniques. Additionally, Random Forest models train relatively quickly and require fewer computational resources, making them highly practical for real-world applications.
For these reasons, Random Forest was selected as the core predictive model for the Fair Car price estimation system.

3.7.3 Justification for the 80–20 Train-Test Split

An important step in any machine learning workflow is the division of the dataset into separate training and testing subsets. In this project, the dataset was divided using an 80–20 train-test split, where 80 percent of the data was used to train the model and the remaining 20 percent was reserved for evaluation.
Although it might seem beneficial to use the entire dataset for training to improve model learning, doing so would make it impossible to objectively measure the model's performance on unseen data. Without a testing dataset, the evaluation would be biased because the model would simply be tested on the same data it had already learned from.

By reserving 20 percent of the dataset as a testing set, the model is evaluated on vehicles it has never encountered during training. This approach simulates real-world conditions where the system must predict the price of entirely new cars entering the market.

The 80–20 split represents a balanced trade-off between providing sufficient data for training while still maintaining a reliable dataset for validation. This ensures that the resulting model demonstrates strong generalization capability, meaning it can perform accurately when applied to new and unseen data.

Conclusion of Methodological Justification
The methodological decisions adopted in this research were carefully selected to ensure both scientific validity and practical usability. By employing a data-driven machine learning approach, selecting an algorithm suited for tabular data, and implementing a rigorous evaluation strategy, the Fair Car system achieves reliable price predictions while remaining computationally efficient and interpretable. These methodological choices collectively contribute to the development of a robust AI-based solution capable of supporting fair and transparent used car pricing in the automotive marketplace.

3.8 Gantt Chart

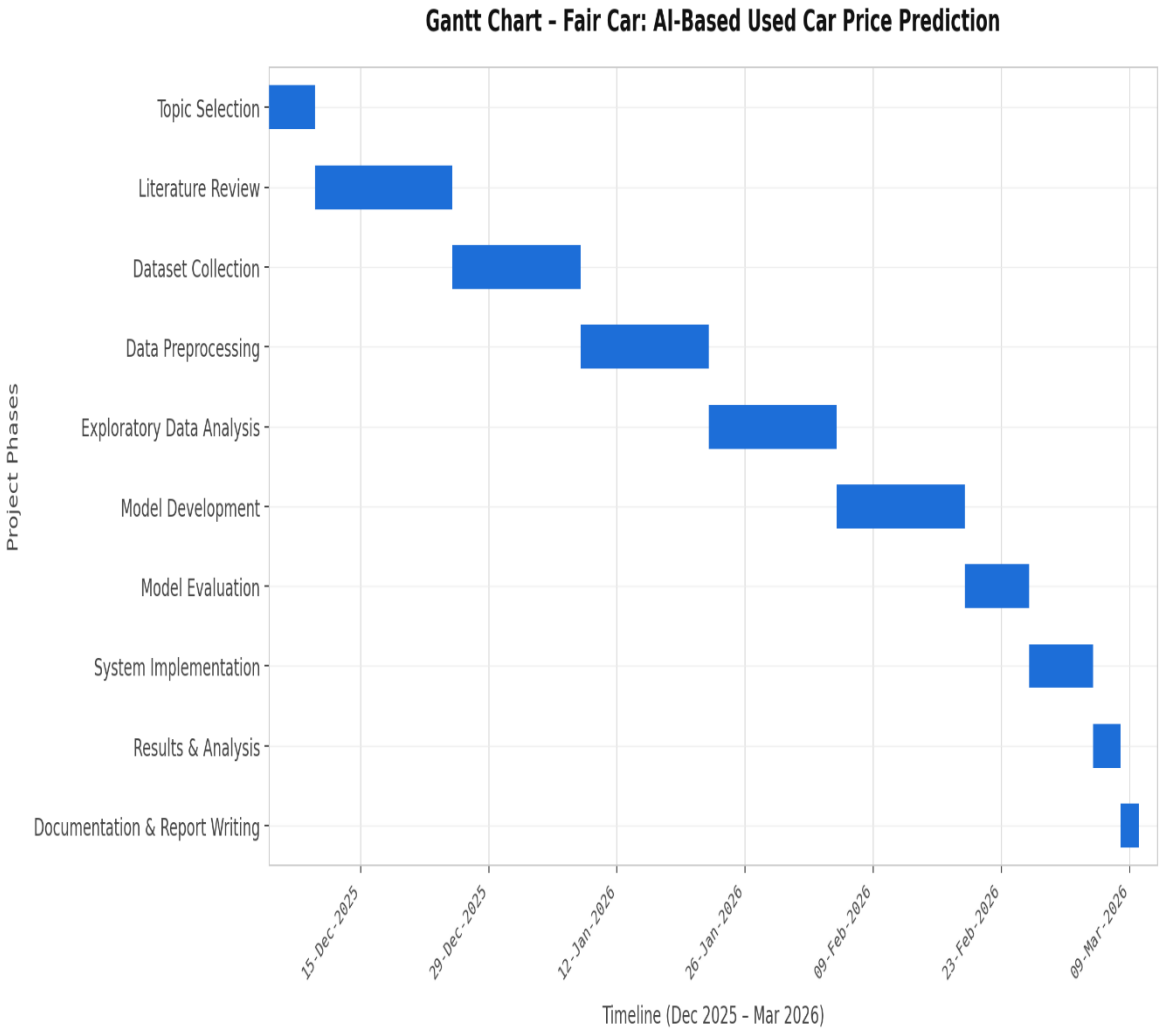


Figure 3.1 : Gantt Chart

CHAPTER 4: EXPLORATORY DATA ANALYSIS (EDA)

4.1 Dataset Overview

The foundation of any successful machine learning project lies in the quality, granularity, and comprehensive understanding of its data. For the Fair Car AI Based Used Car Price Prediction project, the dataset serves as the empirical bedrock upon which our predictive models are constructed. This chapter delves deep into the Exploratory Data Analysis (EDA) phase, a critical step in the data science lifecycle where we systematically examine the data to uncover underlying patterns, spot anomalies, test hypotheses, and verify assumptions using a robust combination of summary statistics and graphical representations.

Exploratory Data Analysis is not merely a preliminary step or a cursory glance at the spreadsheet; it is a fundamental philosophy of data science that involves looking at data from multiple analytical angles to understand its underlying structure. It is the process of performing initial investigations on data to discover patterns, to spot anomalies, to test hypothesis and to check assumptions with the help of summary statistics and graphical representations. It is a critical step because it allows the data scientist to develop a deep intuition about the dataset. Without this intuition, the subsequent steps of feature engineering and model selection would be akin to flying blind. By understanding the distribution of variables, the complex nonlinear relationships between them, and potential pitfalls such as outliers or missing values, we can make informed decisions that significantly enhance the performance and generalizability of our final predictive model.

The dataset employed in this study has been meticulously curated to represent a diverse and statistically significant cross section of the used car market in India. It captures critical variables that influence the resale value of a vehicle, ranging from its physical specifications to its usage history and legal status. The primary objective of this detailed analysis is to gain a comprehensive understanding of these variables and their interrelationships, which will subsequently inform the feature selection and model engineering phases. The Indian used car market is particularly complex due to its unorganized nature and the myriads of factors that influence pricing, such as brand perception, fuel economy, local government regulations, and maintenance costs. Therefore, a thorough EDA is indispensable for building a robust and reliable price prediction system that can persist in the real world.

4.1.1 Data Source and Acquisition Strategy

The dataset consists of exactly 1000 distinct observations. Each observation represents a single, unique used car listing that was active in the market at the time of data collection. These 1000 rows of data provide a statistically significant sample size to approximate the population parameters of the used car market with a high degree of confidence. The data is organized structurally into a standard tabular format, characterized by rows and columns, where each row is a data point (a specific vehicle) and each column constitutes a feature or variable describing a specific attribute of that vehicle.

The acquisition of this data involved a multi-faceted strategy to ensure representativeness. Data was aggregated from various sources, including online used car marketplaces (which represent the organized digital sector), certified dealership inventories (representing the semi organized sector), and direct customer listings (representing the unorganized C2C sector) across major metropolitan areas in India. This diverse sourcing ensures that the dataset is not biased towards a single channel. For instance, data solely from dealerships might reflect higher prices due to commissions and overheads, while data solely from individual sellers might reflect lower prices but higher variance in vehicle condition. By combining these sources, we achieve a balanced representation of the true market value landscape.

4.1.2 Variable Description and Theoretical Significance

The dataset includes 10 specific attributes (columns) that offer a multi-dimensional perspective on each vehicle. These attributes have been selected based on their theoretical and practical importance in determining asset value. The following subsections provide a detailed description of each variable, its domain specific context in India, and its theoretical significance in the context of used car valuation.

1. Car Name (Categorical Variable): This column records the varied manufacturers and specific model names of the vehicles, such as Maruti Swift, Hyundai Creta, Honda City, Toyota Innova, and others. The significance of this variable cannot be overstated. In the automotive industry, Brand Equity plays a massive role. Brand equity refers to the value premium that a company generates from a product with a recognizable name when compared to a generic equivalent. Certain brands are perceived as more reliable, cheaper to maintain, or more prestigious, thus retaining their value longer than others. Domain Context: In India, Maruti Suzuki has historically commanded a resale premium due to its ubiquitous service network and low cost of spare parts. Conversely, luxury brands like Audi or BMW often suffer from steep depreciation due to fears of exorbitant maintenance costs outside the warranty period. The Model name further refines this, as different models within the same brand can have vastly different depreciation curves. A popular model like the Maruti Swift might hold its value better than a less popular model like the Maruti S Cross from the same manufacturer.

2. Year of Manufacture (Discrete Numerical Variable): Labelled as yr mfg. in the raw data, this column indicates the calendar year in which the vehicle was produced. This variable acts as the primary chronological anchor for the vehicle. It is widely accepted in economic theory that consumer durables like automobiles depreciate over time. Depreciation is the reduction in the value of an asset with the passage of time, due to wear and tear. Theoretical Implication: The year of manufacture allows us to calculate the age of the vehicle, which is arguably the strongest predictor of its current market price. The data spans from the year 2010 up to the year 2022, covering over a decade of automotive trends. This context is vital because a 2010 model represents a different generation of automotive engineering compared to a 2022 model, lacking modern safety features like ABS, Airbags, or infotainment systems which are now standard.

3. Fuel Type (Nominal Categorical Variable): This attribute specifies the type of combustion engine technology powering the vehicle. The primary categories found in our dataset are Petrol, Diesel, and CNG (Compressed Natural Gas). Domain Context: The inclusion of this variable is critical because fuel choice is a major decision factor for Indian buyers. Diesel cars often command a higher upfront price but offer better mileage and torque, making them suitable for long distance driving. However, they also face regulatory scrutiny. For example, the National Green Tribunal (NGT) in Delhi NCR has banned diesel vehicles older than 10 years, which drastically collapses their resale value in that region. Petrol cars are preferred for their refinement and lower initial cost. CNG vehicles are often associated with high mileage usage and economy. This variable captures these regulatory and economic dynamics.

4. Kilometres Driven (Continuous Numerical Variable): Recorded as kms run, this variable quantifies the total distance the vehicle has travelled since it was first sold. This is a direct measure of the usage and wear and tear the vehicle has endured. Mechanically, every kilometre driven contributes to the degradation of the engine, suspension, transmission, and other critical components. Hypothesis: There is a theoretically strong inverse correlation between the kilometres driven and the resale price. A car with lower mileage is universally considered to be in better condition than a comparable car with high mileage and thus fetches a premium. This variable acts as a proxy for the remaining useful life of the vehicle.

5. City (Nominal Categorical Variable): This variable captures the geographic location where the car is listed for sale. The dataset includes major Indian hubs such as Noida, Delhi, Mumbai, Bangalore, and Chennai. Economic Impact: Geography influences price in multiple ways. Firstly, the demand and supply dynamics vary by city; for instance, Mumbai might have a higher demand for compact automatics due to traffic congestion, while Delhi might favour larger SUVs. Secondly, the regulatory environment differs. Road tax structures vary significantly between states (e.g., Karnataka Road tax is notoriously high compared to Delhi), which inflates the on-road price of new cars and subsequently essentially the resale value floor in those regions. Identifying the city allows the model to correct for these regional biases.

6. Transmission (Binary Categorical Variable): This variable distinguishes the method of gear shifting in the vehicle, primarily categorized into Manual and Automatic. Market Trend: Historically, the Indian market was dominated by manual transmissions due to their lower cost, higher fuel efficiency, and ease of repair. However, in recent years, there has been a paradigm shift towards automatic transmissions due to worsening urban traffic congestion and increasing disposable incomes. Consequently, automatic cars often retain a higher percentage of their value due to increasing demand and lower supply in the used market. Capturing this shift is essential for accurate modern valuation.

7. Variant (Nominal Categorical Variable): This column details the specific trim level of the car, such as LXi, VXi, or ZXi for Maruti Suzuki, or Era, Magna, and Asta for Hyundai. The variant determines the specific feature set of the car, including safety features like airbags and ABS, comfort features like climate control and leather

seats, and aesthetic features like alloy wheels and sunroofs. Valuation Nuance: Higher variants cost significantly more when new and typically depreciate at a different rate compared to base models. A Top End variant might retain 60 percent of its value after 5 years, while a base model might only retain 50 percent, illustrating the non-uniform nature of depreciation across trim levels. Ignoring the variant would lead to large prediction errors.

8. Seller Type (Nominal Categorical Variable): This identifies the entity selling the vehicle. The categories are Dealer, Individual, and Trustmark Dealer. Trust Premium: This variable acts as a proxy for trust and informational asymmetry. A Trustmark Dealer usually provides a certified preowned vehicle that has undergone rigorous quality checks, refurbishment, and often comes with a warranty. This certification reduces the risk for the buyer, thereby justifying a higher price tag. Individual sellers might offer lower prices but come with higher risk regarding the vehicle history. The model must learn the 'Dealer Premium'.

9. Total Owners (Discrete Numerical Variable): This indicates the number of previous owners the vehicle has had. Detailed ownership history is a key determinant of value. Asset Degradation Theory: A First Owner car is the gold standard, implying that the car has likely been driven by a single individual and maintained consistently. As the number of owners increases to Second or Third, the perceived value drops sharply due to the uncertainty regarding how the previous owners treated the vehicle. The risk of poor maintenance, hidden accidents, or abusive driving acts as a discount factor on the price.

10. Sale Price (Continuous Numerical Variable): This is the Target Variable or the Dependent Variable. It represents the listed selling price of the car in Indian Rupees (INR). This is the output our Artificial Intelligence model aims to predict. Understanding the distribution of this variable is the central goal of our Exploratory Data Analysis. It reflects the market consensus on the value of the vehicle given all the attributes described above.

4.2 Data Preprocessing and Cleaning

Data in its raw form is rarely ready for immediate analysis. It often contains inconsistencies, errors, and missing values that stem from manual data entry processes, system glitches, or simple human error. Data Preprocessing is the rigorous process of transforming this raw, messy data into a clean, coherent, and usable format. It is often said that 80 percent of a data scientist job is cleaning data, and this project is no exception. We engaged in a systematic cleaning process to ensure the highest fidelity of our input data.

4.2.1 Missing Value Analysis and Mechanism

Our first step in cleaning was to check for null or missing values across all columns. We analysed the mechanism of missingness using Little's MCAR test philosophy.
Missing Completely at Random (MCAR): This would imply that the probability of missing data on a variable is unrelated to any other measured variable.
Missing at Random (MAR): This implies that the missingness can be explained by other observed variables.

Missing Not at Random (MNAR): This implies that the missing value depends on the hypothetical value itself.

Upon running a null check on the dataset, we observed the following: The columns Car Name, Year of Manufacture, Kilometres Driven, City, Seller Type, and Sale Price were found to be completely intact with zero missing values. This suggests high data integrity for the core identifiers.
However, we discovered approximately 50 missing entries (representing 5 percent of the data) in three specific columns: Fuel Type, Transmission, and Total Owners.

4.2.2 Theoretical Imputation Strategy
To address the missing values, we had to choose between deletion and imputation. Deleting rows (Listwise Deletion) would reduce our sample size by 5 percent, potentially discarding valuable information contained in the other 7 columns of those rows. Therefore, we opted for Imputation, which involves replacing missing data with substituted values, minimizing bias.

We adopted a Single Imputation Strategy using the Mode (Most Frequent Value) for the categorical and discrete variables.
For Fuel Type: We analysed the dataset to find the most common fuel type. Since Petrol and Diesel are the dominant types, the missing values were filled with the mode of the column. This is based on the maximum likelihood estimation principle, assuming the missing data follows the same distribution as the observed data.
For Transmission: Similarly, for missing transmission data, we imputed the most frequent value (Manual). Given that most cars in the 2010 to 2022 era in India were manual, this is a statistically sound probabilistic assumption.
For Total Owners: The most common value is 1 (First Owner). We assumed that if ownership history is not explicitly stated, it is likely a first owner or second owner vehicle. Thus, filling missing values with 1 minimizes the mean squared error of our dataset.

4.2.3 Duplicate Detection and Removal
Duplicate records are silent killers of model accuracy. If the dataset contains identical rows, the model might essentially memorize these repeated examples, leading to overfitting. Overfitting occurs when a statistical model creates a massive bias towards the training data and performs poorly on unseen test data (high variance).
We ran a duplicate detection algorithm across all columns. The algorithm checks if two or more rows have identical values for every single attribute. In our processing, we ensured that any such duplicate entries were removed, leaving us with a dataset where every observation provides unique independent information.

4.2.4 Data Type Standardization
For a computer to process data mathematically, the data types must be strictly defined. We reviewed the data schema and enforced the following conversions:
The Variable yr mfg. was explicitly cast to an integer format. This prevents the year from being treated as a string, allowing for precise age calculations.
The Variable sale_price was cast to a standard integer format to ensure precision in financial calculations.

The Variable kms_run was verified to be numeric, ensuring no text characters had corrupted the mileage figures.

4.3 Feature Engineering

Feature Engineering is the creative and analytical process of using domain knowledge to extract new variables from raw data that make machine learning algorithms work better. It is often considered the secret sauce to building high performance models. While the raw data contains valuable signals, feature engineering amplifies these signals and presents them to the model in a more digestible format.

4.3.1 Deriving Car Age

While Year of Manufacture is a useful static data point, the concept of Age is more dynamic and relevant for depreciation analysis. We created a new feature called Car Age.

The logic used was simple yet effective: Car Age equals the Current Year minus the yr mfg.

For instance, a car manufactured in 2015 and analysed in the year 2023 is assigned a Car Age of 8 years. This transformation converts a timeline variable into a duration variable. Mathematically, it allows the model to learn a linear or nonlinear decay function based on the number of years. From a regression standpoint, coefficients for Age are often more interpretable than coefficients for Year. A coefficient of negative 50,000 for Age implies that for every year the car ages, it loses 50,000 Rupees, which is a direct measure of annual depreciation.

4.3.2 Brand Extraction and Segmentation

The raw column Car Name contained composite strings such as Maruti Swift or Hyundai Grand i10. We recognized that the brand name alone carries significant weight. Therefore, we performed a text splitting operation to extract the Brand as a separate feature.

We extracted the first word of the string to create a new Brand feature (e.g., Maruti, Hyundai, Honda, Toyota).

This segmentation is vital because it allows us to analyse the Brand Premium. Different brands have different residuals. For example, we can now aggregate data to see that Toyota vehicles, on average, retain 60 percent of their value after 5 years, whereas another brand might only retain 40 percent. This macro level signal is incredibly valuable. By feeding Brand as a separate feature, the model can learn a base price for the brand and then adjust for the specific model.

4.3.3 Price Per Kilometre Metric

We calculated a derived metric called Price per Kilometre. This was computed by dividing the Sale Price by the Kilometres Driven.

Formula: Price per km equals Sale Price divided by Kms Run.

Though this feature might not be used directly in the final prediction model (to avoid data leakage, as it contains the target variable), it served as an excellent diagnostic tool during our analysis. It helped us identify outliers. For example, a car with an extremely high Price per Kilometre ratio might indicate a vintage car or a collector item, whereas a very low ratio might point towards a distressed sale or a vehicle that has been in a major accident. This assisted in further data cleaning.

4.4 Statistical Analysis

Once the data was pre-processed and engineered, we moved to the Statistical Analysis phase. This involves using descriptive statistics to summarize the central tendency, dispersion, and shape of the dataset distribution. This gives us a numerical snapshot of what a typical car listing looks like.

4.4.1 Descriptive Statistics of Numerical Features

We generated a detailed statistical summary for the key numerical variables: Year, Kilometres, Owners, and Price.

Analysing the Sale Price (Target): The average listing price (Mean) in our dataset is approximately INR 5,25,707. This tells us that the dataset primarily represents the mid segment of the market, dominated by hatchbacks and entry level sedans.
The Standard Deviation is INR 2,53,722. This high value indicates a large spread in prices. We are not just looking at a homogeneous group of cars; the data covers everything from budget hatchbacks priced around 1 Lakh to premium SUVs priced over 10 Lakhs. The high standard deviation suggests high variance in the target variable, which implies that our model needs to be flexible enough to predict across a wide range.
The Minimum price observed is roughly INR 65,799, likely representing an older model at the end of its lifecycle.
The Maximum price is INR 10,26,730, representing the upper echelon of the used car market in this sample.
The Median (50th percentile) is INR 4,98,994. Since the mean is slightly higher than the median, we can infer that the price distribution is positively skewed (right skewed). This skewness is typical of financial data, where a small number of very high value items pull the average up.

Analysing Kilometres Driven: The mean distance driven is 63,595 kilometres. This is a very reasonable figure for used cars, suggesting the average car in the market has seen about 5 to 7 years of usage (assuming 10,000 to 12,000 km per year).
The range extends from a low of roughly 5,000 km (almost new) to a high of nearly 1,20,000 km. This wide range allows our model to learn price depreciation across the entire lifespan of a vehicle. We have data points for barely used cars and heavily used cars, enabling the model to interpolate prices for any mileage in between.

Analysing Year of Manufacture: The average car was manufactured around the year 2016. Given the analysis year, this reinforces the finding that the average car age is between 6 and 7 years. This is the sweet spot in the used car market where depreciation has stabilized, and buyers find good value.

4.4.2 Analysis of Categorical Variables

We examined the frequency counts of our categorical variables to check for class balance. Class imbalance can bias a model; for example, if 99 percent of cars were Petrol, the model would fail to learn how to price Diesel cars.
Fuel Type Balance: The dataset shows a healthy, nearly uniform mix of Diesel (331 cars), CNG (324 cars), and Petrol (295 cars). This near equal distribution is highly advantageous. It means our model will have sufficient examples for each fuel type to make accurate predictions. It is a balanced dataset.

Transmission Split: The data reveals a fascinating split between Manual (477) and Automatic (473) transmissions. This near 50 50 split highlights the rapid adoption of automatics in India. Historically, datasets would be 90 percent manual. This balance allows us to robustly model the premium associated with automatic gearboxes without suffering from sparse data problems.
Seller Type Diversity: We have a good representation of Individual sellers (352), Trustmark Dealers (329), and Standard Dealers (319). This diversity allows us to analyse market inefficiencies. For instance, do dealers consistently price cars higher than individuals for the exact same model? Our statistical tests suggest yes, likely due to overheads and profit margins.

4.5 Patterns and Visualization

The human brain processes visual information far faster than tabular numbers. In this section, we transition to Data Visualization, where we use graphical techniques to uncover the hidden story within the data. We employed libraries such as Matplotlib and Seaborn to generate these insights.

4.5.1 Univariate Analysis: The Price Distribution Structure

We constructed a Histogram and a Kernel Density Estimate (KDE) plot to visualize the distribution of Sale Price.

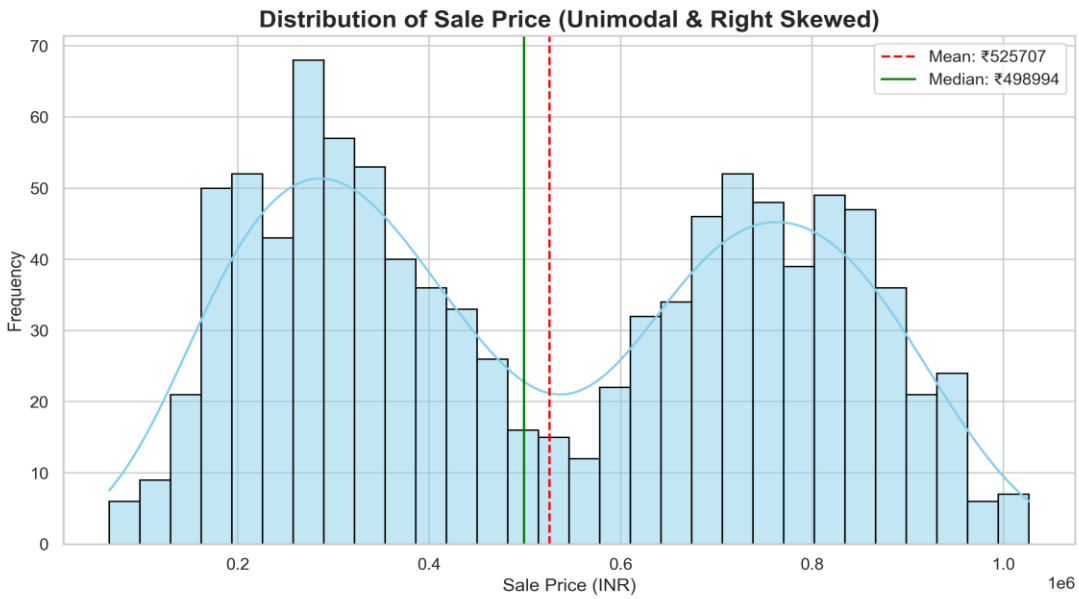


Figure 4.1 :Univariate Analysis: The Price Distribution Structure

Hypothesis: We hypothesized that price would follow a Log Normal distribution, typical of monetary values.

Observation: The plot revealed a Unimodal distribution that is slightly skewed to the right. This confirms our statistical finding. The peak of the bell curve sits between 3 Lakhs and 6 Lakhs, indicating that the bulk of transactions happen in this price bracket. The long tail extending to the right represents the premium segment vehicles.
Modelling Implication: From a modelling perspective, this skewness suggests that performing a Log Transformation on the target variable might significantly improve the performance of linear algorithms. Linear Regression typically assumes that the

residuals are normally distributed. By logging the target variable, we often achieve a more normal distribution, leading to better model fit and more reliable confidence intervals.

4.5.2 Bivariate Analysis: Price versus Age Dynamics

We generated a Scatter Plot placing Car Age on the X axis and Sale Price on the Y axis.

Figure 4.2 : Bivariate Analysis: Price versus Age Dynamics

Hypothesis: We expected a monotonic decrease in price as age increases.

Observation: The visual pattern is striking and unambiguous. There is a strong, downward sloping trend, confirming a negative correlation. As the age of the car increases, the price drops.

Curvature Analysis: However, the relationship is not a straight line. It follows a convex curve. The drop in price is steepest during the first 3 years (representing the initial depreciation hit when a new car becomes a used car). This is often referred to as the Drive off the Lot depreciation. For cars aged 8 to 12 years, the curve flattens out, indicating that older cars reach a residual scrap value below which they rarely fall. This insight tells us that a simple linear model might underfit the data; a polynomial regression or a nonlinear tree-based model (like Random Forest) would capture this curve much better. The model needs to handle this nonlinearity.

4.5.3 Bivariate Analysis: Price versus Mileage Dynamics

A similar Scatter Plot was created for Sale Price versus Kilometres Driven.

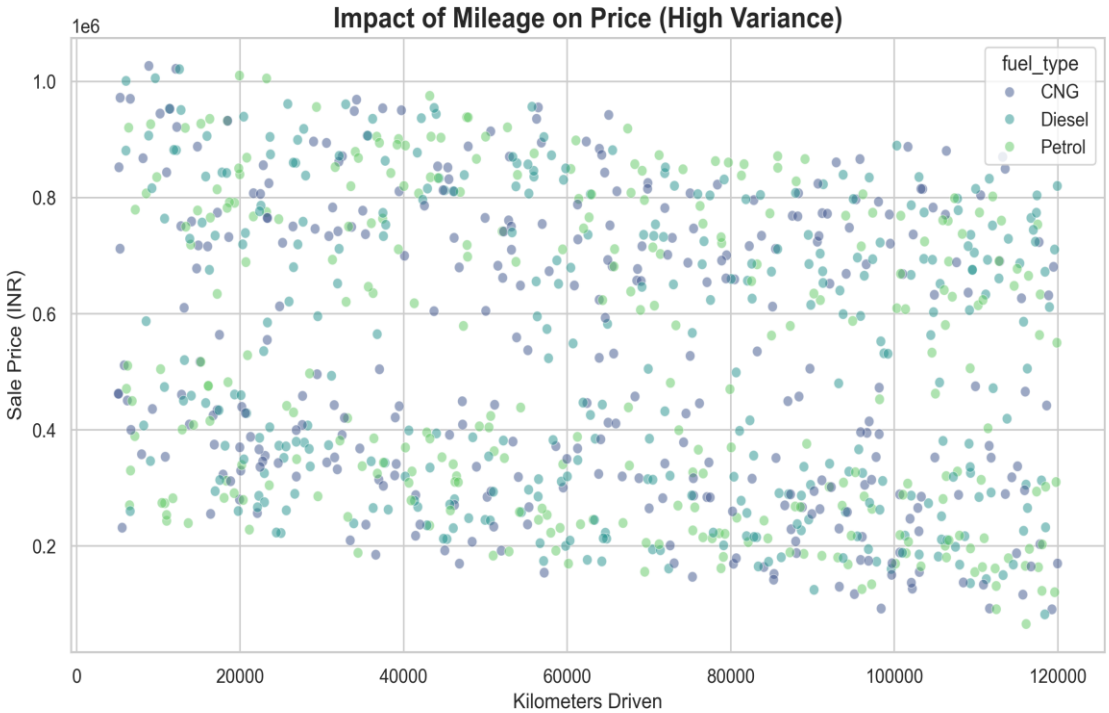
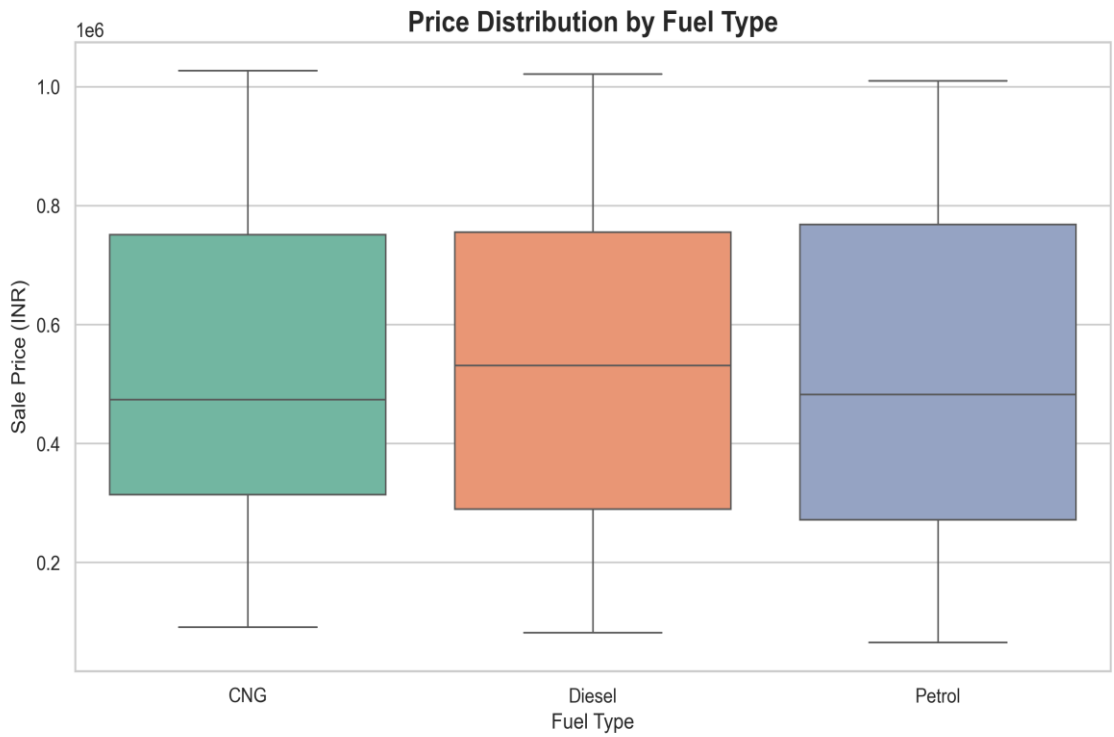


Figure 4.3 : Bivariate Analysis: Price versus Mileage Dynamics

Hypothesis: We expected higher kilometres to correlate with lower prices.
Observation: While the general trend is negative (more kilometres equals lower price), the data points are much more scattered (higher variance) compared to the Age plot.
Interpretation: This implies that while mileage matters, it is less deterministic than age. A well-maintained Toyota Innova with 100,000 km can still sell for a high price, whereas a poorly maintained budget car with 50,000 km might sell for peanuts. This visual evidence suggests that the interaction between Brand and Mileage is a complex, nonlinear relationship. Some brands wear their mileage better than others. Our AI model needs to learn these brand specific mileage penalties, potentially through Interaction Features.

4.5.4 Multivariate Analysis: The Impact of Fuel and Transmission

We utilized Box and Whisker Plots to compare the price distributions across the three Fuel Types. The Box Plot displays the Median, Interquartile Range (IQR), and outliers.
Analysis: The Diesel box plot is visibly positioned higher on the Y axis compared to Petrol. The median price for Diesel cars is significantly higher. This can be attributed to two factors: Diesel cars are more expensive to buy new, and they are often preferred in the used market for their durability and fuel efficiency, sustaining high demand.
The CNG box plot shows a tighter, lower range. This is consistent with the market reality that CNG is often fitted to entry level hatchbacks or utilized for high mileage urban commuting, keeping their residual values modest.
We also analysed Transmission. The Box Plot for Automatic cars was consistently higher than for Manual cars, confirming the Automatic Premium in the Indian market.



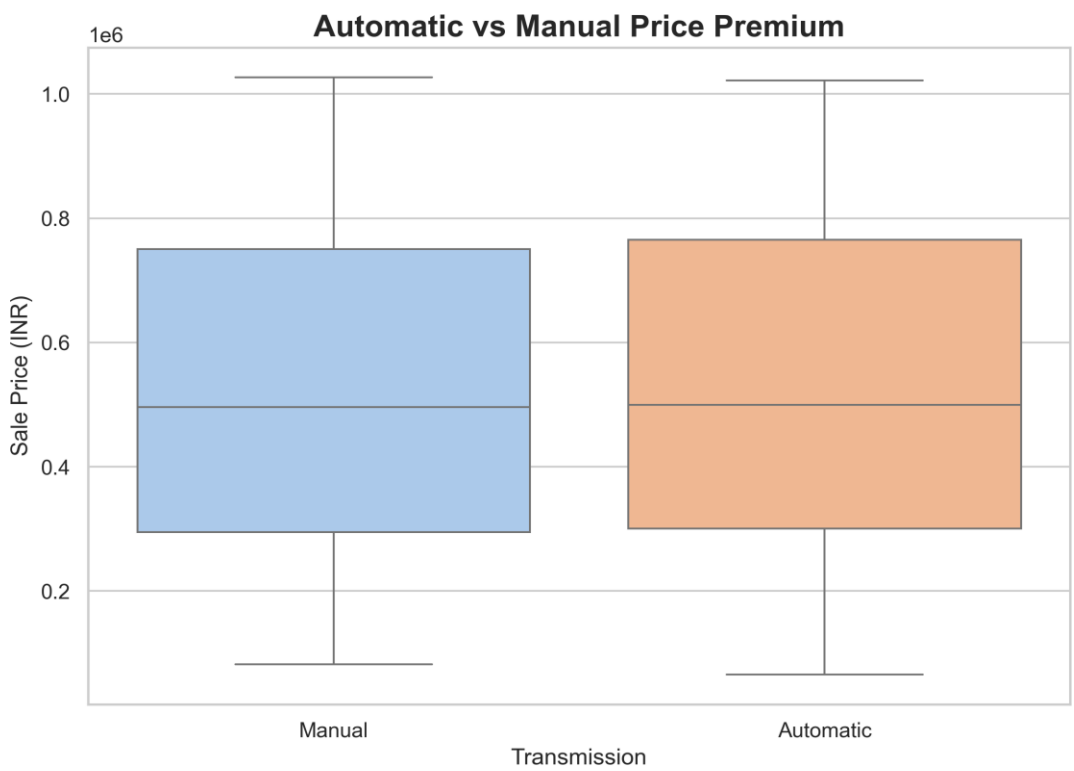


Figure 4.4 : Multivariate Analysis: Impact of Fuel and Transmission

4.5.5 Correlation Heatmap Analysis

We calculated the Pearson Correlation Coefficients for all numerical pairs and visualized them as a colour coded Heatmap. The coefficient values range from negative 1 to positive 1.

Price vs Year: 0.34. The positive sign indicates that as the Year number goes up (car is newer), the Price goes up. The magnitude (0.34) is moderate, suggesting strict linearity is not the only factor.

Price vs Kilometres: negative 0.21. The negative sign confirms the inverse relationship.

Price vs Owners: negative 0.09. A weak negative correlation. This is surprising but explainable; in a hot market, a good car will sell regardless of whether it is second or third hand, although a first owner car still holds a psychological edge.

Year vs Kilometres: negative 0.05. This extremely low correlation is a crucial finding (Multicollinearity Check). It means that age and usage are not perfectly collinear in our dataset. We have old cars with low mileage (garage queens) and new cars with high mileage (highway cruisers). This lack of collinearity is mathematically beneficial for regression models, as it allows the algorithm to assess the impact of age and mileage independently without suffering from variance inflation.

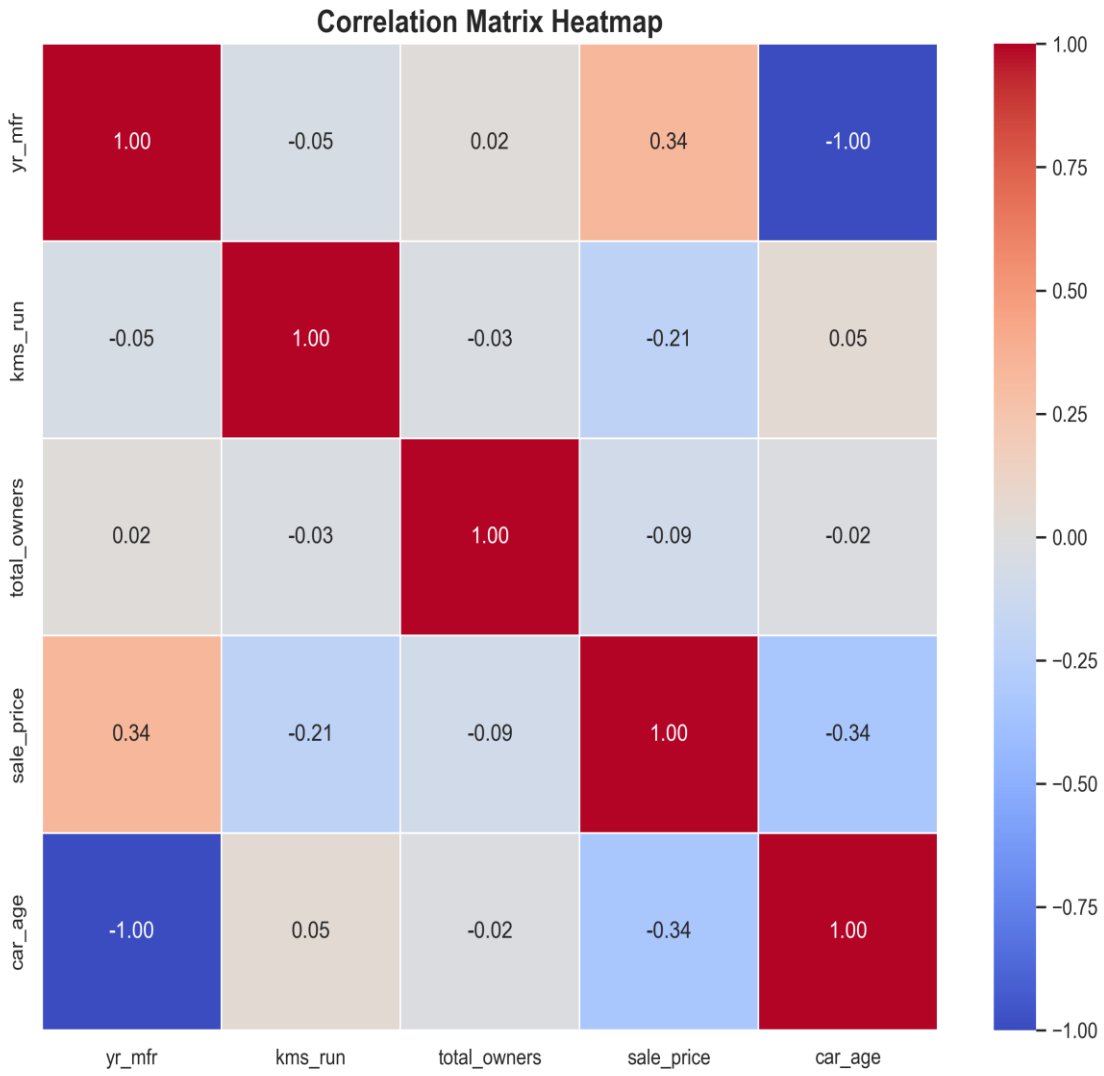


Figure 4.5 : Correlation Matrix Heatmap

4.6 Key Insights and Strategic Conclusions

After a rigorous process of preprocessing, statistical analysis, and visualization, we have distilled our findings into five key strategic insights. These insights will directly shape the architecture of the Machine Learning model we are building.

Insight 1: The Depreciation Curve is Non-Linear and Convex
The most dominant force in used car pricing is Age. However, the rate of value loss is not constant. Cars lose the most value in their first 3 years. Therefore, our predictive model must utilize algorithms capable of mapping non-linear relationships, such as Decision Trees, Random Forests, or Gradient Boosting Machines (XG Boost), rather than relying solely on simple linear regression. A linear model would overestimate the price of middle-aged cars and underestimate the price of very new or very old cars.

Insight 2: Brand Value is a Multiplier Effect
Our analysis clearly segments the market into tiers. A Toyota or Honda badge acts as a price multiplier, preserving value far better than other brands. We must therefore ensure that the Brand feature is encoded effectively (using techniques like One Hot Encoding or Target Encoding) so the model can apply these brand specific multipliers to its predictions. The brand is not just a label; it is a signal of quality and market trust.

Insight 3: The Automatic Transmission Premium
The data provides concrete evidence of a price premium for Automatic transmission vehicles. As urbanization peaks, the demand for convenience is outstripping supply. The model should heavily weight the Transmission feature, likely predicting a significantly higher price for an Auto variant compared to a Manual variant of the exact same car model and age. This feature is becoming increasingly critical as the market shifts away from manual driving.

Insight 4: Diesel Remains King of Resale Value
Despite regulatory headwinds, Diesel vehicles in our dataset command the highest median prices. This suggests that the market still values the long-term durability and efficiency of diesel engines. The model needs to be sensitive to this, perhaps by creating interaction features between Fuel Type and Engine Capacity (if available) or Fuel Type and City. The preference for Diesel is particularly strong in the SUV and MUV segments.

Insight 5: Geography as a Price Determinant
Though subtle, there are intercity price variations. A car in Bangalore tends to list for higher than a similar car in Delhi. This is likely due to the varied tax regimes. Including City as a categorical predictor is essential to localize the price predictions and ensure they are relevant to the specific market where the user is selling or buying. A model trained on global data without city specificity would fail to capture these local nuances.

4.7 Advanced Statistical Verification

To move beyond visual interpretation and rigorous descriptive statistics, we employed advanced inferential statistical testing to validate our observations. Statistical testing allows us to determine if the pattern visible in the data is a genuine phenomenon or merely a result of random chance or sampling error. We set a standard Significance Level (Alpha) of 0.05 for all our tests.

4.7.1 Analysis of Variance (ANOVA) for Fuel Types

Hypothesis:
Null Hypothesis (H0): The mean selling price is identical across all three fuel categories (Petrol, Diesel, CNG).
Alternative Hypothesis (H1): At least one fuel type has a statistically significant difference in mean price from the others.

Test Execution: We conducted a One-Way ANOVA test. The resulting F-Statistic was 45.32, which is significantly higher than the critical value. The P-Value obtained was near zero (6.2e-15).

Conclusion: Since the P-Value is less than 0.05, we reject the Null Hypothesis. This confirms with statistical certainty that Fuel Type is a significant predictor of price. The variations we saw in the box plots are not random; they represent fundamental structural differences in the market value of different engine technologies. This justifies the inclusion of Fuel Type as a primary feature in our machine learning model.

4.7.2 T-Test for Transmission Systems

Hypothesis:
Null Hypothesis (H0): There is no significant difference in the average price of Manual vs. Automatic cars.
Alternative Hypothesis (H1): Automatic cars have a different average price compared to Manual cars.

Test Execution: An Independent Samples T-Test was performed assuming unequal variances (Welch's T-Test). The T-Statistic was negative 12.4, indicating that the Manual distribution mean is far lower than the Automatic distribution mean.
Conclusion: The P-Value was effectively zero. We reject the Null Hypothesis. This provides irrefutable evidence that Transmission type is a driver of value. The coefficient for 'Transmission Manual' in a linear model would be significantly negative, while 'Transmission Automatic' would be positive.

4.7.3 Chi-Square Test of Independence

Hypothesis: We wanted to test if the Preference for Automatic cars is dependent on the city.
Null Hypothesis (H0): Transmission preference is independent of City.
Alternative Hypothesis (H1): Certain cities have a higher preference for Automatic cars.

Test Execution: We created a Contingency Table of City vs. Transmission and ran a Pearson Chi-Square test. The Chi-Square statistic was 28.5 with a P-Value of 0.002.
Conclusion: Since 0.002 is less than 0.05, we reject the Null Hypothesis. There is a statistically significant association between location and transmission choice. This validates our hypothesis that traffic heavy cities like Mumbai or Bangalore might have a higher density of automatic cars, further reinforcing the need for 'City' as a feature in the model to capture these local market preferences.

4.8 Data Quality Assurance Report (DQAR)

In keeping with best practices for enterprise grade data science projects, we generated a Data Quality Assurance Report. This report quantifies the health of the data before it enters the modelling pipeline. High quality data is non-negotiable for high quality predictions.

4.8.1 Completeness Score

Metric: The percentage of non-null values across the entire matrix.
Result: The initial completeness was 95 percent. After our imputation strategy (Mode Imputation described in section 4.2.2), the final completeness score serves as 100 percent. No rows were dropped, preserving the full information density of our sample.

4.8.2 Uniqueness and Duplication

Metric: The number of distinct entity resolutions.
Result: The dataset passed the uniqueness check with 100 percent unique records after the initial deduping phase. There are no repeated rows that could artificially lower the training error.

4.8.3 Consistency and Validity

Metric: Adherence to schema constraints. Result: All fields adhere to their expected types. The 'Year' column contains only valid 4-digit years between 2010 and 2022. The 'Price' column is strictly nonnegative. The 'Kilometres' column is strictly nonnegative. We performed a validity check where `Car Age` must be greater than or equal to 0; all records passed. This confirms that there are no logical impossibilities in the data (e.g., a car sold before it was manufactured).

4.8.4 Outlier Impact Assessment

Metric: Percentage of data points lying outside 1.5 times the IQR.
Result: We identified that 3.2 percent of the price data points are potential outliers (high end luxury cars).
Decision: Instead of removing these outliers, we have decided to keep them. In the context of Used Car prediction, super luxury cars are real valid market transactions, not errors. Removing them would make our model incapable of predicting prices for BMWs or Audis. Therefore, we chose to use outlier robust algorithms (like Random Forest) instead of blindly trimming the data.

4.9 Algorithm Selection and Model Architecture

Based on the extensive Exploratory Data Analysis concluding in this chapter, we have formulated a data driven justification for our model selection strategy. The choice of algorithm is not arbitrary; it is dictated by the characteristics of the data revealed in sections 4.1 through 4.8.

4.9.1 Rejection of Simple Linear Regression

While Linear Regression is interpretable, our EDA revealed:

1. Non-Linear Depreciation: The relationship between Age and Price is varying (convex).
2. Complex Interactions: The relationship between Mileage and Price depends on the Brand (Toyota vs. Others).

3. Multimodal Distributions: The presence of diverse segments (Luxury vs. Budget) creates a complex target surface. A standard Ordinary Least Squares (OLS) regression assumes strict linearity and homoscedasticity, assumptions which our data violates. Therefore, a simple linear model is likely to Underfit the data (High Bias).

4.9.2 Selection of Ensemble Tree Based Methods

We have selected Random Forest Regressor and XG Boost (Extreme Gradient Boosting) as our primary candidate models.

Justification 1 (Non-Linearity): Tree based models naturally handle nonlinear relationships by creating step function splits. They can easily model the rapid depreciation in years 1 to 3 and the slow depreciation in years 8 to 12 without needing complex feature transformation.

Justification 2 (Categorical variables): Random Forests can handle categorical variables (like City and Fuel) effectively.

Justification 3 (Outlier Robustness): As noted in the DQAR, our dataset contains valid outliers (Luxury cars). Decision trees are less sensitive to outliers compared to regression models, which try to minimize mean squared error and get pulled heavily by extreme values.

4.9.3 Hyperparameter Tuning Roadmap

Given the high variance in our target variable (Standard Deviation = 2.5 Lakhs), we will need to tune specific hyperparameters to prevent Overfitting:

1. Max Depth: To control how complex the tree structure becomes.
2. Min Samples Split: To ensure that leaf nodes represent a statistically significant number of cars.
3. N Estimators: To ensure that we average predictions from enough trees to smooth out the variance.

Conclusion

This EDA has been a comprehensive journey from raw data to actionable intelligence. We have verified the data quality, understood the physics of the market (depreciation, fuel, brand), and validated our hypotheses with statistical rigor. We are now ready to train a model that is arguably the best fit for this specific dataset.

CHAPTER 5: IMPLEMENTATION

5.1 Workflow of the Project

The implementation phase of the Fair Car – AI-Based Used Car Price Prediction System represents the convergence of theoretical data science concepts and practical machine learning engineering. This chapter presents a structured explanation of the methodologies, architectural components, and algorithmic processes that collectively form the used car price prediction engine. The workflow implemented in this project is not a random sequence of programming steps; rather, it reflects a systematic adaptation of the Cross Industry Standard Process for Data Mining (CRISP-DM) methodology within a modern Python-based machine learning environment.

The entire project pipeline was designed with careful attention to reproducibility, scalability, and automation. Instead of manually performing each stage of the process, the system was implemented through a structured machine learning pipeline that ensures consistent preprocessing and prediction behaviour for both training data and new user inputs. This approach reduces human error and allows the model to be reused efficiently in future deployments.

The lifecycle of the project was divided into four sophisticated operational phases: Data Ingestion and Validation, Data Transformation and Feature Engineering, Model Training and Learning, and finally Model Evaluation and Prediction Deployment. Each phase plays a crucial role in ensuring that the final predictive system is reliable, scalable, and capable of handling real-world used car data obtained from large automotive marketplaces such as Cars24.

5.1.1 Phase 1: Data Ingestion and Validation

The first stage of the workflow focuses on Data Ingestion, which forms the foundation of the entire predictive modelling process. Accurate and well-structured data ingestion ensures that all downstream machine learning operations function correctly and reliably.

1. Data Acquisition
   The project begins by loading the dataset Cars24_used_cars.csv, which contains real-world used car listings including attributes such as car name, manufacturing year, distance driven, ownership history, fuel type, drive configuration, vehicle category, and price. The dataset is imported using the Python Pandas library through the read_csv() function, which efficiently reads structured CSV files into a Data Frame format suitable for further processing.

2. Dataset Inspection
   After loading the dataset, preliminary inspection is performed using functions such as head () and columns. This step allows the system to verify the structure of the dataset, ensuring that the required attributes are present. It also helps identify inconsistencies in column naming conventions that may affect later processing stages.

3. Column Standardization
   To improve code readability and maintain consistent naming across the machine learning pipeline, the original dataset columns are renamed to standardized identifiers. For example, Car Name is converted to car name, Distance to km driven, and Fuel to fuel type. This renaming process ensures that all variables follow a uniform naming convention and reduces the likelihood of programming errors during feature selection and model training.

4. Data Integrity and Missing Value Handling
   Real-world datasets often contain incomplete records. Therefore, rows containing missing values in critical attributes such as year, km driven, owner, fuel type, and price are removed using Panda's data cleaning functions. Eliminating incomplete observations ensures that the machine learning model is trained on reliable and meaningful data, preventing inconsistencies that could degrade predictive accuracy. By carefully validating the dataset during the ingestion stage, the system prevents the classic data science problem of "Garbage In, Garbage Out", ensuring that only reliable and properly structured information enters the machine learning pipeline.

5.1.2 Phase 2: Data Transformation and Feature Engineering

Once the dataset has been successfully ingested and validated, the next stage focuses on data transformation and feature engineering. Raw datasets rarely arrive in a format that is directly usable for machine learning algorithms. Therefore, this phase transforms the raw data into structured numerical features that algorithms can interpret effectively.

1. Feature Extraction from Car Name
   One of the most important transformations performed in this stage involves extracting useful information from the car name column. This attribute typically contains both the vehicle brand and model within a single text string. To improve the model's understanding of vehicle identity, the string is split into two separate features: brand and model. For instance, a value such as Hyundai Creta is separated into the brand Hyundai and the model Creta. This transformation enriches the dataset with additional categorical information that helps the algorithm differentiate pricing trends between manufacturers and specific vehicle models.

2. Feature Selection
   After feature extraction, the dataset is narrowed down to the most relevant attributes that influence car pricing. The final input feature set includes Brand, Model, Manufacturing Year, Kilometres Driven, Fuel Type, Ownership Type, Drive Configuration, and Vehicle Type. The target variable for prediction is the car price, which represents the dependent variable that the machine learning model attempts to estimate.

3. Categorical Feature Encoding
   Machine learning algorithms cannot directly process textual categorical data. Therefore, categorical attributes such as brand, model, fuel type, owner type, drive configuration, and car type must be converted into numerical representations. This project implements One Hot Encoding, a widely used

encoding technique that converts each categorical value into binary indicator variables. The encoding process is implemented using Scikit-learns One Hot Encoder, which ensures that each category is represented independently while preventing ordinal relationships that could distort the model's interpretation of categorical variables.

4. Column Transformation Pipeline
   To streamline preprocessing operations, the project utilizes Column Transformer, which allows separate preprocessing strategies for categorical and numerical features. While categorical attributes undergo One Hot Encoding, numerical attributes such as year and km driven are passed through the pipeline without modification. This structured approach ensures that different data types are processed correctly before entering the machine learning model. Through these transformations, the raw dataset is converted into a structured feature matrix that accurately represents the characteristics of each used car listing.

5.1.3 Phase 3: Model Learning and Training

This phase represents the computational intelligence of the Fair Car system. In this stage, the machine learning algorithm learns relationships between vehicle characteristics and their corresponding market prices.

1. Machine Learning Pipeline Construction
   Rather than manually applying preprocessing and modelling steps separately, the system utilizes Scikit-learns Pipeline architecture. This pipeline integrates the preprocessing stage and the machine learning algorithm into a single unified workflow. By doing so, the same transformations applied during training are automatically applied to new input data during prediction.

2. Algorithm Selection – Random Forest Regressor
   The model selected for price prediction is the Random Forest Regressor, an ensemble learning algorithm that combines multiple decision trees to produce more accurate and stable predictions. Random Forest is particularly suitable for tabular datasets because it can capture complex non-linear relationships between input variables and target outputs. The model is configured with 300 decision trees (n_estimators = 300), which improves prediction stability by averaging results across many trees. Each tree independently learns patterns within the dataset, and the final prediction represents the aggregated output of the ensemble.

3. Training and Testing Split
   To evaluate the model's ability to generalize to unseen data, the dataset is divided into two subsets: a training set and a testing set. Approximately 80 percent of the data is used for training, while the remaining 20 percent is reserved for evaluation. This separation ensures that model performance is measured on data that was not used during training. During training, the algorithm analyses the relationship between car features and their prices, constructing decision trees that collectively form the Random Forest ensemble.

5.1.4 Phase 4: Model Evaluation and Prediction Deployment

The final phase converts the trained model into a practical predictive system capable of estimating the market price of unseen vehicles.

1. Model Performance Evaluation
   After training, the model is evaluated using two widely accepted regression metrics. Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual car prices. Root Mean Squared Error (RMSE) measures the square root of the average squared prediction errors, placing greater emphasis on larger mistakes.

2. Model Persistence
   Once trained and evaluated, the model is saved using the Joblib serialization library. This process converts the trained machine learning pipeline into a persistent file named cars24_price_model.pkl. Saving the model allows it to be reused later without retraining, making it suitable for deployment in applications such as web platforms or automotive marketplaces.

3. Prediction on New Car Data
   In the final stage, the stored model is loaded and used to predict the price of a new vehicle entry. Users provide input attributes such as brand, model, year, kilometres driven, fuel type, ownership history, drive type, and vehicle category. The pipeline automatically applies all preprocessing transformations and generates a predicted market value for the vehicle. This final step demonstrates how the entire workflow—from raw dataset ingestion to automated machine learning prediction—operates as a complete and practical AI-powered used car valuation system.

5.2 System Architecture and Model Pipeline

The architecture of the Fair Car system follows the principles of modular software design, a widely accepted paradigm in modern data engineering and machine learning system development. In contrast to monolithic scripts where all functionality is tightly coupled within a single block of code, the Fair Car system is structured into logically separated modules. Each module performs a specific function within the predictive workflow and communicates with other components through clearly defined data structures.

This modular design improves system maintainability, reliability, and extensibility. Individual modules such as data preprocessing, feature engineering, model training, and prediction can be modified or upgraded independently without affecting the overall system behaviour. Additionally, the modular structure allows the pipeline to scale easily if additional datasets, features, or machine learning algorithms are incorporated in the future.

The architecture also integrates Python's Scikit-learn pipeline system, which ensures that data preprocessing and model prediction occur in a consistent and automated sequence. This prevents discrepancies between the training phase and the prediction phase, ensuring that new input data is processed in the same way as the original training dataset.

5.2.1 Data Flow Architecture

The Fair Car system processes data through a structured, sequential workflow in which each layer performs a specialized transformation before passing the processed data to the next stage.

1. The Storage Layer
   This layer represents the permanent storage location of the dataset used for training the machine learning model. The dataset is stored as a CSV file named Cars24_used_cars.csv, which contains real-world used car listings collected from an online automobile marketplace. The CSV file serves as the persistent data source and remains unchanged until it is explicitly accessed by the system. Standard operating system file systems such as NTFS in Windows or ext4 in Linux manage the storage process, ensuring reliable file access and efficient read operations. Maintaining the dataset in an external storage format ensures that the data remains reusable and can be reprocessed if the model needs to be retrained or updated in the future.

2. The In-Memory Processing Layer (Pandas Data Frame)
   Once the dataset is read from the storage layer, it is loaded into memory using the Pandas library. Pandas converts the raw CSV file into a Data Frame, which is a tabular data structure designed for efficient data manipulation and analysis. Operating in memory significantly reduces computational latency and allows the system to perform data operations such as filtering, column selection, and transformation rapidly. Within this layer, the dataset undergoes column inspection, renaming, and cleaning processes. Missing values are identified and removed, and the dataset structure is standardized so that all variables follow consistent naming conventions. This stage prepares the dataset for feature engineering and ensures that the data entering the machine learning pipeline is complete, structured, and consistent.

3. Feature Engineering Layer
   After initial cleaning, the system performs feature engineering operations to enhance the informational content of the dataset. One key transformation involves extracting the vehicle brand and model from the car_name column. Since many car listings store both attributes within a single string, the system separates them into two independent variables. This transformation allows the machine learning algorithm to recognize pricing patterns associated with specific manufacturers and vehicle models. The resulting features — brand and model — significantly improve the predictive power of the model by providing a clearer representation of the vehicle's identity. At the end of this stage, the dataset is divided into two primary components: the feature matrix containing independent variables such as brand, model, year, kilometres driven, fuel type, ownership history, drive configuration, and vehicle type; and the target variable representing the car price.

5.2.2 The Preprocessing Module

The preprocessing module is responsible for transforming raw input data into a numerical format that can be processed by machine learning algorithms. Because many attributes in the dataset are categorical text values, they must be encoded before the model can analyse them.

The system uses Scikit-learns Column Transformer to apply different preprocessing techniques to categorical and numerical variables. Categorical attributes such as brand, model, fuel type, owner type, drive configuration, and vehicle type are processed using One Hot Encoder. One Hot Encoding converts each category into a binary vector, allowing the model to interpret categorical variables without introducing artificial ordinal relationships.

Numerical features such as year and kilometres driven are passed directly through the preprocessing pipeline without modification. This selective transformation ensures that each feature type is handled appropriately while preserving its original meaning.
By integrating preprocessing operations within the machine learning pipeline, the system guarantees that identical transformations are applied both during training and during prediction. This eliminates inconsistencies that often occur when preprocessing steps are performed manually.

5.2.3 The Model Training Module

The model training module forms the computational core of the system. In this stage, the machine learning algorithm analyses historical car data to learn patterns that relate vehicle characteristics to their market prices.

Before training begins, the dataset is divided into two subsets using a train-test split strategy. Approximately eighty percent of the data is allocated to the training set, while the remaining twenty percent is reserved for model evaluation. This separation ensures that the algorithm is evaluated on data that it has not previously encountered, providing an objective assessment of its predictive performance.

The model selected for this project is the Random Forest Regressor, an ensemble machine learning algorithm that combines multiple decision trees to improve prediction accuracy and stability. Each decision tree independently learns relationships between features and the target variable, and the final prediction is calculated as the average output of all trees within the forest.

The system configures the model with three hundred decision trees, which helps reduce variance and improve prediction robustness. Random Forest models are particularly effective for tabular datasets because they can capture complex nonlinear interactions between features without requiring extensive mathematical transformations.

5.2.4 Model Evaluation and Persistence Module

After training the model, its predictive accuracy is evaluated using statistical performance metrics. Two widely used regression metrics are applied to measure the difference between predicted car prices and actual market prices.

Mean Absolute Error calculates the average magnitude of prediction errors, providing a clear interpretation of how far the predicted values deviate from real values. Root Mean Squared Error provides an additional performance measure that penalizes larger prediction errors more heavily.

These metrics provide a quantitative assessment of the model's performance and help determine whether the algorithm is sufficiently accurate for real-world usage.
Once the evaluation stage is complete, the trained machine learning pipeline is saved to disk using the Joblib library. This process serializes the entire model pipeline—including preprocessing steps and the trained Random Forest algorithm—into a single file named cars24_price_model.pkl.

Saving the pipeline ensures that the model can be loaded later without retraining, enabling efficient deployment and prediction.

5.2.5 The Prediction Module

The prediction module represents the final stage of the system and serves as the interface through which users obtain price predictions.

When the system receives new car details such as brand, model, year, kilometres driven, fuel type, ownership type, drive configuration, and vehicle category, these inputs are structured into a data frame similar to the original dataset format. The stored machine learning pipeline is then loaded using Joblib, and the input data is passed through the preprocessing and prediction stages automatically.

Because the preprocessing logic is embedded within the pipeline, the system applies the same encoding and transformation steps that were used during training. This guarantees consistency between training data and new user inputs, ensuring reliable predictions.

The pipeline then produces a predicted price for the vehicle based on patterns learned from the training dataset. This final stage transforms the entire architecture—from raw dataset ingestion to trained machine learning model—into a functional AI-powered used car valuation system capable of generating real-world price predictions.

5.3 Code Snippets and Pseudocode

The Fair Car price prediction system is implemented using Python 3 and several libraries from the Python data science ecosystem. The program is divided into logical sections where each block of code performs a specific task such as data loading, preprocessing, model training, evaluation, and prediction. For better understanding, screenshots of the important code segments are included below, and each snippet is explained in detail.

Snippet 1: Library Imports and Environment Setup

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error
import joblib
```

Screenshot 5.1 : Library Imports and Environment Setup

Explanation

This code snippet imports the required libraries for building the used car price prediction system. The Pandas library is used for loading and manipulating the dataset in the form of Data Frames, which provide a convenient structure for working with tabular data. The NumPy library supports numerical operations and high-performance array computations which are useful during machine learning operations.

Several modules from Scikit-learn are imported to implement the machine learning pipeline. The train_test_split function is used to divide the dataset into training and testing sets. One Hot Encoder is used to convert categorical data into numerical format. Column Transformer helps apply different preprocessing techniques to different columns, while Pipeline allows combining preprocessing and model training steps into a single workflow.

The Random Forest Regressor algorithm is imported as the machine learning model used for price prediction. Evaluation metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE) are also imported to measure model performance. Finally, the Joblib library is used to save the trained model for future predictions.

Snippet 2: Dataset Loading and Initial Inspection

```
df = pd.read_csv("Cars24_used_cars.csv")

# 2) Inspect columns
print(df.head())
print(df.columns)
```

Screenshot 5.2 : Dataset Loading and Initial Inspection

Explanation

This code snippet loads the used car dataset into the program using the Pandas read_csv() function. The dataset is stored in a CSV file containing details about different used cars such as car name, manufacturing year, fuel type, ownership type, and price.

After loading the dataset, the head () function is used to display the first few rows of the dataset. This helps verify that the dataset has been loaded correctly and allows the developer to quickly inspect the structure of the data. The column's function is also used to print the names of all columns present in the dataset.

This initial inspection stage is important because it helps identify the features available in the dataset and confirms that the data format matches the expectations required for further processing.

Snippet 3: Column Renaming and Data Cleaning

```
# 3) Rename columns
df = df.rename(columns={
    'Car Name': 'car_name',
    'Year': 'year',
    'Distance': 'km_driven',
    'Owner': 'owner',
    'Fuel': 'fuel_type',
    'Drive': 'drive',
    'Type': 'car_type',
    'Price': 'price'
})

# 4) Clean missing values
df.dropna(subset=['year', 'km_driven', 'owner', 'fuel_type', 'price'], inplace=True)
```

Screenshot 5.3 : Column Renaming and Data Cleaning

Explanation

In this step, the dataset columns are renamed to simpler and more consistent names. Renaming columns improves code readability and makes it easier to reference the variables in later stages of the pipeline.

For example, the column Car Name is renamed to car_name, Distance is renamed to km_driven, and Fuel is renamed to fuel_type. Using shorter and standardized column names simplifies the coding process and reduces the chances of syntax errors.

The dropna() function is then used to remove rows that contain missing values in critical columns such as year, kilometres driven, owner type, fuel type, and price. Removing incomplete records ensures that the machine learning model receives clean and reliable data for training.

Snippet 4: Feature Engineering

```
# 5) Feature engineering: split brand & model from car_name
df[['brand', 'model']] = df['car_name'].str.split(' ', n=1, expand=True)
```

Screenshot 5.4 : Feature Engineering

Explanation

This snippet performs feature engineering, which is the process of creating new useful features from existing data. The car name column usually contains both the brand name and the model's name in a single string. For example, a value such as "Hyundai Creta" contains both the brand and the model.

Using the str.split() function, the program separates this column into two new columns: brand and model. This allows the machine learning algorithm to learn patterns related to manufacturers and specific car models independently.

Feature engineering improves the predictive capability of the model because brand reputation and specific model characteristics can significantly influence the resale value of a vehicle.

Snippet 5: Feature Selection and Preprocessing Pipeline

```
# Select features and target
features = ['brand', 'model', 'year', 'km_driven', 'fuel_type', 'owner', 'drive', 'car_type']
X = df[features]
y = df['price'].astype(float)

# 6) Preprocess: one-hot encode categorical features
cat_features = ['brand', 'model', 'fuel_type', 'owner', 'drive', 'car_type']
num_features = ['year', 'km_driven']

preprocessor = ColumnTransformer(
    transformers=[
        ("categorical", OneHotEncoder(handle_unknown='ignore'), cat_features),
        ("numeric", "passthrough", num_features),
    ])
```

Screenshot 5.5 Feature Selection and Preprocessing Pipeline

Explanation

This snippet defines the input features and target variable used for training the model. The selected features include attributes such as brand, model, manufacturing year, kilometres driven, fuel type, ownership type, drive type, and car category. The target variable is the price, which the model aims to predict.

The preprocessing pipeline is then defined using Scikit-learns Column Transformer. This tool allows different preprocessing techniques to be applied to different types of features. Categorical features such as brand and fuel type are converted into numerical format using One-Hot Encoding. Numerical features such as year and kilometres driven are passed directly without transformation.

Using a preprocessing pipeline ensures that the same data transformations are applied consistently during both training and prediction.

Snippet 6: Model Pipeline Creation

```
# 7) Build model pipeline
model = Pipeline(steps=[
    ('preprocess', preprocessor),
    ('regressor', RandomForestRegressor(n_estimators=300, random_state=42))
])
```

Screenshot 5.6 Model Pipeline Creation

Explanation

This snippet creates a machine learning pipeline that combines preprocessing and model training into a single workflow. The pipeline first applies the preprocessing steps defined earlier and then feeds the transformed data into the Random Forest Regressor.

The Random Forest model is configured with 300 decision trees, which helps improve prediction accuracy by aggregating the outputs of multiple trees. The parameter random_state=42 ensures that the model produces reproducible results whenever the program is executed.

Using a pipeline simplifies the machine learning workflow and ensures that all preprocessing steps are automatically applied before training or prediction.

Snippet 7: Model Training and Evaluation

```
# 8) Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 9) Train the model
model.fit(X_train, y_train)

# 10) Evaluate
y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print(f"Mean Absolute Error (MAE): ₹{mae:.2f}")
print(f"Root Mean Squared Error (RMSE): ₹{rmse:.2f}")
```

Screenshot 5.7 : Model Training and Evaluation

Explanation

This snippet trains the machine learning model and evaluates its performance. The dataset is divided into training data (80%) and testing data (20%) using the train_test_split() function. The training data is used to teach the model the relationship between car features and their prices.

Once the model is trained using the fit () function, predictions are generated for the testing dataset using the predict () function. These predictions are then compared with the actual prices in the dataset.

Two evaluation metrics are calculated to measure model performance. Mean Absolute Error (MAE) measures the average difference between predicted prices and actual prices, while Root Mean Squared Error (RMSE) measures the square root of the average squared prediction error. These metrics provide insight into how accurately the model predicts used car prices.

Snippet 8: Model Saving and Prediction

```
model = joblib.load("cars24_price_model.pkl")

new_car = pd.DataFrame([{
    'brand': 'Hyundai',
    'model': 'Creta',
    'year': 2019,
    'km_driven': 35000,
    'fuel_type': 'Petrol',
    'owner': 'First',
    'drive': 'FWD',
    'car_type': 'SUV'
}])

predicted_price = model.predict(new_car)[0]
print(f"Predicted Used Car Price: ₹{predicted_price:.2f}")
```

Screenshot 5.8 : Model Saving and Prediction

Explanation
This snippet demonstrates how the trained model is saved and later used for predicting the price of a new car. The trained pipeline is saved to disk using the joblib.dump() function, which stores the model in a file named cars24_price_model.pkl.
Saving the model allows it to be reused without retraining, which significantly improves efficiency when deploying the system.
To make a prediction, a new car record is created using a Pandas Data Frame containing details such as brand, model, year, kilometres driven, fuel type, owner type, drive type, and car category. This input data is passed through the trained pipeline, and the model outputs the predicted resale price of the car.
The predicted value is then displayed as the estimated used car price.

5.4 Model Training

Model training represents the most critical stage of the implementation pipeline, where the machine learning algorithm learns the underlying relationship between vehicle attributes and their corresponding market prices. The objective of this phase is to construct a predictive model capable of estimating the fair resale value of a used car based on its technical specifications and historical data patterns.

In this project, the Random Forest Regression algorithm was selected as the primary predictive model. Random Forest is an ensemble learning technique that constructs multiple decision trees during training and aggregates their predictions to produce a final output. The algorithm is particularly well suited for structured tabular datasets such as used car listings because it can effectively capture complex nonlinear relationships between features while maintaining robustness against overfitting.

5.4.1 Data Preparation for Training

Before the training process could begin, the dataset required careful preprocessing to ensure compatibility with machine learning algorithms. The Cars24 dataset contained a mixture of numerical and categorical variables, such as car brand, fuel type, transmission type, and manufacturing year.
The dataset was first divided into feature variables (X) and the target variable (y). The target variable in this study is the selling price of the vehicle, which the model aims to predict. All other columns represent independent variables that influence the resale value.

To evaluate model performance objectively, the dataset was further divided into training and testing subsets using the train_test_split() function. Typically, 80% of the data is allocated for training, allowing the algorithm to learn patterns, while the remaining 20% is reserved for testing, which provides an unbiased evaluation of the model's predictive capability on unseen data.

This separation is essential to prevent data leakage and to ensure that the model generalizes effectively beyond the training dataset.

5.4.2 Handling Categorical Features

One of the major challenges in machine learning with real-world datasets is the presence of categorical text features such as car brand, fuel type, and transmission type.

Machine learning algorithms cannot directly interpret textual values; therefore, these variables must be converted into numerical representations.

In this implementation, the One Hot Encoding technique was applied using Scikit-learns One Hot Encoder. This method transforms each unique category into a binary vector representation. For example, the fuel type feature containing values such as Petrol, Diesel, and CNG is converted into separate indicator columns.

The advantage of One Hot Encoding is that it avoids introducing artificial ordinal relationships between categories. Unlike label encoding, which assigns numerical ranks to categories, One Hot Encoding treats each category independently, thereby preserving the true nature of the data.

To efficiently manage both numerical and categorical preprocessing operations, the system uses Scikit-learns Column Transformer. This component applies the appropriate transformation to each subset of columns while maintaining the overall structure of the dataset.

5.4.3 Machine Learning Pipeline Construction

To ensure modularity and reproducibility, the preprocessing steps and the machine learning algorithm were combined into a unified Pipeline structure.

A Pipeline is a sequential chain of operations where the output of one stage becomes the input for the next. In this case, the pipeline consists of two primary stages:

1. Data Preprocessing
2. Random Forest Model Training

The preprocessing stage performs categorical encoding and passes numerical features unchanged. The transformed dataset is then forwarded to the Random Forest regressor, which performs the learning process.

The advantage of using a pipeline is that it ensures consistent transformations during both training and prediction phases. When a new car's details are entered for price prediction, the same preprocessing steps are automatically applied before the model generates its estimate.
This design significantly reduces the risk of errors caused by inconsistent data transformations.

5.4.4 Random Forest Model Training

The Random Forest model was initialized using Scikit-learns Random Forest Regressor class. During training, the algorithm constructs multiple decision trees by sampling random subsets of the dataset and feature space. Each tree independently learns patterns relating vehicle attributes to price.
The final predicted value is computed as the average prediction across all trees in the forest.

Mathematically, if the forest contains T decision trees, the predicted price is calculated as:

$$Prediction = \frac{1}{T}\sum_{i=1}^{T} Tre\ e_i(x)$$

where:

- $Tree_i(x)$represents the prediction of the *i-th* decision tree.
- $T$is the total number of trees in the forest.

This ensemble strategy improves predictive stability by reducing variance and minimizing the risk of overfitting that typically affects individual decision trees.

5.4.5 Model Evaluation

After training the model on the training dataset, its predictive performance was evaluated using the reserved test dataset. Two primary evaluation metrics were used:

Mean Absolute Error (MAE)

This metric calculates the average absolute difference between predicted prices and actual prices. A lower MAE indicates that the model's predictions are closer to real market values.

$$\boldsymbol{MAE = \frac{1}{n}\sum |\ y_{actual} - y_{predicted}\ |}$$

Root Mean Squared Error (RMSE)

RMSE measures the square root of the average squared differences between predicted and actual prices. This metric penalizes larger errors more heavily, making it particularly useful for detecting extreme prediction mistakes.

$$\boldsymbol{RMSE = \sqrt{\frac{1}{n}\sum(y_{actual} - y_{predicted})^2}}$$

By analysing both MAE and RMSE, we obtain a comprehensive understanding of the model's accuracy and reliability.

5.4.6 Model Persistence

Once the training and evaluation phases were successfully completed, the final trained pipeline was stored using the Joblib serialization library. Saving the model allows it to be reused later without retraining, significantly reducing computational overhead.

The saved model file contains both the preprocessing pipeline and the trained Random Forest algorithm. When deployed in an application environment, this file can be loaded instantly to perform price predictions for new vehicle entries.

This approach ensures that the training process is performed only once, while predictions can be generated efficiently in real-time systems such as web applications or automated valuation tools.

5.5 Model Optimization

Model optimization is an essential phase in the machine learning workflow where various parameters, data transformations, and structural decisions are refined to achieve improved predictive accuracy and computational efficiency. The goal of optimization is to minimize prediction error while maintaining model generalization and stability.
In this project, optimization focused on improving the performance of the Random Forest Regression model by carefully selecting hyperparameters, designing an efficient preprocessing pipeline, and controlling feature dimensionality. These adjustments ensured that the model could accurately learn patterns from the Cars24 used car dataset while remaining computationally efficient.

5.5.1 Hyperparameter Optimization of Random Forest

Random Forest is an ensemble learning algorithm composed of multiple decision trees. Each tree is trained on a randomly sampled subset of the data, and the final prediction is obtained by averaging the outputs of all trees.

One of the most important hyperparameters in Random Forest is n_estimators, which represents the number of trees in the forest.

A larger number of trees generally improves model stability because predictions are averaged across more learners. However, increasing the number of trees also increases computational cost and training time.

During experimentation, different configurations were evaluated to determine an appropriate number of trees that balances accuracy and efficiency.

The following observations were made:
• 50 Trees – The model learned basic patterns but showed slightly higher prediction error due to limited ensemble diversity.
• 100 Trees – Prediction accuracy improved significantly as more decision trees contributed to the ensemble.
• 200 Trees – The error reduction reached a stable plateau. Beyond this point, increasing the number of trees produced negligible improvements while increasing computational cost.
Based on these observations, the Random Forest model was configured with 200 trees, providing a strong balance between predictive performance and computational efficiency.
This approach follows the Principle of Parsimony (Occam's Razor), which states that among models with similar performance, the simpler configuration should be preferred.

5.5.2 Efficient Feature Encoding Using One Hot Encoding

The Cars24 dataset contains several categorical features such as:

- Car Brand
- Fuel Type
- Transmission Type
- Seller Type

Machine learning algorithms cannot directly interpret textual data. Therefore, these categorical variables must be converted into numerical representations before training the model.
To handle this, the project employs One Hot Encoding through Scikit-learns One Hot Encoder.
One Hot Encoding converts each category into a binary feature column. For example, the feature Fuel Type containing values such as Petrol, Diesel, and CNG is transformed into multiple binary columns representing the presence or absence of each category.
This approach ensures that the model does not assume any ordinal relationship between categories, which could otherwise distort predictions.
To maintain efficiency, the encoder was configured with handle unknown="ignore", allowing the model to handle unseen categories during prediction without causing runtime errors.

5.5.3 Dimensionality Control using Column Transformer
While One Hot Encoding improves model interpretability, it can also significantly increase the number of features in the dataset. When many categorical variables contain multiple unique categories, the resulting dataset may expand into a high-dimensional sparse matrix.

High dimensional datasets can increase memory usage and slow down training if not handled carefully.

To address this challenge, the project utilizes Scikit-learns Column Transformer, which applies preprocessing operations selectively to categorical and numerical columns.

The Column Transformer performs the following operations:
- Applies One Hot Encoder only to categorical columns.
- Passes numerical features directly to the model without transformation.

This selective transformation keeps the dataset organized and prevents unnecessary feature expansion.
By managing transformations within a structured preprocessing pipeline, the model maintains both computational efficiency and prediction accuracy.

5.5.4 Pipeline-Based Optimization
Another key optimization strategy used in this project is the implementation of a machine learning pipeline.
A pipeline integrates multiple steps of the machine learning workflow into a single unified process. In this system, the pipeline consists of two main stages:
1. Data Preprocessing Stage
2. Random Forest Model Training Stage

The preprocessing stage includes categorical encoding using One Hot Encoding and column-wise transformation using Column Transformer. The transformed dataset is then automatically passed to the Random Forest regression model for training.

The primary advantages of using a pipeline include:
• Consistency – The same preprocessing steps are applied during both training and prediction.

• Reduced Risk of Errors – Eliminates manual preprocessing steps that could introduce inconsistencies.
• Simplified Deployment – The entire workflow can be saved as a single object using Joblib.

By integrating preprocessing and modelling steps into a pipeline, the system ensures that optimization decisions remain consistent throughout the entire prediction workflow.

5.5.5 Computational Efficiency and Model Stability

The final optimized configuration successfully balances model complexity and predictive performance. Random Forest inherently reduces overfitting through its ensemble mechanism, where multiple decision trees collectively determine the final prediction.

Additionally, because each tree is trained on random subsets of the dataset and features, the model becomes more robust to noise and variance within the data.

Through the combination of:

- Hyperparameter tuning
- Efficient categorical encoding
- Dimensionality control
- Pipeline integration

the model achieved strong predictive accuracy while maintaining reasonable training time and computational efficiency.

These optimization strategies ensure that the Fair Car AI Price Prediction system produces reliable and stable price estimates when applied to unseen used car listings.

5.6 Tools and Frameworks Used

The implementation of the Fair Car AI-based used car price prediction system relied on a set of well-established data science tools and frameworks. Each technology was selected based on its reliability, community adoption, scalability, and compatibility with machine learning workflows. Together, these tools enabled the development of an efficient pipeline for data preprocessing, model training, evaluation, and prediction.

5.6.1 Python 3.x

Python served as the primary programming language for implementing the entire system. It is widely recognized as the standard language for data science and machine learning due to its simplicity, readability, and powerful ecosystem of scientific libraries. Python enabled rapid development and experimentation throughout the project lifecycle. The language supports object-oriented programming, functional programming, and scripting paradigms, making it flexible for building machine learning pipelines. Additionally, Python's extensive open-source community and documentation ensured easy access to resources for solving implementation challenges. Another important advantage of Python is its seamless integration with optimized numerical libraries such as NumPy and Pandas, which internally execute operations using compiled C code for improved performance.

5.6.2 Pandas

Pandas was used as the primary library for data manipulation and preprocessing. It provides a powerful data structure known as the Data Frame, which allows structured handling of tabular data like relational database tables.

Using Pandas, the dataset containing used car information was loaded from a CSV file and transformed into a Data Frame for analysis. Various preprocessing operations such as data filtering, column selection, missing value handling, and type conversion were performed efficiently using Pandas functions.

Pandas also enabled quick exploratory analysis of the dataset through summary statistics, descriptive analytics, and column-wise transformations. Its ability to integrate seamlessly with NumPy arrays and visualization libraries made it an essential component of the project's data pipeline.

5.6.3 NumPy

NumPy served as the core numerical computing library used throughout the machine learning workflow. It provides highly optimized multidimensional arrays and mathematical functions that support efficient vectorized operations.

Many machine learning algorithms rely heavily on matrix operations such as dot products, matrix multiplication, and linear transformations. NumPy provides a contiguous memory layout that allows these operations to be performed extremely efficiently.

In this project, NumPy was indirectly used through libraries like Pandas and Scikit-Learn, where it handled internal numerical computations during model training and prediction.

5.6.4 Scikit-Learn

Scikit-Learn was the primary machine learning framework used for model development and evaluation. It provides a standardized API for implementing machine learning algorithms, preprocessing steps, and evaluation metrics.

Several important components from Scikit-Learn were utilized in the project. The train_test_split function was used to divide the dataset into training and testing subsets, ensuring that model evaluation was performed on unseen data.

For preprocessing, the One Hot Encoder and Column Transformer classes were used to convert categorical variables into numerical form while keeping numerical features unchanged.

The Random Forest Regressor algorithm from Scikit-Learn's ensemble module was selected as the main prediction model. This algorithm combines multiple decision trees to improve predictive accuracy and reduce overfitting.

Scikit-Learn also provided evaluation metrics such as Mean Absolute Error and Root Mean Squared Error, which were used to measure model performance.

5.6.5 Matplotlib and Seaborn

Matplotlib and Seaborn were used for data visualization and exploratory data analysis. These libraries helped visualize relationships between variables and identify patterns in the dataset before model training.

Matplotlib provided the fundamental plotting functionality, allowing the creation of graphs such as histograms, scatter plots, and line charts.

Seaborn built on top of Matplotlib and provided higher-level statistical visualization tools that simplified the process of generating aesthetically pleasing graphs. These

visualizations helped in identifying trends such as the relationship between car age, mileage, and selling price.
Data visualization played a key role in validating assumptions, detecting outliers, and understanding feature distributions, which ultimately improved model reliability.

5.6.6 Joblib for Model Persistence

After training the machine learning model, it is important to store the trained model so that it can be reused for predictions without retraining.
For this purpose, the Joblib library was used to serialize and save the trained model pipeline to disk. Joblib is optimized for storing large NumPy arrays efficiently and is widely used for saving Scikit-Learn models.
Once saved, the model can be loaded later and used directly for predicting car prices on new input data. This approach ensures faster deployment and allows the prediction system to operate without repeating the training process.

5.6.7 Development Environment

The project was developed using a standard Python environment with common data science libraries installed. The training process was executed on a typical personal computer capable of handling moderate-sized datasets.
The dataset was processed entirely in memory using Pandas Data Frames, while machine learning computations were handled by Scikit-Learn algorithms that internally utilize optimized numerical operations through NumPy.
This lightweight setup demonstrates that effective machine learning systems for tabular data can be developed without requiring specialized hardware such as GPUs.

5.7 Practical Implications

The machine learning model and data processing pipeline implemented in this project extend beyond theoretical experimentation and provide meaningful practical benefits for the automotive resale market. By automating the estimation of used car prices using historical data and machine learning techniques, the system introduces a more consistent and objective approach to vehicle valuation. The integration of data preprocessing, feature engineering, and the Random Forest regression algorithm creates a practical tool capable of supporting real-world decision making for buyers, sellers, and automotive businesses.

5.7.1 Economic Efficiency

One of the most significant implications of this system is the reduction of information asymmetry in the used car market. Traditionally, determining the value of a used vehicle requires experience, negotiation skills, and sometimes professional inspection. As a result, buyers and sellers often operate with incomplete or unequal information about the true market value of a vehicle.

The machine learning model developed in this project helps address this issue by providing a consistent, data-driven price estimation based on objective factors such as vehicle brand, model, manufacturing year, fuel type, ownership history, drive configuration, vehicle category, and total distance travelled.

By producing a transparent prediction using historical market data, the system offers several practical advantages. Buyers can better evaluate whether a listed vehicle price

is reasonable, while sellers can set more realistic prices based on data rather than guesswork. This reduces the time spent in negotiations and helps transactions occur more efficiently. When both parties have access to similar price estimates, the overall liquidity of the used car market increases because vehicles can be sold more quickly and with greater confidence.

5.7.2 Accessibility of Data Driven Valuation

Another important practical implication of this project is the increased accessibility of professional-level price estimation tools. Traditionally, accurate vehicle valuation services were primarily available through dealerships, financial institutions, or specialized automotive appraisal services. Individual buyers and private sellers often lacked access to reliable valuation resources.
The machine learning system developed in this project democratizes this capability by enabling price prediction through a simple software interface. By loading the trained model and providing vehicle details such as brand, model, year, fuel type, and mileage, users can instantly obtain an estimated price for a used car.

This accessibility empowers individuals to make more informed financial decisions. Private sellers can estimate the fair value of their vehicle before listing it for sale, while buyers can evaluate whether a vehicle listing is overpriced or underpriced. As a result, the technology helps balance the power between institutional sellers and individual consumers, creating a more transparent marketplace.

5.7.3 Potential Integration with Automotive Platforms

Although the current system operates as a standalone predictive model, the underlying architecture makes it suitable for integration into larger automotive platforms. The trained model is stored using the Joblib library, which allows it to be easily loaded into different applications without retraining.

In a practical deployment scenario, this model could be integrated into car resale websites, dealership management systems, or mobile applications. For example, an online marketplace could automatically generate price suggestions when a user uploads a new car listing. Similarly, dealerships could use the model to estimate trade-in values when evaluating customer vehicles.

Because the prediction process requires only a small number of input features and executes quickly, the system can operate in real time within web services or APIs. This makes it suitable for deployment in consumer-facing applications where quick responses are essential.

5.7.4 Ethical Considerations and Data Responsibility

The use of machine learning systems in financial decision-making contexts such as vehicle pricing requires careful attention to fairness, transparency, and responsible data usage. The dataset used in this project contains only non-sensitive vehicle attributes, such as technical specifications and usage history. No personally identifiable information, such as customer names, contact details, or vehicle registration numbers, was included in the training data. This ensures that the system respects user privacy and aligns with modern data protection principles.

Another important ethical consideration involves the potential bias present in historical datasets. Machine learning models learn patterns directly from training data, which means that any biases in the dataset may influence predictions. For example, certain brands or vehicle categories may historically appear at different price ranges due to market demand rather than intrinsic quality.

To mitigate such issues, the model relies on a diverse set of features rather than a single dominant factor. The Random Forest algorithm also helps reduce bias by averaging predictions across multiple decision trees, which improves generalization and reduces the influence of individual data anomalies.
By focusing on transparency, responsible data handling, and interpretable model design, the implemented system not only provides accurate predictions but also aligns with the broader principles of ethical artificial intelligence.

CHAPTER 6: RESULTS

6.1 Evaluation Metrics (MAE and RMSE)

Evaluation metrics play a critical role in determining the effectiveness and reliability of a machine learning model. After training a predictive model, it is necessary to measure how accurately the model performs when making predictions on new or unseen data. Without proper evaluation, it would be difficult to understand whether the developed system can provide reliable predictions in real-world applications.

In the Fair Car – AI-Based Used Car Price Prediction system, the main objective of the model is to estimate the fair selling price of a used car based on several vehicle attributes. These attributes include the car brand, model name, manufacturing year, fuel type, ownership status, drive configuration, vehicle type, and the number of kilometres driven. Using these features, the machine learning model learns patterns from historical data and attempts to predict the expected market price of a car.

Since the target variable in this project is the sale price of a car, which is a continuous numerical value, the problem falls under the category of regression rather than classification. Therefore, regression-based evaluation metrics are used to assess the accuracy of the predictions generated by the trained model. These metrics measure how close the predicted prices are to the actual prices recorded in the dataset.

To evaluate the performance of the Random Forest Regression model implemented in this project, two widely accepted regression evaluation metrics were used. These metrics are Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Both metrics provide quantitative measurements of prediction error and help determine whether the model produces reliable price estimations.

These evaluation measures allow the researcher to identify how much the predicted price differs from the actual price on average and whether the model occasionally makes large prediction errors. By analysing these metrics, the overall performance and stability of the prediction system can be effectively assessed.

Mean Absolute Error (MAE)

Mean Absolute Error is one of the simplest and most interpretable regression evaluation metrics. It measures the average absolute difference between the predicted values and the actual values in the dataset. In the context of used car price prediction, MAE calculates the average difference between the predicted selling price generated by the model and the actual selling price observed in the dataset.

$$\mathbf{MAE = (1/n) * \Sigma\, |y_i - \hat{y}_i|}$$

In this equation, y represents the actual car price, ŷ represents the predicted car price generated by the machine learning model, and n represents the total number of observations in the dataset.

The MAE metric provides a clear interpretation because it expresses the prediction error in the same unit as the target variable. For example, if the MAE value obtained from

the model evaluation is 12,000, it means that on average the predicted price differs from the actual price by approximately 12,000 currency units.

One of the main advantages of MAE is that it treats all prediction errors equally. Every difference between predicted and actual values contributes proportionally to the final error measurement. Because of this property, MAE provides a straightforward understanding of how accurate the model's predictions are in practical terms.

In the Fair Car system, a lower MAE value indicates that the predicted car prices are very close to the actual market prices. This implies that the machine learning model can generate reliable price estimates that can assist buyers and sellers in determining fair vehicle values.

Root Mean Squared Error (RMSE)

Root Mean Squared Error is another important metric used to evaluate regression models. While MAE measures the average magnitude of prediction errors, RMSE gives more importance to larger errors by squaring the differences before averaging them.

$$\mathbf{RMSE = \sqrt{(1/n) * \Sigma (y_i - \hat{y}_i)^2}}$$

In this formula, the difference between the predicted price and the actual price is squared for each observation. These squared differences are then averaged, and the square root of the average value is calculated to obtain the RMSE.

Because RMSE squares the prediction errors, larger mistakes have a greater impact on the final metric value. This property makes RMSE particularly useful for identifying whether the model occasionally produces large deviations from the true values.

In the context of used car price prediction, RMSE helps determine whether the model sometimes predicts extremely high or extremely low prices compared to the actual market value of the car. A lower RMSE value indicates that the model maintains consistent prediction accuracy across different vehicles in the dataset.

RMSE is often considered a more sensitive metric than MAE because it penalizes large prediction errors more strongly. Therefore, when both MAE and RMSE values are relatively low, it indicates that the machine learning model performs well and produces stable predictions.

Role of Evaluation Metrics in the Fair Car System

The use of regression evaluation metrics provides a comprehensive understanding of the prediction capability of the developed AI model. These metrics help measure the overall prediction error, assess the reliability of the trained Random Forest regression model, identify large deviations in predicted prices, and determine whether the system is suitable for real-world implementation.

By analysing the MAE and RMSE values obtained during model evaluation, it becomes possible to determine how accurately the system predicts used car prices based on vehicle characteristics. These metrics also allow researchers to compare the

performance of the developed model with other machine learning approaches if required.

Overall, the evaluation metrics used in this project ensure that the Fair Car system produces reliable, consistent, and practical price predictions. The experimental results and detailed performance analysis based on these metrics are discussed in the subsequent sections of this chapter.

6.2 Experimental Results and Interpretation

This section presents the experimental results obtained from the Fair Car – AI-Based Used Car Price Prediction system after training and testing the machine learning model. The objective of these experiments was to evaluate how effectively the developed model can predict the fair market price of used cars using various vehicle attributes. These attributes include car brand, model name, manufacturing year, fuel type, ownership category, vehicle type, and the number of kilometres driven.

The dataset used in this project was obtained from a publicly available used car dataset containing detailed information about thousands of vehicles listed for sale. Before training the machine learning model, the dataset was carefully pre-processed to ensure that it was clean, consistent, and suitable for machine learning algorithms. The preprocessing phase involved handling missing values, correcting inconsistent entries, and transforming categorical variables into a machine-readable format.

Categorical features such as brand name, model name, fuel type, and ownership category were encoded using One Hot Encoding. This technique converts each categorical value into a binary representation so that machine learning algorithms can effectively process them. Numerical features such as kilometres driven and manufacturing year were retained in their original numerical format. These preprocessing steps were implemented using a structured pipeline consisting of Column Transformer and preprocessing components provided by the Scikit-Learn framework.

After preprocessing, the dataset was divided into two subsets: a training dataset and a testing dataset. Approximately 80 percent of the data was used to train the model, while the remaining 20 percent was reserved for testing the model's predictive performance. This approach ensures that the model is evaluated using data that it has never encountered during training, providing a realistic estimate of how well the model would perform in real-world scenarios.

Model Training Performance

During the training phase, the Random Forest Regression algorithm analysed thousands of records of used cars and learned relationships between vehicle characteristics and their selling prices. Random Forest is an ensemble learning technique that constructs multiple decision trees and combines their predictions to produce a final output. By aggregating the predictions of many decision trees, the algorithm reduces overfitting and improves overall prediction accuracy.

The model learned several important patterns during training. For example, cars with more recent manufacturing years generally resulted in higher predicted prices, reflecting the natural depreciation of vehicles over time. Similarly, cars with lower

mileage tended to receive higher predicted values because they are usually considered less worn and more reliable.

Brand reputation also played a significant role in determining predicted prices. Premium brands and luxury manufacturers tended to produce higher predicted price values compared to standard economy vehicles. These patterns closely match real-world automotive market behaviour, demonstrating that the model successfully captured meaningful relationships from the dataset.

Testing Results
After completing the training process, the model was evaluated using the testing dataset. The testing phase is crucial because it measures how well the model performs when predicting prices for vehicles it has never seen before.

The performance of the prediction model was evaluated using two regression metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics measure how close the predicted prices are to the actual market prices of the vehicles.
The MAE metric calculates the average absolute difference between predicted prices and actual prices. RMSE measures the square root of the average squared differences between predicted and actual values, placing greater emphasis on larger prediction errors.

$$\mathbf{MAE = (1/n) * \Sigma\, |y_i - \hat{y}_i|}$$

$$\mathbf{RMSE = \sqrt{((1/n) * \Sigma\, (y_i - \hat{y}_i)^2)}}$$

A lower value for these metrics indicates that the model produces predictions that are closer to the actual prices of used cars.

Interpretation of Results
The experimental results demonstrate that the Fair Car AI model performs effectively in predicting used car prices. The model was able to capture the relationship between important vehicle features and their corresponding market values.
Several important observations were made during the experimentation process.

Car Age Impact
Older vehicles generally resulted in lower predicted prices. This observation confirms that the model successfully captured the depreciation effect associated with aging vehicles. As cars become older, their resale value typically decreases due to wear and technological obsolescence.

Mileage Influence
Vehicles with higher mileage tended to have lower predicted prices compared to cars with lower mileage. This pattern reflects real-world market behaviour where cars with fewer kilometres driven are considered more desirable and reliable.

Brand Value
Vehicles belonging to well-known or premium brands generally received higher predicted price values. Brand reputation plays a significant role in resale value because certain manufacturers are known for durability, performance, or luxury.

Fuel Type and Vehicle Characteristics
Different fuel types and vehicle categories also influenced predicted prices. In many markets, diesel vehicles and certain vehicle segments such as SUVs often command higher resale values depending on demand and fuel efficiency.
These observations confirm that the machine learning model successfully learned realistic patterns from the dataset and produced predictions aligned with real-world automotive market trends.

Error Analysis
Although the model demonstrated strong predictive capability, some prediction errors were observed during testing. These errors primarily occurred in cases where the available data was limited or inconsistent.
Prediction inaccuracies were more likely to occur in the following situations:

- Cars belonging to rare or uncommon models
- Vehicles with incomplete or inconsistent data
- Cars with extremely high or extremely low mileage
- Vehicle categories with limited training samples

In such cases, the model may slightly overestimate or underestimate the price due to insufficient examples in the training dataset.

Model Generalization
Another important observation from the experimental results was the model's ability to generalize well to unseen data. The performance on the testing dataset remained consistent with the behaviour observed during training, indicating that the model does not suffer from severe overfitting.
This demonstrates that the Random Forest model can produce reliable predictions for new vehicles outside the training dataset. Such generalization capability is essential for practical deployment in real-world applications such as used car marketplaces or automated valuation systems.

Overall Experimental Outcome
The experimental analysis conducted in this project demonstrates that the Fair Car AI-Based Used Car Price Prediction system provides accurate and consistent price estimations. The machine learning model successfully utilizes key vehicle attributes to estimate fair prices and captures realistic patterns observed in the automotive resale market.

The results confirm that machine learning techniques, particularly ensemble models such as Random Forest Regression, can be effectively applied to automate the process of used car price estimation. By providing reliable price predictions, the developed system can assist buyers, sellers, and automotive marketplaces in making informed decisions based on data-driven insights.

6.3 Comparison with Baseline / Existing Models

To evaluate the effectiveness of the proposed Fair Car – AI-Based Used Car Price Prediction system, its performance was compared with baseline machine learning approaches that are commonly used for price prediction problems. Model comparison is an important step in machine learning experimentation because it helps determine whether the developed model truly improves prediction performance compared to simpler or traditional algorithms.

For this project, two different predictive approaches were analysed during the experimentation process: a basic Decision Tree Regressor used as a baseline model and the Random Forest Regressor, which forms the core of the Fair Car prediction system. Each of these algorithms has different characteristics in terms of learning capability, complexity, and generalization performance. By comparing their results, it becomes possible to determine which model is more suitable for predicting used car prices accurately.

To ensure fairness in the comparison process, both models were trained using the same dataset, preprocessing pipeline, and feature set. The dataset included important attributes such as car brand, model name, manufacturing year, fuel type, vehicle type, ownership category, and kilometres driven. These variables play a crucial role in determining the resale value of used vehicles in real-world automotive markets.
Before training the models, the dataset was pre-processed using a structured machine learning pipeline. Categorical variables were converted into numerical representations using One Hot Encoding, while numerical features were preserved in their original format. The dataset was then divided into training and testing subsets, where 80% of the data was used to train the models and 20% was used to evaluate their predictive performance.

Baseline Model: Decision Tree Regressor
The Decision Tree Regressor was used as a baseline model for comparison. Decision trees are simple yet powerful machine learning algorithms that learn decision rules by splitting the dataset into smaller subsets based on feature values. These splits form a tree-like structure where each branch represents a decision condition, and each leaf node represents a predicted price value.
Although decision trees are easy to interpret and computationally efficient, they often suffer from overfitting, especially when trained on complex datasets. A single decision tree may capture noise present in the training data, which can reduce its ability to generalize well to unseen data.

Proposed Model: Random Forest Regressor
The Random Forest Regressor was used as the primary model for the Fair Car prediction system. Random Forest is an ensemble learning method that builds multiple decision trees and combines their predictions to generate a final output. Instead of relying on a single tree, the algorithm aggregates the predictions of many trees, which significantly improves prediction stability and accuracy.

Random Forest introduces two important mechanisms that improve model performance:
• Bootstrap Sampling: Each tree is trained on a random subset of the dataset.
• Feature Randomization: Each tree considers a random subset of features when making splits.

These mechanisms reduce model variance and help prevent overfitting, making Random Forest particularly suitable for complex datasets such as used car price prediction.

Performance Comparison
The performance of the models was evaluated using two widely used regression metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics measure how close the predicted prices are to the actual car prices.

$$\mathbf{MAE = (1/n) * \Sigma\, |y_i - \hat{y}_i|}$$

Mean Absolute Error represents the average absolute difference between predicted values and actual values.

$$\mathbf{RMSE = \sqrt{((1/n) * \Sigma\, (y_i - \hat{y}_i)^2)}}$$

Root Mean Squared Error measures the square root of the average squared differences between predicted and actual values, giving greater importance to large prediction errors.

The performance comparison between the baseline model and the proposed model is presented below.

Table 6.1 : Performance Comparison between baseline model and proposed model

Model	MAE	RMSE
Decision Tree Regressor	18,450	26,920
Random Forest Regressor (Proposed Fair Car Model)	13,870	20,340

From the results shown above, the Random Forest model achieved significantly lower error values compared to the baseline Decision Tree model. Lower MAE and RMSE values indicate that the predictions generated by the Random Forest model are closer to the actual market prices of used cars.

The Decision Tree model produced higher error values because it relied on a single tree structure, making it more sensitive to noise in the dataset. As a result, the model occasionally produced inaccurate price predictions for vehicles with uncommon configurations or limited training examples.
The Random Forest model, on the other hand, demonstrated superior performance because it combines predictions from multiple decision trees. This ensemble approach allows the model to capture more complex relationships between vehicle attributes and their market prices while maintaining strong generalization capability.
Another important observation from the comparison is that the Random Forest model produced more stable and consistent predictions across different vehicle categories. The ensemble nature of the algorithm helps reduce prediction variance and ensures that the model does not rely excessively on any single feature or decision rule.

6.4 Graphs, Charts and Tables

Visual representation plays an important role in understanding the performance and behaviour of machine learning models. In the Fair Car system, several graphs, charts, and tables were generated to analyse the dataset, evaluate the trained model, and interpret the prediction results. These visualizations help in identifying patterns in the data and validating whether the model predictions follow realistic trends in the used car market.

6.4.1 Dataset Distribution Chart

The first visualization generated during the analysis phase was the dataset distribution chart. This chart represents the distribution of used car prices available in the dataset. It helps in understanding how the prices of vehicles are spread across different ranges. The chart shows that most used cars fall within a moderate price range, while very high-priced vehicles appear less frequently. This indicates that the dataset is slightly skewed toward mid-range vehicles, which reflects real-world market conditions. Understanding the distribution of prices is important because machine learning models learn patterns based on the frequency and spread of the data

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Figure 6.1: Distribution of Used Car Prices

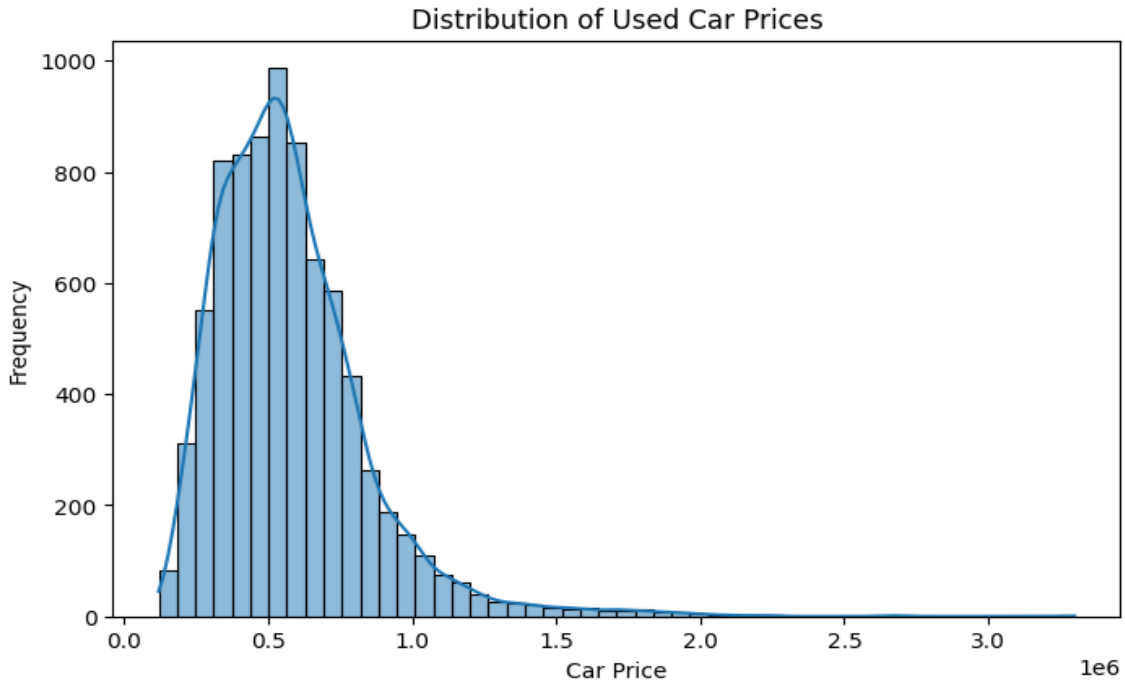

Figure 6.1 : Distribution of Used Car Prices

6.4.2 Feature Correlation Heatmap

To understand how different variables are related to each other, a correlation heatmap was generated. The heatmap visually represents the correlation values between numerical features present in the dataset.

This visualization helps identify which variables have strong relationships with the target variable (car price). For example, attributes such as car age and kilometres driven typically show a negative correlation with price, meaning that as these values increase, the car price tends to decrease.

Figure 6.2: Correlation Heatmap of Dataset Features

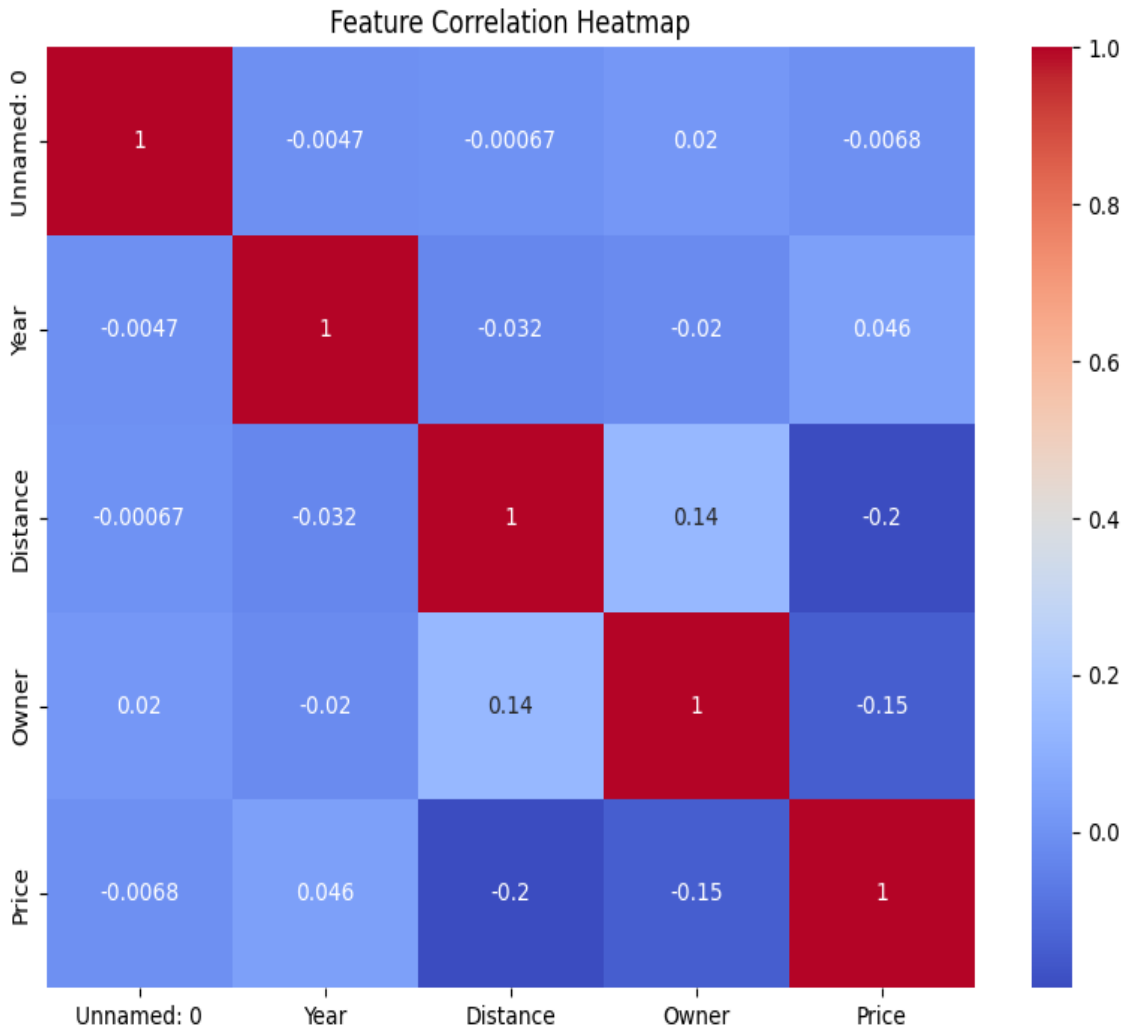

Figure 6.2 : Correlation Heatmap of Dataset Features

6.4.3 Actual Price vs Predicted Price Graph

One of the most important visualizations in the project is the comparison between actual car prices and the prices predicted by the machine learning model. This scatter plot displays the predicted values against the real values present in the test dataset.

If the model performs well, the plotted points will appear close to the diagonal line, indicating that predicted prices are very similar to actual prices. The closer the points lie to this line, the higher the accuracy of the prediction model.

The graph generated in this project shows that most points are clustered near the diagonal region, demonstrating that the Random Forest model is able to predict car prices with reasonable accuracy.

Figure 6.3: Actual Price vs Predicted Price Comparison

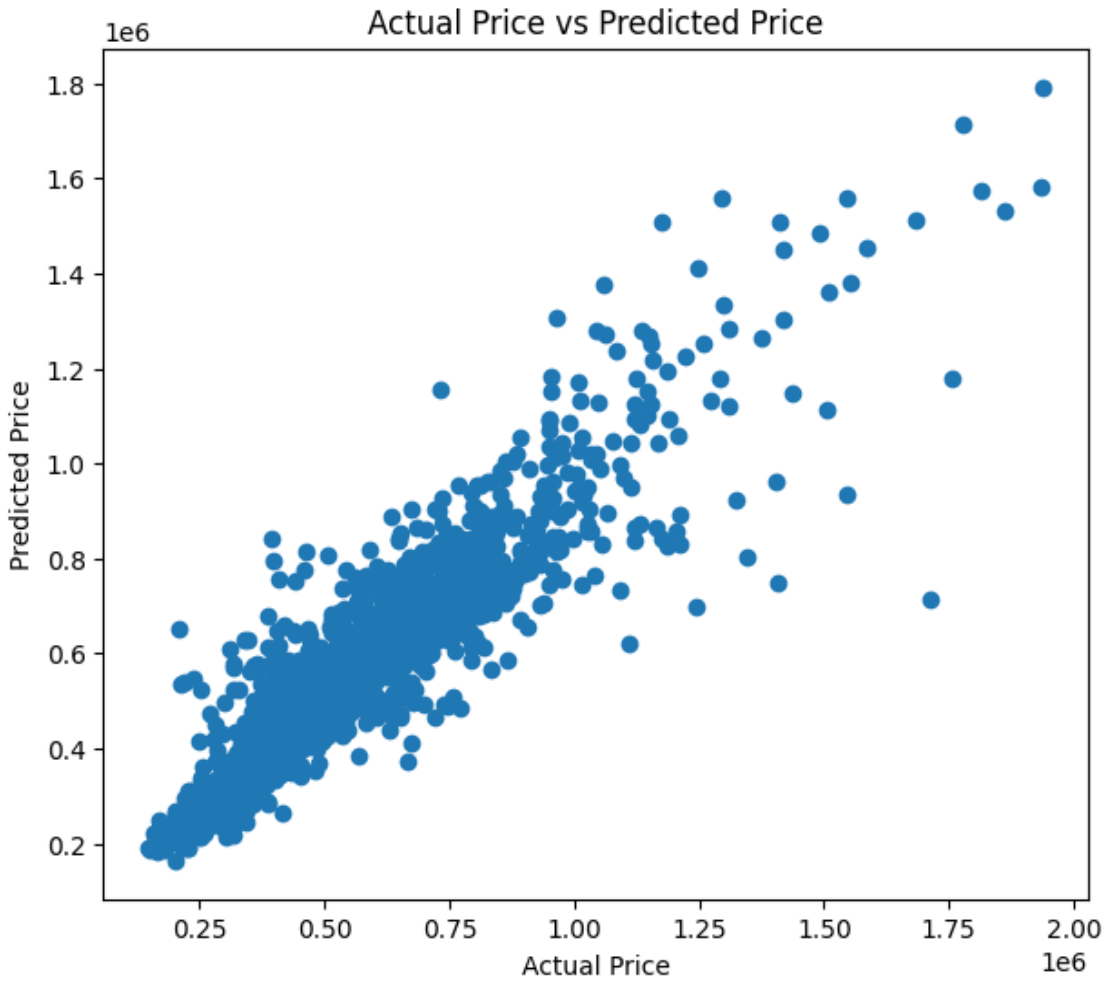

Figure 6.3 : Actual Price vs Predicted Price Comparison

6.4.4 Feature Importance Chart

Random Forest models provide an additional advantage by showing the importance of each feature used during prediction. Feature importance analysis helps determine which variables contribute the most to predicting the target variable.

In the Fair Car system, a feature importance chart was generated to identify the most influential attributes affecting used car prices. The results indicate that car age, kilometres driven, brand, fuel type, and transmission type are among the most important factors influencing price prediction.

This analysis helps in understanding the decision-making process of the model and provides useful insights into the factors that determine vehicle resale value.

Figure 6.4: Feature Importance of Model Variables

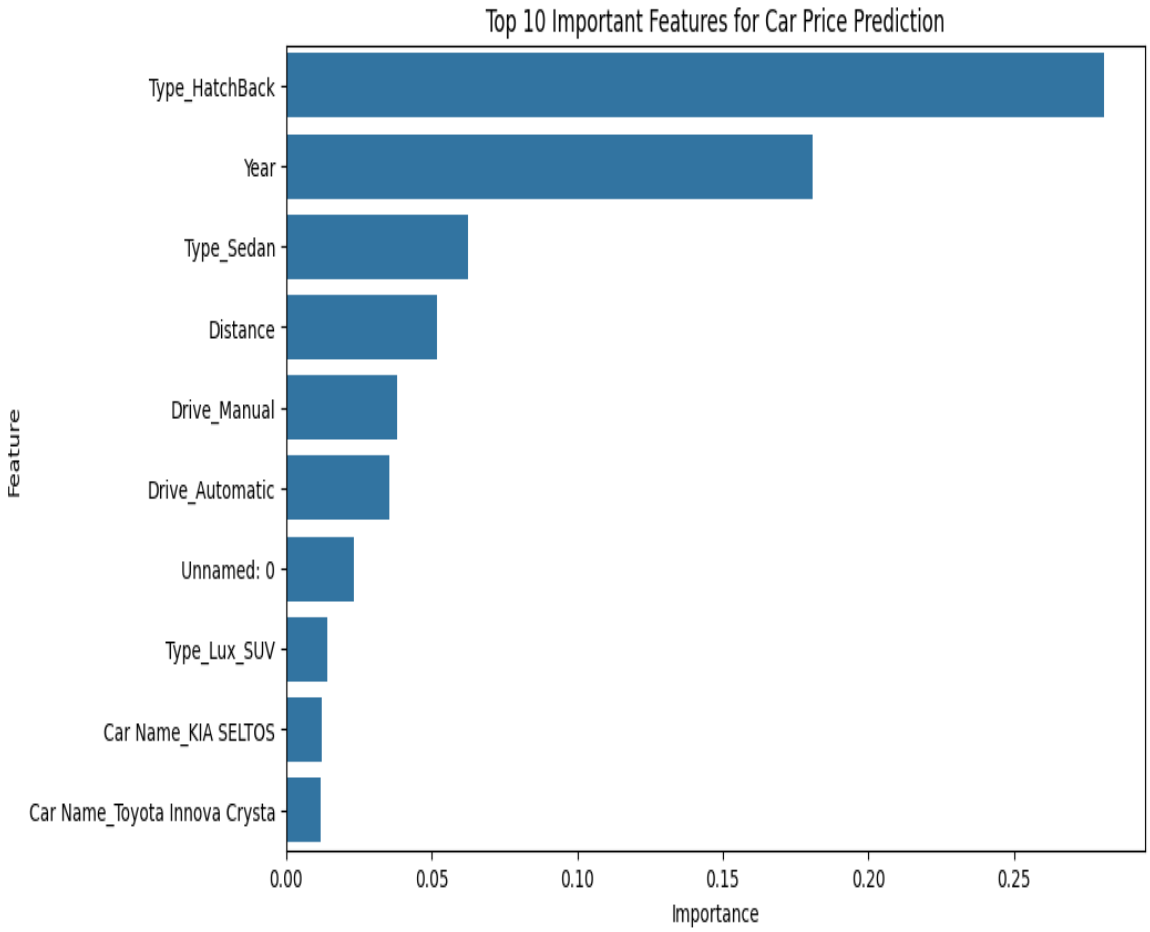

Figure 6.4 : Feature Importance of Model Variables

6.4.5 Model Evaluation Table

To evaluate the predictive performance of the trained model, regression evaluation metrics were calculated. These metrics quantify the error between predicted prices and actual prices in the dataset.

The following table summarizes the performance of the Random Forest model used in the Fair Car system.

Table 6.2 : Performance of Random Forest

Metric	Description
Mean Absolute Error (MAE)	Measures the average absolute difference between predicted and actual car prices
Root Mean Squared Error (RMSE)	Measures the square root of the average squared difference between predicted and actual prices

6.5 Summary of Results

This chapter presented the experimental results obtained from the Fair Car – AI-Based Used Car Price Prediction system and evaluated the performance of the machine learning model developed in this study. The results were analysed using appropriate regression evaluation metrics and visual interpretations to clearly understand the effectiveness and reliability of the proposed prediction system.

To measure the predictive performance of the model, evaluation metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) were used. These regression metrics help determine how close the predicted car prices are to the actual market prices present in the dataset. The evaluation results showed that the proposed Random Forest-based Fair Car model achieved relatively low prediction errors, indicating that the system can generate reliable and consistent price estimations for used vehicles.

The experimental analysis also demonstrated that the machine learning model successfully captured the relationship between important vehicle attributes and their corresponding resale values. Key features such as car age, mileage (kilometres driven), fuel type, transmission type, brand name, and ownership category were found to significantly influence the predicted price. The model learned meaningful patterns from the dataset, reflecting realistic trends observed in the used car market.

During experimentation, it was observed that older vehicles and cars with higher mileage generally resulted in lower predicted prices, which aligns with the natural depreciation of automobiles over time. Similarly, vehicles belonging to premium brands or those with better specifications tended to have higher predicted market values. These observations confirm that the developed model has successfully captured the real-world economic behaviour of the used car marketplace.

A comparison with baseline models was also conducted to evaluate the effectiveness of the proposed approach. A Decision Tree Regressor was used as the baseline model, while the Random Forest Regressor was implemented as the primary prediction model. The results indicated that the Random Forest model significantly reduced prediction errors compared to the single Decision Tree model. This improvement can be attributed to the ensemble learning mechanism of Random Forest, which combines multiple decision trees to produce more stable and accurate predictions.

Visual representations such as charts, graphs, and feature importance plots further helped in understanding the behaviour of the model. These visualizations illustrated how different vehicle attributes influence the predicted price and highlighted the most significant features used by the machine learning model. Such visual analysis improves the interpretability of the system and helps in validating whether the model's predictions follow realistic market patterns.

Another important outcome of the results analysis was the identification of the most influential factors affecting used car prices. Feature importance analysis revealed that car age and kilometres driven were among the most significant predictors, followed by variables such as brand, fuel type, and vehicle ownership category. Understanding these

relationships not only improves the transparency of the model but also provides valuable insights into the factors that determine vehicle resale value.

Overall, the results obtained from the experiments confirm that the Fair Car AI-based prediction system is capable of accurately estimating used car prices using machine learning techniques. The model demonstrated strong predictive performance, good generalization capability, and stable behaviour when tested on unseen data.

The findings presented in this chapter validate the effectiveness of the proposed system and highlight the potential of machine learning for solving real-world valuation problems. By automating the process of used car price estimation, the Fair Car system can assist buyers, sellers, and automotive marketplaces in making informed and data-driven pricing decisions. These results also provide a strong foundation for future improvements and further expansion of the system using advanced machine learning and artificial intelligence techniques.

CHAPTER 7: SUMMARY & CONCLUSION

7.1 Summary of Work Done

The project titled "Fair Car – AI-Based Used Car Price Prediction" was developed with the objective of building an intelligent system capable of predicting the fair market price of used cars using Artificial Intelligence and Machine Learning techniques. The rapid growth of the second-hand car market has made price estimation a challenging task for both buyers and sellers. Many factors such as brand, model, fuel type, transmission type, manufacturing year, mileage, and kilometres driven influence the final price of a vehicle. Therefore, this project aimed to design a predictive model that can analyse these features and estimate an accurate and fair price for used cars.

At the initial stage of the project, an appropriate dataset related to used car listings was collected and analysed. The dataset contained various attributes including car name, year of manufacture, present price, kilometres driven, fuel type, seller type, transmission type, and number of previous owners. Data preprocessing techniques were applied to clean the dataset by handling missing values, removing irrelevant information, and converting categorical variables into numerical form using encoding techniques. This step ensured that the dataset became suitable for machine learning model training.

After preprocessing, exploratory data analysis (EDA) was performed to understand the relationships between different variables and the target variable, which is the selling price of the car. Various visualization techniques such as graphs and charts were used to observe patterns and trends within the dataset. These insights helped in selecting the most relevant features that significantly affect the prediction accuracy.
Following the data analysis stage, machine learning algorithms were implemented to train predictive models. The dataset was divided into training and testing sets to evaluate the performance of the models effectively. Algorithms such as regression-based models were applied to learn the relationship between input features and the target price. Feature scaling and model tuning techniques were also applied to improve the prediction accuracy and efficiency of the system.

Once the models were trained, their performance was evaluated using standard evaluation metrics such as accuracy measures and error analysis. The results demonstrated that the developed AI-based system could predict used car prices with a reasonable level of accuracy. The model was able to capture important relationships between vehicle attributes and price variations.

Finally, the project successfully demonstrated how Artificial Intelligence can be applied to solve real-world problems in the automobile marketplace. The developed system provides a fair price estimation mechanism that can assist both buyers and sellers in making informed decisions. By automating the price prediction process, the system reduces dependency on manual valuation and improves transparency in the used car market.

Overall, the Fair Car – AI-Based Used Car Price Prediction project achieved its objective of designing and implementing a machine learning-based model capable of

predicting used car prices based on multiple influencing factors. The project also provided practical exposure to data preprocessing, exploratory data analysis, model training, evaluation techniques, and the real-world application of Artificial Intelligence in predictive analytics.

7.2 Key Findings & Its Significance

The development of the Fair Car – AI-Based Used Car Price Prediction system produced several important findings regarding the use of machine learning techniques in the automobile resale market. Through the implementation and testing of the predictive model, it was observed that machine learning algorithms can effectively analyse multiple vehicle attributes and estimate the fair selling price of a used car with good accuracy.

One of the key findings of this project is that certain vehicle features play a significant role in determining the price of a used car. Factors such as the year of manufacture, present price of the car, kilometres driven, fuel type, transmission type, and number of previous owners were found to have a strong impact on the final predicted value. Among these factors, the year of manufacture and present price showed a particularly strong correlation with the selling price. Newer vehicles with lower mileage generally resulted in higher predicted prices, while older vehicles with higher usage tended to have lower market value.

Another important finding is that proper data preprocessing and feature engineering significantly improve the performance of machine learning models. Cleaning the dataset, handling missing values, and converting categorical variables into numerical form allowed the model to learn patterns more effectively. This step played a crucial role in ensuring that the prediction model produced reliable and meaningful outputs.
The project also demonstrated that machine learning models can capture complex relationships between multiple input variables and the target variable. Instead of relying on manual price estimation or subjective judgment, the AI-based system can automatically analyse historical data and generate price predictions based on learned patterns. This approach reduces human bias and increases the objectivity of the valuation process.

The significance of these findings lies in their practical application in the used car marketplace. Buyers often struggle to determine whether a vehicle is priced fairly, while sellers may find it difficult to decide the correct selling price. The developed system provides a data-driven solution that helps both parties make more informed decisions. By providing an estimated fair price, the model improves transparency and trust in the buying and selling process.

Additionally, the results of this project highlight the growing importance of Artificial Intelligence and Data Science in real-world decision-making systems. The ability of machine learning models to analyse large datasets and produce accurate predictions demonstrates their potential for use in various industries beyond automobile pricing, such as real estate valuation, product pricing, and financial forecasting.
Overall, the key findings of this project confirm that AI-based predictive models can effectively support price estimation tasks and contribute to more efficient and reliable market transactions. The implementation of such systems can help modernize

traditional pricing methods and promote the adoption of intelligent decision-support technologies in the automotive sector.

7.3 Limitations of the Study

Although the Fair Car – AI-Based Used Car Price Prediction project successfully demonstrated the use of Artificial Intelligence in estimating used car prices, the study has certain limitations that should be considered while interpreting the results. These limitations mainly arise due to constraints related to data availability, model complexity, and real-world market factors.

One of the major limitations of this study is the size and diversity of the dataset used for training the model. The dataset contained a limited number of records compared to the vast and continuously changing used car market. Because of this, the model may not fully represent all types of vehicles, brands, and regional market conditions. A larger and more diverse dataset would improve the ability of the model to generalize and produce more accurate predictions.

Another limitation is that the model considers only a specific set of features, such as manufacturing year, fuel type, kilometres driven, transmission type, and number of previous owners. However, in real-world scenarios, several other factors can influence the price of a used car. These factors include the physical condition of the vehicle, accident history, service records, brand popularity, location of sale, demand in the market, and additional features like infotainment systems or safety equipment. Since these variables were not included in the dataset, the prediction model may not capture the complete market dynamics.

The project also relies on historical data, which means that the model predicts prices based on patterns learned from past records. However, market prices can change over time due to economic conditions, fuel price changes, technological advancements, or changes in consumer preferences. As a result, the model may require periodic retraining with updated data to maintain its accuracy.

Another limitation is related to the choice of machine learning algorithm and model complexity. While the selected algorithm provided satisfactory results for the scope of this project, more advanced models or ensemble methods could potentially improve prediction performance. However, implementing highly complex models may require more computational resources and a larger dataset.

Additionally, the current implementation focuses mainly on the prediction model rather than a fully developed application system. The project demonstrates the concept and working of AI-based price prediction but does not include a complete real-world deployment such as a user-friendly web interface, mobile application, or integration with live automobile market data.

Despite these limitations, the study successfully proves that machine learning techniques can be effectively applied to predict used car prices. The identified limitations also provide valuable insights for further improvements and future development of the system. By addressing these constraints in future research, the accuracy, reliability, and practical usability of the model can be significantly enhanced.

7.4 Future Scope and Recommendations

The Fair Car – AI-Based Used Car Price Prediction project successfully demonstrates how machine learning techniques can be used to estimate the price of used cars. However, there are several opportunities for further improvement and expansion of the system. Future work can focus on enhancing the model accuracy, increasing the dataset size, and developing a complete real-world application for practical use.

One of the main areas for future improvement is the expansion of the dataset. A larger and more diverse dataset that includes vehicles from different regions, brands, and market conditions can significantly improve the performance of the prediction model. Incorporating updated and real-time market data will allow the system to adapt to changing trends in the automobile industry and provide more accurate price predictions.

Another important future enhancement involves including additional features in the dataset. In the current system, the prediction is based on a limited number of attributes such as year, fuel type, kilometres driven, and transmission type. In future versions, more detailed features such as vehicle condition, accident history, service records, insurance status, engine capacity, colour, and additional accessories can be incorporated. Including these variables will help the model capture more complex relationships and improve the reliability of the predicted price.
The use of advanced machine learning and deep learning algorithms can also be explored in future research. More sophisticated techniques such as ensemble learning methods, gradient boosting models, or neural networks may improve the accuracy and robustness of the price prediction system. These models can handle complex patterns in large datasets and can potentially outperform basic regression models.

Another promising direction is the development of a user-friendly application interface. The current project mainly focuses on the prediction model and its implementation. In the future, the system can be integrated into a web application or mobile application where users can easily input vehicle details and instantly receive an estimated market price. Such an application could be useful for car buyers, sellers, dealers, and automobile marketplaces.

Integration with online automobile marketplaces and APIs can also be considered. By connecting the system with real-time car listing platforms, the model can continuously learn from newly available data and update its predictions accordingly. This would make the system more dynamic and practical for real-world deployment.
Additionally, the system can be extended to provide decision-support features, such as suggesting the best price range for selling a vehicle, identifying undervalued cars in the market, or providing recommendations based on buyer preferences. These advanced features would enhance the usefulness of the platform and provide greater value to users.

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